

AI DC White Paper 2026



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A Reference for CIOs on Planning and Building AI DCs

Released By

China Construction Bank, Huawei Technologies Co., Ltd.



Yang Chaobin

—Executive Director of the Board, CEO of the ICT Business Group, Huawei

In the past two years, the average daily token consumption worldwide has surged nearly 300-fold, with over 30 million AI agents now deployed across manufacturing, finance, energy, and other industries. AI is expanding from technological exploration to large-scale production, and the infrastructure underpinning this change—AI DCs—is undergoing a fundamental transformation.

The design logic of traditional DCs can no longer meet the demands of the agentic world. Every aspect of AI DCs is being redefined, from the power supply architecture to cooling solutions, from computing infrastructure to critical product forms, and from construction models to O&M. This is not merely a technological upgrade; it is a revolution of business logic.

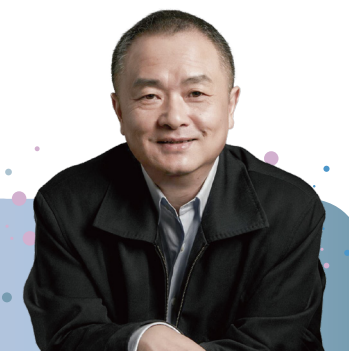
This white paper draws on Huawei's practical experience and systematic considerations in AI computing infrastructure. It offers a reference for enterprises to build an AI computing foundation for the agentic world, covering everything from architecture planning and project implementation to technology selection and ongoing operations. As the intelligent transformation advances, we hope this white paper will contribute to the innovative development of AI DCs worldwide.



Lei Ming

—Executive Vice President, China Construction Bank (CCB)

AI is rapidly reshaping all industries. AI DCs, as the core computing foundation for the digital economy, provide critical support for the real economy to go intelligent. CCB is a large state-owned commercial bank and has been actively implementing the "AI+" initiative. CCB has deployed the computing infrastructure with a moderately advanced approach and built a high-availability AI computing cluster with five centers in four cities. We are driving the digital and intelligent transformation of our entire business value chain with a solid computing foundation. This has in turn deepened our understanding of the strategic value of computing infrastructure. This white paper analyzes the underlying technical logic with a focus on practical industry applications, systematically explaining the roadmap for the collaborative development between AI and AI DCs. It offers both forward-looking insights and practical guidance, making it a valuable reference for industry peers. We look forward to collaborating with all stakeholders to build a thriving AI computing ecosystem, powering new quality productive forces with solid computing infrastructure, and driving high-quality growth in the digital economy.



Mei Hong

—Academician, Chinese Academy of Sciences; Professor, Peking University; Former President, China Computer Federation

Cutting-edge ITs, like the Internet, cloud computing, big data, and AI, are reshaping the way we live and work with unprecedented depth and breadth. These technologies are the core driving force for the latest technological revolution and industrial transformation that define our times. Recent breakthroughs in large model technologies and the rise of AI agents mark the dawn of agentic AI, a new era characterized by autonomous planning and task execution, unlocking a massive leap in AI's ability to assist in production and our daily life. All of these technological advancements are underpinned by the systematic upgrade of computing infrastructure. As a new generation of DCs designed to support AI workloads, AI DCs are the computing power factories that support the iteration of large models and the large-scale application of AI agents. This white paper explains the definition, technical architecture, construction path, and development trends of AI DCs, providing practical and forward-looking insights for industry, academia, research, and real-world application.



Li Peng

—Director of the Board, President of ICT Sales & Service, President of the China Region, Huawei

Huawei has been working closely with customers in finance, manufacturing, electric power, and many other industries to innovate in digital and intelligent transformation. We are pushing the boundaries of conventional experience in every aspect, from physical facilities to the computing infrastructure and intelligent O&M. AI is evolving from an auxiliary tool to a key driver of productivity, and AI DCs are advancing from conceptual exploration to systematic implementation.

Through extensive practice, we have realized that AI DCs are no longer an optional technical upgrade from traditional DCs, but rather a strategic foundation driving enterprises' digital and intelligent transformation. This is a major trend in industry evolution and a significant opportunity for pioneers.

This white paper aggregates the experience and insights that Huawei and industry customers have accumulated from real-world practice, aiming to provide a forward-looking yet practical reference for more enterprises to build and use AI DCs effectively. We look forward to using this as an opportunity to unite the industry, pooling our strengths to jointly create sustainable, efficient, and open AI infrastructure, accelerate the high-quality development of AI computing infrastructure, and share success in the agentic world.



Ma Haixu

—Corporate Vice President, President of ICT Product Portfolio Management & Solutions, Huawei

The advent of the agentic world is profoundly changing the underlying logic of AI infrastructure. SuperPoDs have become standard for building training and inference computing clusters, hierarchical KV cache has emerged as a key method to boost token efficiency, and the integration of general-purpose and AI computing, as well as data-AI integration, now forms the core foundation supporting tens of thousands of agents and EB-scale multimodal data. At the same time, traditional DC operations and O&M models are becoming obsolete, signaling a new chapter for AI DC planning and construction.

Huawei released its first *AI DC White Paper* in 2024, which introduced Huawei's preliminary exploration and insights in AI DCs. Over the past two years, we have participated in numerous customer projects as industries adopted agentic AI, gaining extensive experience and a deeper understanding of the future of AI DCs. This white paper is a complete update to the previous version. It examines the trends, innovation models, planning and construction, and operations and O&M for AI DCs in the agentic world. We hope that it will serve as a valuable reference for global AI DC planning and construction, and contribute to the high-quality development of AI computing infrastructure.



Wang Lei

—Chief Information Officer, China Pacific Insurance Group

Large models and agents are increasingly used in core business areas like intelligent services and risk-related decision-making. The explosive growth of intelligent applications has made computing power a major bottleneck for enterprises' intelligent transformation, elevating AI DCs to a strategic production system that underpins business innovation. Today, rapid model iterations and continuously upgrading application scenarios demand higher standards of dynamic adaptability and evolution ability from AI DCs. For enterprises' intelligent development, deploying infrastructure is just step one. The key to large-scale intelligent transformation lies in keeping pace with model iterations, ensuring that computing power continuously and efficiently enables business development and delivers tangible value. This white paper takes an end-to-end perspective that spans from resource investment to business output to break down the construction logic and operational paradigms of AI DCs. It provides a highly actionable reference for enterprises to accurately evaluate AI computing investments, build computing foundations designed for the intelligent world, and drive long-term business growth.



Wei Kai

—Director of the Artificial Intelligence Institute, China Academy of Information and Communications Technology (CAICT); Secretary General, Artificial Intelligence Standardization Technical Committee of the Ministry of Industry and Information Technology (MIIT/TC1)

With the rapid evolution of AI and the expanding application of large models and agents, AI DCs built for AI workloads are becoming the primary direction for computing infrastructure upgrades. This white paper, grounded in industry practices, outlines the technical architecture and construction path of AI DCs. It defines the key technical requirements of core elements such as AI facilities, AI infrastructure, and AI agents, and delves into cutting-edge topics like the characteristics of computing demand in the agentic world and the evaluation of token service capabilities. With its systematic framework and practical insights, this white paper will play a pivotal role in fostering industry consensus and driving industry development.



Liu Kang

—President of ICT Marketing & Solution Sales, Huawei

From hyperscale model training to production-grade real-time inference, token output efficiency is becoming a key metric for measuring the value of AI computing infrastructure. AI DCs are not merely about scaling up traditional IDCs with more computing power. Instead, they involve a systematic engineering approach that encompasses compute scheduling, storage-compute-network synergy, power supply, and heat dissipation. This directly determines the efficiency of converting chip computing power into business value and serves as the core foundation for deploying large models and agents at scale. This white paper outlines the technical evolution logic and feasible architecture solutions for AI DCs, and presents forward-looking, practical benchmark cases from various industries, including finance, healthcare, and large-scale AI DCs. We hope these insights will provide valuable references for enterprises of all sizes at different stages, and we look forward to furthering our exploration and refining our practice with industry peers.

Contents

01

Innovation and Development of AI DCs



1.1 Five Major Trends of AI DC Development — 08

- Agents Emerge as the Primary Variable in AI DC Loads — 09
- Models Evolve Toward General Intelligence — 10
- Scaling Laws Transform Inference Scenarios — 10
- The Era of SuperPoDs Is Now — 10
- Energy Supply Becomes a Strategic Constraint for AI DCs — 10

1.2 New Paradigm of AI DCs in the Agentic Era — 11

- Traditional Planning Methods Are Facing Failures — 11
- A New AI DC Paradigm Centering on "3T" and "3P" — 12

02

Planning and Construction of AI DCs



2.1 Agent-Centric Applications and Ecosystems — 15

- Planning Based on the AKDM Reference Framework — 15
- Enterprise Agent Implementation in Four Phases — 20
- Agents Are Driving AI Infrastructure Boundary Reform — 21

2.2 Token-Centric AI Infrastructure — 22

- Core Changes of AI Infrastructure Compared with Cloud Infrastructure — 22
- Token Generation Efficiency Evaluation System — 22
- Key Constraints and Core Technologies for Token Generation Efficiency — 25
- Target Reference Architecture for Token-Centric AI Infrastructure — 30
- Four Capability Circles Reshaping Token-Centric AI Infrastructure — 32
- Key Directions for Enterprise AI Infrastructure Planning and Construction — 49

2.3 Watt-Centric AI Facility — 51

- Systematic Challenges of AI Facilities — 51
- Suggestions on AI Facility Planning and Construction — 53
- Implementation of AI Facilities in New and Existing Scenarios — 56

03

AI DC Operations, O&M, and Security



3.1 AI DC Intelligent Production Operations — 59

- From Compute Resource Operations to AI Production Operations (AIPO) — 59
- Five Core Capabilities of AIPO — 62
- Case Study: AI DC Value Operations Upgrade for Banking — 64
- Evolution Path of AIPO — 66

3.2 AI DC Autonomous O&M Assurance — 67

- From Device O&M to Intelligent Production O&M Assurance — 67
- Six Core Capabilities of Agentic Autonomous O&M — 68
- Agentic Autonomous O&M Reference Architecture — 72
- Case Study: Intelligent Fault Demarcation and Quick Recovery Based on Agentic O&M — 74
- Evolution Path of Agentic Autonomous O&M — 75

3.3 AI DC Security Defense — 76

- Internal and External Security Challenges to AI DCs — 76
- Core Security Protections for AI DCs — 77
- AI-Powered AI Defense Security Reference Architecture — 81

04

Success Stories



4.1 Huawei Cloud Wuhu 3D DC — 85

4.2 CCB's AI DC Construction and Operational Transformation — 87

4.3 Zhongshan Hospital's AI DC Practice in the Healthcare Industry — 92

01

Innovation and Development of AI DCs



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As AI becomes a core productivity for the digital economy, DCs are making a profound and critical shift. They have transcended their traditional role of merely holding computing power and data, and have become strategic production resources in the intelligent world. Old paradigms can no longer explain new workloads, and the curtain has been raised on a paradigm shift.

Throughout human history, each leap in infrastructure has fundamentally transformed civilization. The steam engine liberated power, surmounting physical constraints; electricity unlocked energy, illuminating industries and homes; the internet revolutionized information flow, shattering barriers of knowledge and time.

Now, the fourth leap is upon us, set to redefine the nature of intelligence. From advanced pre-trained models to widespread distributed inference, and from passive tools to autonomous agents that sense, plan, and execute tasks, intelligence is permeating enterprises and devices with unparalleled speed and depth.

DCs are undergoing a profound transformation: they are evolving from mere repositories of compute storage, and networking resources into dynamic hubs that continuously generate intelligent capabilities. This shift isn't just a technological overhaul of IT architecture; it's a systemic revolution poised to shape productivity for decades to come.

1.1 Five Major Trends of AI DC Development

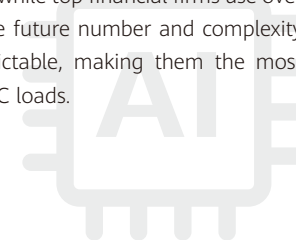
Agents Emerge as the Primary Variable in AI DC Loads

Agents are propelling AI from "cognitive intelligence" to "action intelligence," turning it into an "actor" capable of autonomous planning, tool invocation, and closed-loop execution. This shift transforms AI from an auxiliary "dialog box" into a "digital employee" that can independently manage business processes, fundamentally reshaping traditional enterprise applications.

From early 2024 to 2026, the average daily token usage in the Chinese market skyrocketed from 100 billion to 140 trillion. This surge is driven by the exponential increase in token demand from agents. Complex agent tasks often require multiple rounds of reasoning and tool invocations, triggering dozens or even hundreds of Large Language Model (LLM) calls, with context windows spanning millions of tokens.

Simultaneously, the inference cost per million tokens has dropped by about an order of magnitude in two years. This dramatic reduction in costs has further spurred demand for agents and tokens, aligning with the Jevons paradox: Increased AI efficiency doesn't reduce computing power demand; it amplifies it.

Today, it's common for AI developers to consume hundreds of millions of tokens daily, while top financial firms use over 10 billion tokens daily. The future number and complexity of agents remain unpredictable, making them the most significant variable in AI DC loads.



Models Evolve Toward General Intelligence

The development path of foundation models toward Artificial General Intelligence (AGI) is marked by these key trends:

- Ultra-large parameter models are pushing capability boundaries, shifting from "fast thinking" to "slow thinking" and matching or exceeding human expert levels in programming, mathematical reasoning, and scientific research.
- Small-and medium-sized parameter models are enhancing intelligence efficiency and lowering inference costs. These models optimize intelligence output per token and per unit of computing power through techniques like model architecture optimization, compression, Mixture of

Experts (MoE), and algorithmic innovation.

- Models are transitioning from static training to continuous learning and self-optimization, refining their capabilities based on data, feedback, and environmental changes. For instance, in OpenAI's GPT-5.6 series, the lightweight Luna model's post-training is autonomously managed by the flagship Sol model.
- The evolution of model capabilities demands more from AI DCs, requiring AI DCs to support ultra-large computing loads and efficiently manage resources in hybrid training and inference scenarios.

Scaling Laws Transform Inference Scenarios

Traditional scaling laws dictate that performance improves predictably with increases in model parameters, training data, and computational power. Now, these principles are expanding into the inference phase. Inference outcomes are no longer dictated solely by model size; the amount of computational power used during inference has emerged as a critical factor.

Boosting inference computing power yields benefits akin to scaling up model parameters, enhancing capabilities through extended reasoning, more intricate searches, and robust self-verification. Consequently, AI DCs face simultaneous increases in computational demand per request and inference latency.

The Era of SuperPoDs Is Now

As single-chip performance nears its physical limits, a comprehensive reconstruction of the computing power optimization chain is essential. Huawei's Tau (τ) law exemplifies this, achieving system-level breakthroughs through collaboration across components, circuits, chips, and systems.

Systematic engineering innovations interconnect computing, networking, and storage infrastructure via

buses, further boosting overall performance through resource pooling. Both Huawei Ascend SuperPoDs and its industry peers embrace this mega-trend.

At the DC level, new systematic solutions are emerging. Beyond SuperPoDs, innovations like Prefill-Decode (PD) disaggregation and multi-level Key-Value (KV) cache are reshaping the design paradigms of AI DCs.

Energy Supply Becomes a Strategic Constraint for AI DCs

The power density of computing power continues to rise. The power of a single cabinet has increased from tens of kW to hundreds of kW, and will exceed 300 kW in the medium and long term. The traditional air cooling solution can only support a maximum of 20 kW per cabinet, which is insufficient to support high-density AI computing power.

As a result, **liquid cooling has shifted from an option to a necessity.**

Ultra-large DCs have also scaled from the MW level to the GW level. The power consumption of a GW-level AI DC is comparable to that of a city with a million population.

Therefore, energy supply is no longer just a cost issue but a strategic constraint.

These five trends point to the same direction: DCs are standing at a critical point of paradigm shift.

1.2 New Paradigm of AI DCs in the Agentic Era

Tokens are now the fundamental measure of computing power. DCs have evolved beyond mere server storage; they're transitioning from resource suppliers to computing power producers. The metric has shifted from available

XPU resources to token output. Enterprise CIOs confront a significant challenge: Can traditional methods for planning and building AI DCs adapt to these AI-driven shifts.

Traditional Planning Methods Are Facing Failures

Service planning fails, making load capacity difficult to predict.

In the agentic AI era, traditional software applications and tools are shifting to the background, becoming the underlying infrastructure for AI agents. A new agent or model can rocket from obscurity to viral fame across the network in just three months, sparking a hundredfold surge in token demand.

Linear thinking based on historical service development no longer suffices for predicting future service planning. Forecasting service growth and load over the next three to five years has become increasingly challenging.

For the first time, DCs cannot predict their future scale. Consider Google (Alphabet): Its AI capital expenditure (CAPEX) surged from US\$31 billion in 2022 to an estimated US\$200 billion by 2026, a sixfold increase in just four years.

The architecture paradigm fails, and the era of layered optimization ends.

The architecture of traditional DCs assumes slow technological change and stable systems, making layered optimization of cloud resources, computing/storage/network layers, and facilities seem logical.

However, the rapid pace of technology iteration in the AI era has shattered this assumption. Model updates now occur quarterly or even monthly, and chip microarchitecture cycles have shortened from 24 months to 12-18 months.

The traditional layered decoupling architecture is no longer viable. AI service loads require simultaneous and coupled increases in computing, storage, and networking resources, demanding a system-level collaborative optimization design. The concept of a "stable system" is yielding to a "continuously changing system."

At the physical infrastructure level, the power consumption of a single cabinet has skyrocketed from 5-8 kW to 40-120 kW. Air cooling is inadequate, making liquid cooling essential. This shift requires redesigning floor load-bearing capacity, ceiling height, and power supply. More critically, there is a time mismatch: enterprises aim to deploy NPUs within 3-6 months of acquisition, but traditional DC construction takes 12-24 months. Equipment room preparation cannot keep up with the pace of IT equipment iteration.

O&M assurance fails, and AI breaks through the original O&M boundaries.

Traditional DC operations are device-centric, focusing on server availability, network connectivity, and environmental conditions to ensure uninterrupted resource delivery.

In the AI era, the focus of DC O&M has fundamentally shifted. Service links are now longer and involve cross-domain calls, where any anomaly in model inference, tool execution, or data access can degrade service quality. The traditional device-centric monitoring system is no longer effective.

DCs must also build enterprise token operations capabilities to support the surging AI demands of business units.

AI introduces significant security threats. Autonomous agents within DCs can plan and execute tasks, leading to

risks like unauthorized access and data leaks. External hackers are leveraging AI foundation models to launch sophisticated automated attacks, leaving traditional defense systems at a generational disadvantage.

A New AI DC Paradigm Centering on "3T" and "3P"

AI computing power has become a key production factor, transforming DCs from cost centers to value centers. Their worth is now measured by continuous value generation rather than resource quantity. Industry public data shows that from 2025 to 2026, AI-related capital expenditure (Capex) of China's leading tech companies generally increased by more than 50% year-over-year; among them, ByteDance's investment in AI infrastructure in 2026 has been raised to over CNY200 billion.

Leading enterprises' AI infrastructure teams are evolving

from "O&M cost centers" to "AI productivity operations departments." The focus has shifted from server count to daily effective token output.

Planning and designing AI DCs now require a new systematic approach, encapsulated in the "3T" and "3P" reference framework:

"3T" Reference Framework

T1: Agent(application-driven) → T2: Token(computing power operations) → T3: Wat (physical support)

The three elements form a complete value chain: Agents drive token consumption, token consumption determines computing power requirements, and computing power requirements dictate watt consumption. Planning and design must balance trade-offs at each phase. Ignoring any of the three "T" elements renders AI DC design incomplete.

"3P" Reference Framework

P1: Production(intelligent production and operations) →
P2: Provisioning(autonomous O&M assurance) →
P3: Protection(security defense)

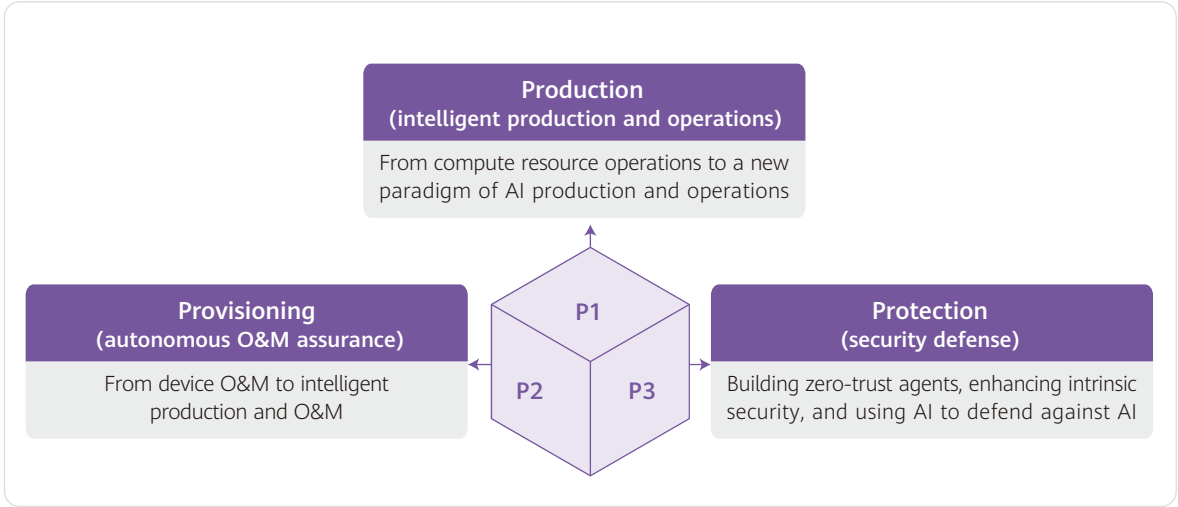


Figure 1-2 "3P" reference framework

Production and operations drive intelligent value creation, using tokens as the unit of measurement. O&M assurance ensures system continuity and reliability through autonomous methods. Security control safeguards services

follows the zero trust principle, and uses AI to defend against AI attacks. These three elements support and collaborate to ensure the stable transformation of AI DCs from computing power resources to intelligent value.

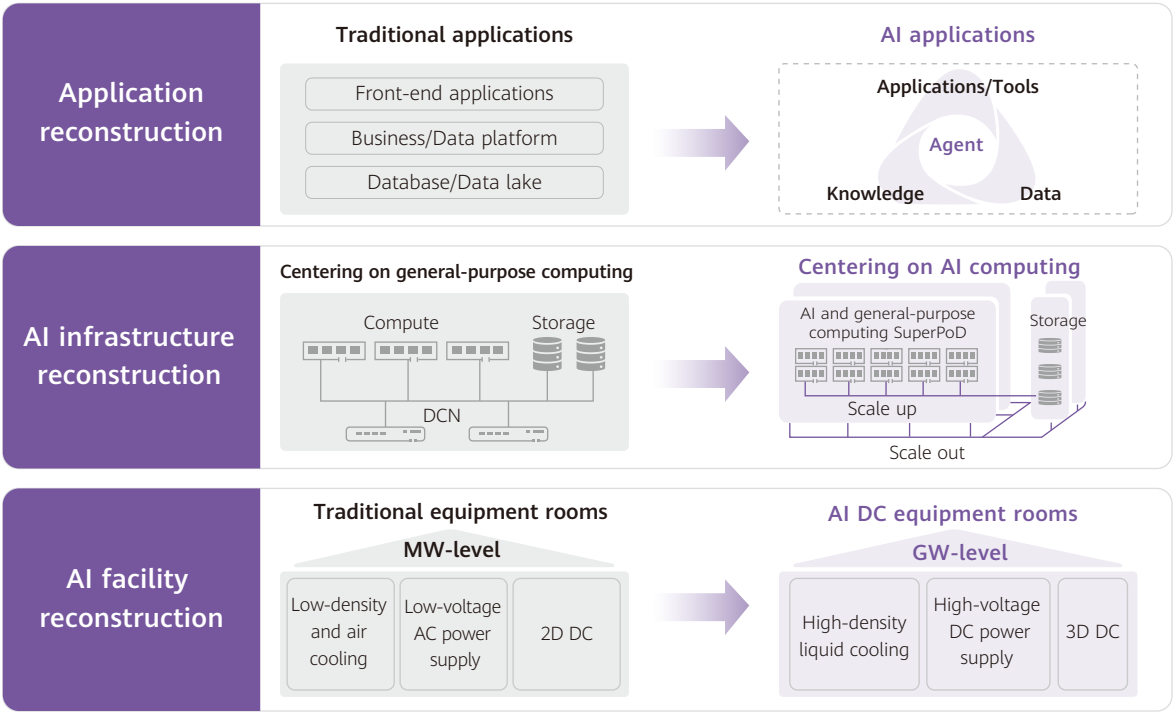


Figure 1-1 "3T" reference framework

02

Planning and Construction of AI DCs



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Every industrial revolution starts with rebuilding infrastructure. When AI demand evolves faster than construction cycles allow, DC planning and engineering rules must be completely rewritten. A true paradigm shift isn't drawn on paper; it must materialize as an engineering system that can be implemented and delivered.

2.1 Agent-Centric Applications and Ecosystems

In the agentic AI era, enterprise AI applications are evolving from single model invocations to the comprehensive delivery of long-term, complex tasks, marking a fundamental leap from "tokens" to "tasks." This shift requires enterprises

to build agent-centric applications and ecosystems, integrating AI into core production scenarios.

Planning Based on the AKDM Reference Framework

When agents are integrated into the core production processes of enterprises, model capabilities alone are insufficient. Agents need knowledge to understand business boundaries, data to support accurate execution, and models to drive inference and decision-making. Building enterprise AI applications and ecosystems based on the AKDM framework is key to large-scale AI implementation:

Agent: Serves as the execution center, converting organizational intent into tasks and actions.

Knowledge: Provides enterprise cognition, transforming business rules, process experience, and ontologies into reusable assets.

Data: Ensures trusted facts, guaranteeing every step is executed based on accurate, real-time, and traceable information.

Model: Acts as the inference engine, supporting understanding, generation, judgment, and planning.

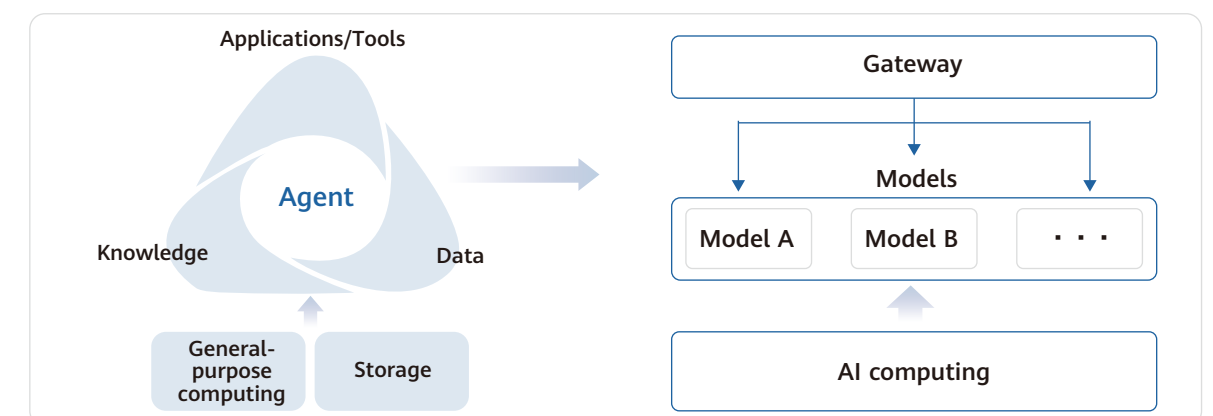


Figure 2-1 Building agent-centric applications and ecosystems

Agent: Building a Production-Grade Platform for Multi-Agent Collaboration

Agents are becoming the new execution hub and business entry point for enterprises. The core of a production-grade multi-agent system lies not in the sheer number of agents, but in establishing clear task division, collaboration protocols, and cyclic optimization mechanisms.

In a typical multi-agent system, a Lead Agent serves as the task scheduling hub, responsible for task decomposition, priority management, context distribution, and result aggregation. Multiple Sub-Agents handle detailed tasks such as architecture design, code development, and quality evaluation. Each Sub-Agent operates within defined responsibilities, permissions, and tool boundaries to prevent excessive authority and context overload. Enterprises need to define a clear collaboration

contract: Who breaks down tasks, who has execution permissions, who validates results, and who is accountable for the final output.

To improve the success rate of long-duration, complex tasks, a multi-agent system must introduce a three-level cyclic mechanism, forming a self-evolving Harness closed loop. The Lead Agent optimizes task organization, Sub-Agents refine execution, and the Agent Platform enhances the runtime environment and collaboration mechanisms. Continuous learning from tasks to the platform reduces reliance on manual scheduling, ensuring stable, scalable, and resilient execution of complex tasks under high uncertainty.

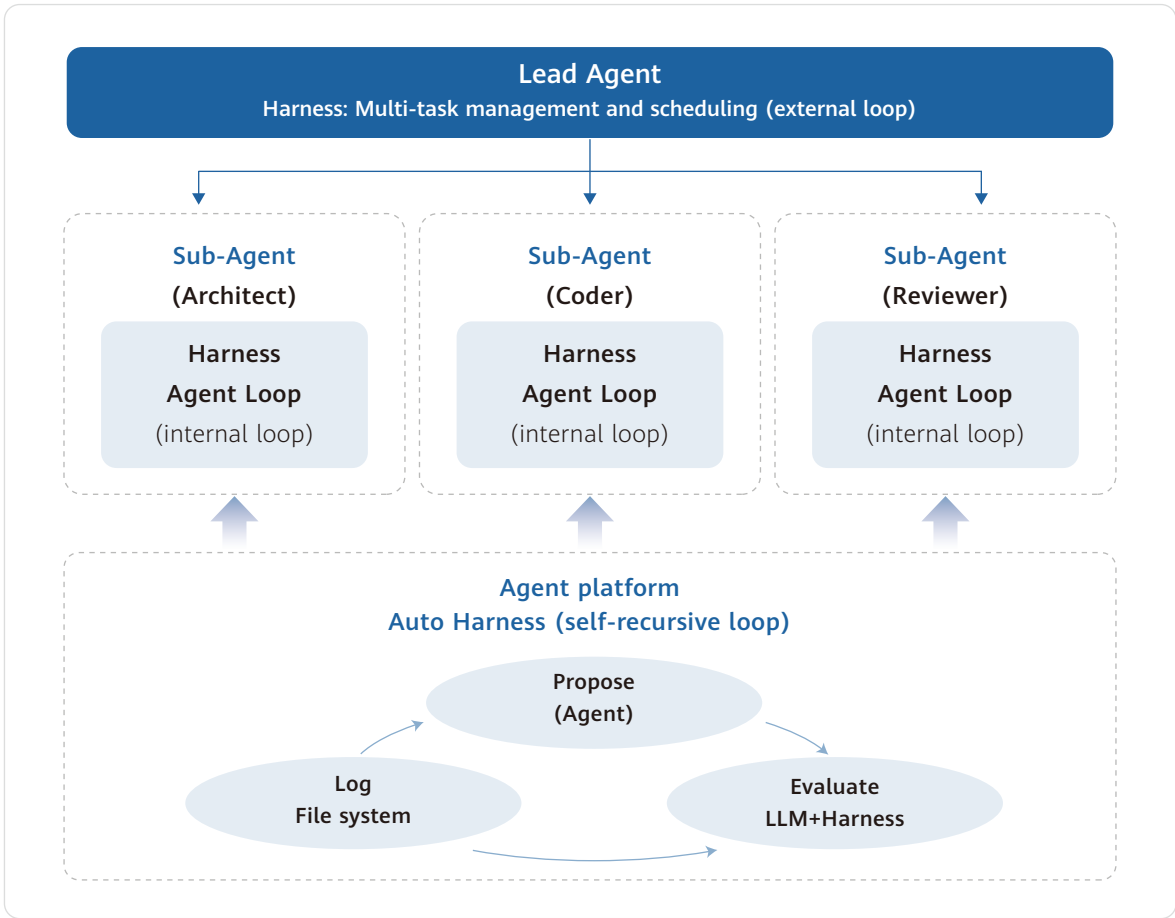


Figure 2-2 Multi-Agent Harness loop mechanism

The production-grade operation of multiple agents requires a three-layer Agent platform, each addressing specific challenges:

Agent framework: Handles intelligent task orchestration, encompassing core capabilities such as context engineering, hierarchical memory, multi-agent collaboration, self-evaluation and evolution, tool orchestration, and task planning.

Agent distributed runtime: Provides distributed, large-scale, and reliable running capabilities for agents, covering distributed state management, task queues,

checkpoints, cross-agent communication, resource scheduling, and compute-affinity scheduling. The core value is ensuring reliable execution even under faults, concurrency, tenant isolation, and resource fluctuations.

Agent system service: Offers secure and reliable system services for agents, including system sandboxing, Command-Line Interface-native execution, security isolation, file and process access control, code execution environment, and tool invocation boundaries. Its primary function is to ensure agents do not overstep boundaries when invoking system capabilities.

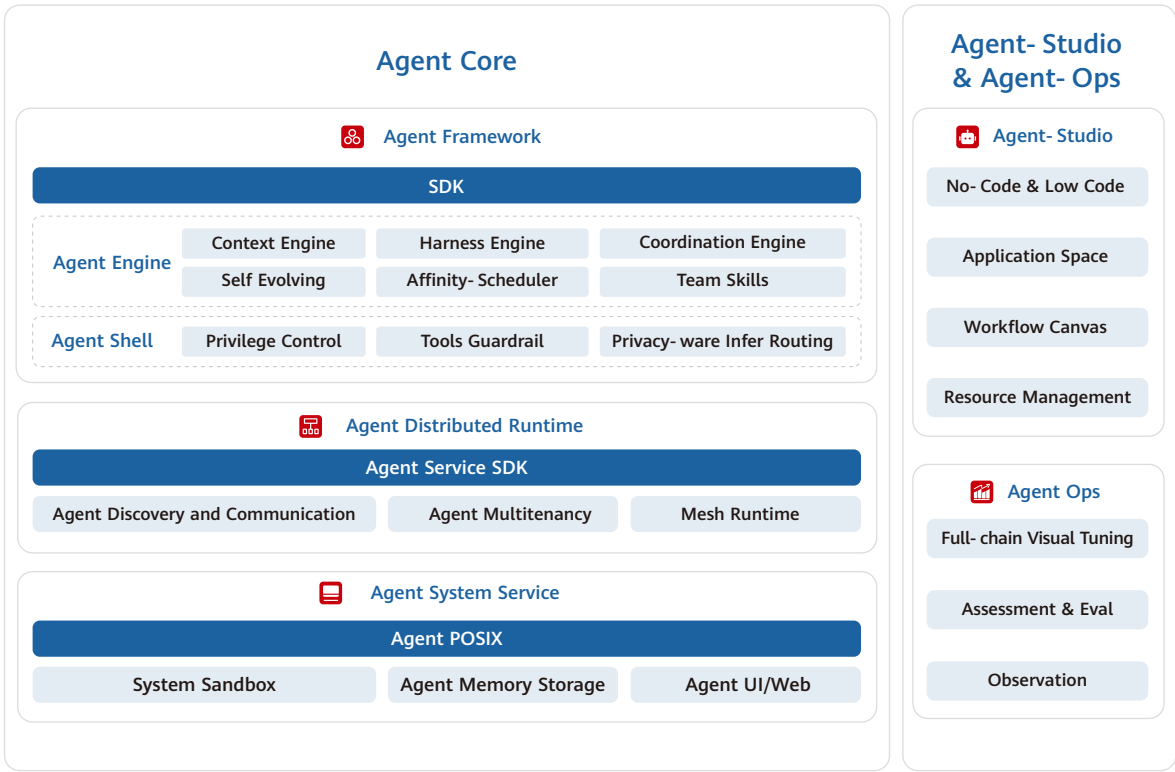


Figure 2-3 Agent platform reference architecture

Knowledge: Enabling Agents to Plan and Execute Tasks Based on Enterprise Logic

Enterprise knowledge is a cognitive asset shared, accumulated, and governed over time. It includes institutional documents, business specifications, industry ontology, knowledge graphs, process experience, executable business rules, and more, enabling agents to understand industries, organizations, and processes.

The construction of enterprise knowledge faces a fundamental contradiction: knowledge is scattered across documents, processes, systems, and human minds, while agents require structured and inferable cognitive assets. To resolve this, a knowledge lake must be built. On a unified data foundation, the knowledge lake transforms implicit enterprise experience into structured knowledge assets consumable by agents through semantic processing. The construction of the knowledge lake follows the progressive path of "collection—extraction—modeling—service."

The collection layer interconnects with multiple knowledge carriers, including institutional documents, operation manuals, FAQs, historical service tickets, and unstructured data, as well as structured rules in business systems.

The extraction layer uses foundational models to automatically extract entities, relationships, and business rules from unstructured documents. After manual review and verification, high-quality "fact-reasoning" triplets are generated.

The modeling layer utilizes ontology as the semantic framework to define the core concept system and constraint rules of an enterprise. Without ontology, a knowledge graph would be a scattered mess, treating "supplier" and "vendor" as different concepts, and "Jason" and "JASON" as two separate entities. Ontology embeds concepts and relationships defined by standards such as RDF and OWL into the entire lifecycle of agent perception, inference, and decision-making, fundamentally solving the problem of "what does the data mean?"

The construction, operation, and governance of enterprise ontology assets can be implemented on the agent platform, with three roles—ontology modeling agent, business experts, and application verification agent—collaborating to drive high-quality ontology models from design to application.

The ontology modeling agent converts domain knowledge into ontology models that comply with semantic specifications, offering automated modeling and consistency verification.



Business experts participate in modeling from a business semantics and domain rules perspective, reviewing and optimizing concept definitions and relationship semantics to ensure accuracy in reflecting real business logic.



The application verification agent conducts semantic inference verification and application testing in specific scenarios, assessing the availability and effectiveness of ontology models in intelligent applications.



Through the platform's collaboration mechanism, these three roles form a closed loop of knowledge modeling, business verification, and application verification, continuously enhancing the quality of ontology assets.

The service layer provides agents with query, verification, and inference services through semantic mapping, rule engines, and standard APIs, enabling agents to form explainable judgments along the path of "entity-relationship-constraint."

Data: Building Traceable and Verifiable Factual Data for Agents

Data is the factual foundation for agents to understand the environment, assess status, and execute tasks. In the agentic era, data lakes, data warehouses, BI, master data, and data governance platforms will continue to exist. The main change in data management is the shift from 'human beings consume data' to 'models consume data,' a transformation that also restructures the design logic of the enterprise data foundation.

Enterprise data foundation needs to evolve from a data lakehouse primarily storing structured data to an AI-oriented multimodal data lake.

Structured data provides precise facts, such as order amounts, inventory quantities, and customer tiers; unstructured documents offer business context, like contract clauses, due diligence reports, and compliance files; images, voice, videos, and logs capture onsite processes and behavioral evidence; while agent traces and user feedback supply the operational data needed for continuous optimization. The core of the multimodal data lake is to achieve unified governance and authorization

across different data types, providing stable and trustworthy data supply for knowledge and agents.

Data and knowledge work in synergy on a unified data foundation. Data represents the raw facts within the multimodal data lake, while knowledge is the semantic layer built upon it—extracting entities, relationships, and rules from the same data through ontology modeling and knowledge extraction. The data lake answers 'what data is available,' and the knowledge lake explains 'what the data means.' During inference, the agent discovers 'which datasets can be consumed' through the data asset catalog, understands 'what business meaning these data correspond to' via the knowledge catalog, and verifies 'where the data is and whether it is trustworthy' through the metadata catalog. These three catalogs collectively influence the accuracy of agent reasoning.

Therefore, enterprise data construction should follow a progressive path: Data ingestion into the lake → semantic labeling → knowledge building → agent consumption.

Model: Flexible Model Services for Task-Oriented Use

In the Agent era, the focus of enterprise AI shifts from selecting the strongest model to operating a diverse set of model services tailored to specific tasks. General, domain-specific, small, multimodal, and inference models each excel in unique scenarios. Enterprises balance capability, cost, latency, and security through strategic model combinations. Two key engineering components are essential: The model gateway and the observation system.

The model gateway serves as the unified entry point for agents to access model services. All agents must access the model pool via the model gateway, preventing direct binding to a single model API.

When selecting a model gateway, enterprises prioritize these four capabilities: **Unified access** for managing open-source and public cloud models; **fine-grained metering** for token-level tracking by agent, task, department, model, and tenant; **policy-based routing** for dynamic model selection based on task complexity, latency, cost, security, and availability; **high availability** assurance with rate limiting, circuit breaking, retries,

fallbacks, gray releases, and quick rollbacks.

The observation system is crucial for model service engineering, covering both model-level and agent request-level observations.

- **Model-level observation:** Focuses on the performance, cost, and availability of model services. Key metrics collected include call volume, token throughput, response latency, error rate, per-token cost, and tenants' token usage. It also covers statistics on anomalies, such as rate limit and service degradation.

- **Agent request-level observation:** Targets end-to-end full-link observation, recording link IDs, model and tool invocation paths, token consumption, task status, and business outcomes. It also covers anomalies, such as manual intervention.

This answers three key questions: Why did a task succeed or fail, where were the main costs incurred, and is optimization needed for model routing, context management, or tool policies.

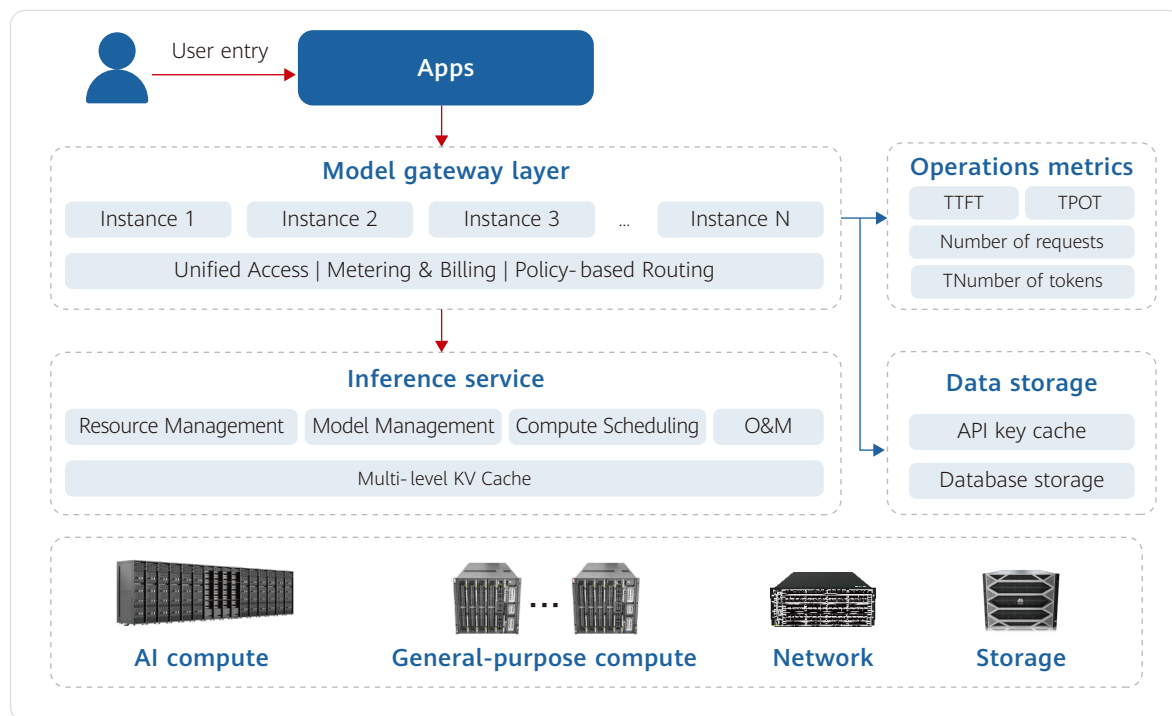


Figure 2-4 Reference architecture and open-source software selection for enterprise private token services

The engineering deployment of model services must form a seamless closed loop. Uniform agent requests are routed through the model gateway for authentication, routing, metering, and governance. The observation

system tracks metrics at both model and task levels. The evaluation system refines model selection and routing policies using success rates, costs, latency, and service outcomes.

Enterprise Agent Implementation in Four Phases

To build agent-centric enterprise AI applications and ecosystems, we should follow the principle of "value-driven, platform-supported, asset accumulation, and continuous operations," and steadily advance the construction in four phases:

Phase 1: Scenario preparation

Confirm the business value and whether data, semantics, knowledge, tools, and rules are ready. Then, set business objectives such as the task success rate, manual takeover rate, result adoption rate, and cost per task.

Phase 2: Production verification

Build a unified platform and standard interfaces to ensure that key actions can be verified, replayed, and audited.

Confirm that the Agent's quality, cost, permissions, security, task recovery, and business continuity meet production requirements.

Phase 3: Capability reuse

Continuously accumulate execution paths, exception patterns, best practices, job skills, business rules, and semantic assets, enabling knowledge, tools, and skills to be reused across agents, departments, and scenarios.

Phase 4: Large-Scale operations

Form a unified application portfolio, platform capabilities, intelligent assets, and organizational mechanisms, driving agent development from scattered projects toward an enterprise-level ecosystem.

In the long run, enterprise agentic AI development is a continuously running flywheel: Agents enter the business to create quantifiable value. During execution, they accumulate data, semantics, knowledge, skills, and experience. These accumulated capabilities are then

reused in more scenarios, further expanding business value. Enterprises can measure the flywheel's efficiency through task completion rates, capability reuse rates, time

Agents Are Driving AI Infrastructure Boundary Reform

To support enterprise AI applications, AI infrastructure is evolving its boundary to encompass three critical platform service capabilities:

Agent running environment that supports long-term tasks and multi-agent collaboration:

The agent running environment must be able to handle long-term tasks, from hours to days, ensuring continuity with persistent states and checkpoints.

For multi-agent parallelism, the platform should be able to offer secure, on-demand sandboxes that can scale elastically, with cold start latencies for lightweight containers kept in milliseconds at the 95th percentile.

A task-level observability system is essential, monitoring concurrent sessions, task success rates, data access, and tool usage, generating comprehensive task traces for fault localization and security audits.

Model running and services that provide composable and governable inference capabilities:

Model running and services deliver unified, orchestratable

model production and high-performance token inference, supporting on-demand features like PD disaggregation, large-scale Expert Parallelism (EP) for MoE models, and multi-level KV cache.

Data foundation services that centrally manage knowledge, data, long-term memory, and training and evaluation datasets

Data foundation services manage two key functions: Accumulating enterprise intelligence and supplying data for model training. Vector databases, graph relationship databases, and long-term memory services support the creation of enterprise knowledge lakes. Meanwhile, the multi-modal data lakehouse handles tasks like training dataset management, data cleansing and labeling, and lineage tracing, offering traceable and reproducible data for model training and evaluation.

Overall, AI infrastructure must evolve to a new paradigm that integrates these three platform capabilities and a peer-to-peer interconnection architecture for heterogeneous resources.

2.2 Token-Centric AI Infrastructure

Core Changes of AI Infrastructure Compared with Cloud Infrastructure

Workloads have shifted from "CBD" to "AKDM." Cloud infrastructure primarily handles business workloads like cloud computing, big data, and databases (CBD). With CPU at its core, cloud infrastructure is adapted to all types of data from various sources and compatible with diverse general-purpose business operations. In contrast, AI

infrastructure supports new workloads such as agent applications, enterprise knowledge bases, data lakes, and model inference and training. Relying on NPUs, these workloads are characterized by compute-intensity, high bandwidth, long duration, and significant power consumption.

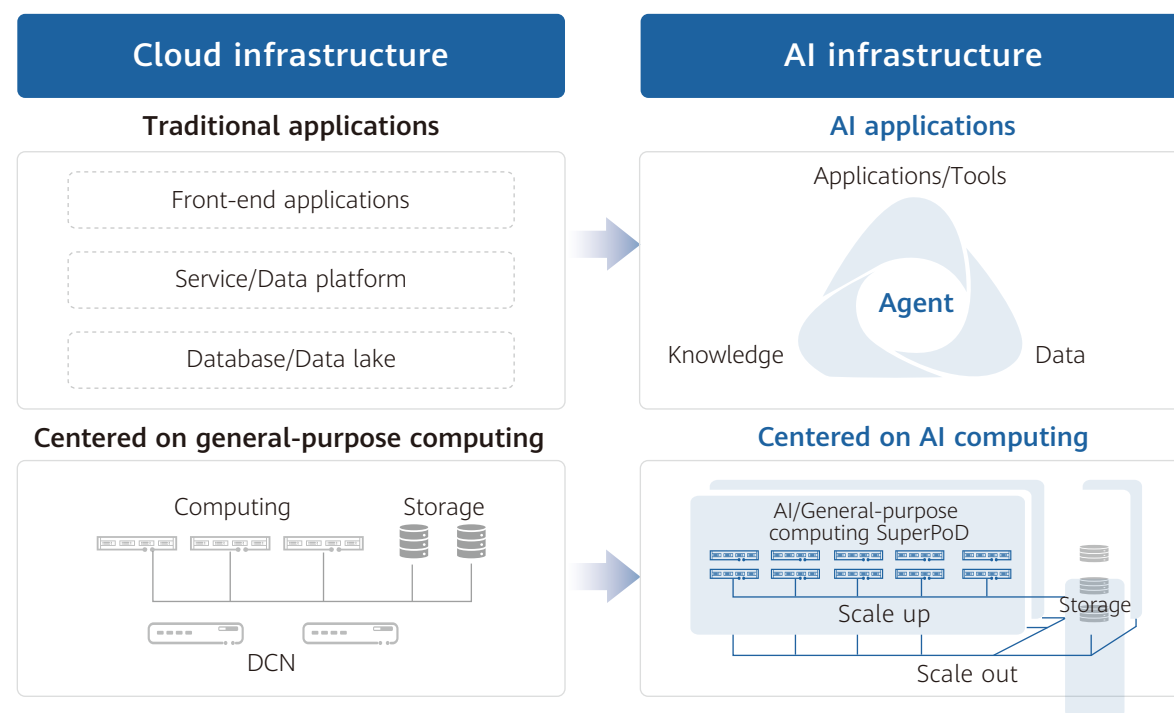


Figure 2-5 Core changes of AI infrastructure compared with cloud infrastructure

Tokens now serve as the value carrier connecting computing power, models, and user experience. The core mission of AI infrastructure is to ensure the continuous, efficient, and stable generation of high-quality tokens.

Token Generation Efficiency Evaluation System

Service scenarios vary with each other in token generation efficiency requirements.

Scenario Type	Typical Application	Service Characteristics	Token Production Efficiency Requirements	Typical Indicator
Internet 2C	Intelligent customer service, AI search, generative recommendation, and AI assistant	Hundreds of millions of users; tens of thousands of requests per peak period; typically hundreds of tokens per request, complex Q&A and inference consuming thousands of tokens	Low latency, high throughput, and low cost	TTFT<500 ms; tens of thousands of QPS per peak period; continuously increasing output TPS
Real-time decision-making	Financial risk control, transaction analysis, industrial diagnosis, and real-time operations	A single request is typically completed within seconds; hundreds to thousands of concurrent requests per peak period; result directly impacting business decisions	Low latency, high reliability, and high quality	TTFT < 500 ms; SLA ≥ 99.99%
Knowledge Q&A	Government Q&A, enterprise knowledge base, medical assistant, and professional consultation	The context typically ranges from tens of thousands to hundreds of thousands of tokens, involving multiple rounds of interaction, with significant knowledge retrieval and context reuse.	High quality, high consistency, and trustworthiness and controllability	TTFT<1s; knowledge accuracy>90%; result traceability
AI coding	Software development, code review, and automated test	A single task can involve 10 to 100+ model calls, with the context reaching thousands to hundreds to millions of tokens, and the task duration lasting hours or more.	Continuous generation, status reuse, and reliable tasks	TPOT<50ms; task completion rate>90%; support for long tasks lasting hours

Token production efficiency can be evaluated across four dimensions: performance, concurrency, cost, and quality.

Performance:

A user experience metric measuring the time from request initiation to receiving a valid response. Key indicators include:

Time to First Token (TTFT): The time from when a user request enters the system to when the first output token is returned.

Time Per Output Token (TPOT): The average time required to generate each subsequent token after the first. TPOT ensures smooth continuous output and is crucial for measuring the streaming interaction experience.

Concurrency:

A performance metric evaluating the AI system's ability to produce tokens continuously while meeting user experience constraints. Key indicators include:

Tokens Per Second (TPS): Measures token processing per unit time, split into output TPS and total TPS. The output TPS reflects the number of tokens generated per second, indicating the user-perceived speed. The total TPS shows the combined input and output tokens processed per second, reflecting overall system throughput.

Queries Per Second (QPS): Measures the number of requests processed per unit time.

The true engineering goal is not merely to generate more tokens, but to continuously produce more valid tokens while meeting user experience constraints. Effective TPS, the total number of tokens generated under user experience constraints like TTFT and TPOT, is the core metric for assessing real production capability.

To evaluate concurrent throughput, consider the specific service load, not just TPS. Optimization priorities vary across scenarios:

- In short Q&A settings like intelligent customer service, the focus is on high QPS and concurrent handling.
- In long text or code generation, the priority is improving the efficiency of token generation per request.

Cost efficiency:

Measures the resource cost required to produce a certain number of valid tokens, including hardware depreciation, power consumption, network transmission, storage, and software optimization.

Cost per million tokens: The core metric for economic efficiency, this decreases with model optimization, hardware upgrades, and improved inference efficiency.

Quality:

A Service Effectiveness metric evaluates the accuracy, stability, and service compliance of tokens. While speed and scale represent production capability, highly reliable and available tokens are the true carriers of service value. This includes the accuracy and consistency of content generation.

Token Production Efficiency Evaluation Indicator System

Evaluation Dimension	Evaluation Indicator	Indicator Description
Performance	TTFT	Response time of the first token, which determines the user waiting experience
	TPOT	Speed of continuous token generation, which determines the streaming interaction experience
Concurrency	TPS	Number of tokens processed per second, measuring the token processing capability per unit time, including the output TPS and total TPS
	QPS	Number of requests processed per second, measuring the number of requests the system processes per unit time
Cost Efficiency	Cost per million tokens	Cost of generating a single token
Quality	Accuracy	Whether the output meets business requirements
	Consistency	Stability and reliability of the output
	Trustworthiness	Whether the measurement results are verifiable and auditable

The goal of enterprise AI infrastructure development is not just to maximize computing power, but to find the optimal balance between performance, throughput, cost, and quality, building a token production system that best meets service requirements.

Key Constraints and Core Technologies for Token Generation Efficiency

Improving token generation efficiency is essentially about continuous optimization of the entire inference pipeline of large models. As model scale increases, context length grows, and real-time interaction requirements rise, the key performance bottleneck in the inference process is shifting from compute performance to data flow efficiency.

Therefore, the entire technology stack, from AI chip microarchitecture to system-level co-design across compute, storage, and high-speed interconnects, is evolving to improve the efficiency of inference data access and transfer. This includes improving compute unit

utilization, increasing memory bandwidth, optimizing communication efficiency, and efficiently managing, storing, and accessing the KV cache.

Data flow efficiency becomes the key bottleneck of inference

A complete large model inference process consists of two stages: Prefill and decode. These two stages have fundamentally different requirements on the underlying chip and system architecture, as shown in the following table:

	Prefill	Decode
Processing	After the user inputs a prompt, all tokens are processed in parallel at once, completing the attention computation and generating the KV cache.	The model predicts the next token one by one in an autoregressive manner.
Computing characteristics	Tokens are highly parallel, with intensive matrix computation.	Tokens are generated one at a time, requiring sequential computation and frequent data access.
Performance bottleneck	Large-scale matrix computation (Compute-Bound)	Data supply is limited by memory bandwidth (Memory-Bound)
Optimization directions	Compute optimization: • Improve the compute performance of AI chips. • Optimize parallel policies such as Tensor Parallelism. • Improve the Model FLOPs Utilization (MFU). • Match workloads with resources offering high compute performance after PD disaggregation.	Data flow optimization: • Compress, share, and hierarchically manage KV cache. • Improve memory capacity and bandwidth. • Optimize high-speed connections between chips. • Match workloads with high-bandwidth resources after PD disaggregation.

Long context increases the pressure on data transfer

Taking the DeepSeek V4 Flash model (284B total parameters) processing a 128k-length text as an example, a single request requires approximately 288 GB of resident model weights and KV cache state. This means that during the decode stage, every newly generated token requires scheduling and accessing **resident data with a total size of several hundred GB**. During this process, the chip's compute units may need less than 0.01 milliseconds to complete the computation, while more than 99.9% of the remaining time is spent waiting for data supply. The core challenge in the decode stage has shifted completely from computing faster to moving data faster.

The memory-compute gap further amplifies the bottleneck

In recent years, the growth rate of AI chip compute performance has far exceeded that of memory bandwidth, further intensifying the memory wall. For example, from NVIDIA's H100 to B200, the low-precision inference compute has increased by 4.5 times, while the memory bandwidth has only increased by approximately 2 times. The leap in compute performance can significantly shorten the prefill stage, but the decode stage struggles to achieve equivalent benefits due to bandwidth limitations, and the gap between computing and data supply continues to widen.

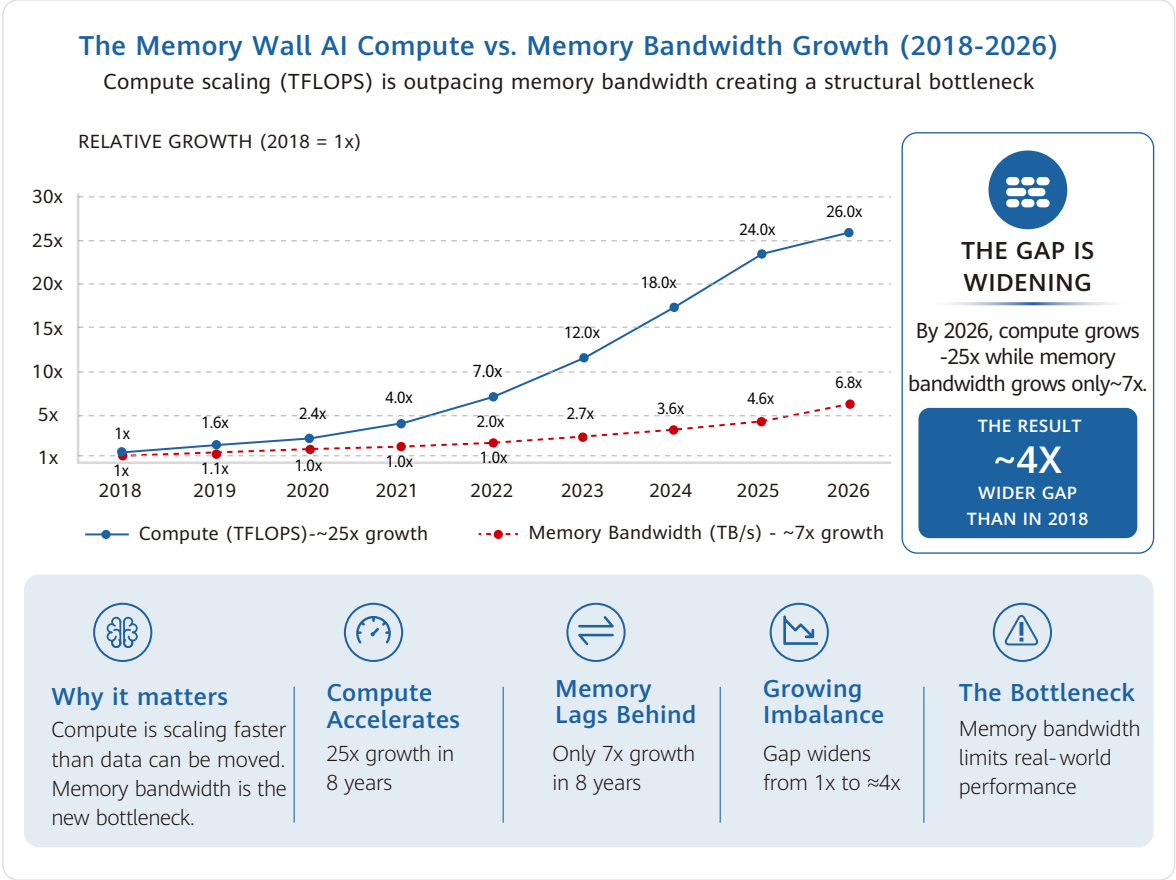


Figure 2-6 The Memory Wall AI Compute vs. Memory Bandwidth Growth (2018-2026)

Therefore, the key bottleneck of large model inference is comprehensively shifting from improving peak compute performance to improving data transfer efficiency. The future competition in token production efficiency is

essentially a comprehensive competition in the collaborative scheduling capabilities of chip memory access, interconnection transmission, and system-level data flow.

Key indicators of AI inference chips have become the key hard constraints on token production efficiency

For AI chips, the following four metrics affect token production efficiency:

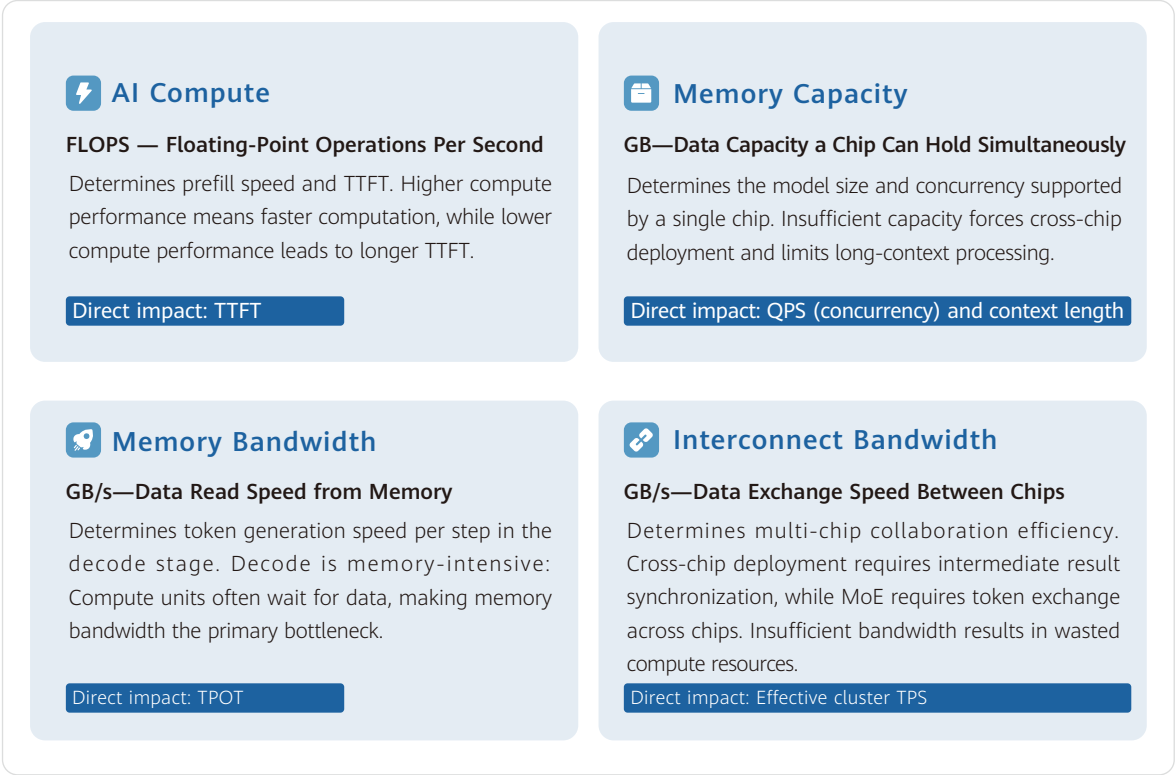
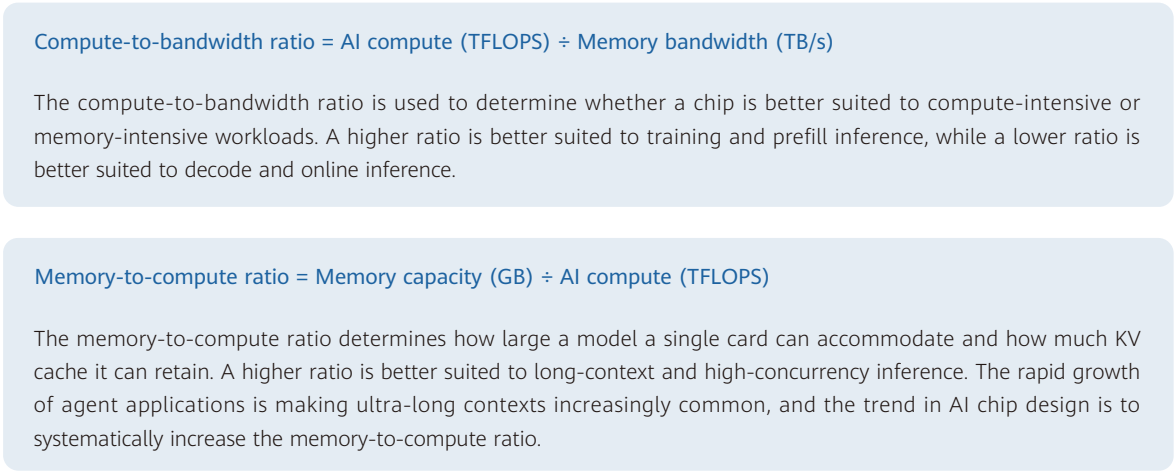


Figure 2-7 Four metrics determining token production efficiency

Large-model inference performance is constrained by the weakest of these factors. Therefore, the ratios among these four metrics can better reflect the balance of AI chip inference performance, including:



For AI chips, even with high compute performance, insufficient memory bandwidth will still constrain decode. Likewise, even with large memory capacity, insufficient interconnect bandwidth cannot support large-scale MoE collaboration.

Key technologies to break through token production bottlenecks

In addition to innovations in AI chip microarchitecture, the core evolution of AI infrastructure is also focused on data flow efficiency: **Matching different stages with different resources and moving data less frequently and delivering it to compute units faster.**

1. Architecture-level reconstruction: PD disaggregation

Traditional solutions deploy prefill and decode together in the same cluster, which can easily cause resource misallocation and performance interference (Head-of-Line Blocking):

- Resource requirement conflicts: Prefill is a compute-intensive workload, while decode is a bandwidth-intensive workload.
- Mutual interference in hybrid deployment: Long-prompt prefill requests consume substantial compute resources and affect the real-time response of decode. At the same time, to ensure low latency and SLA requirements for decode, the system must reserve memory and compute resources, resulting in low compute utilization for prefill.

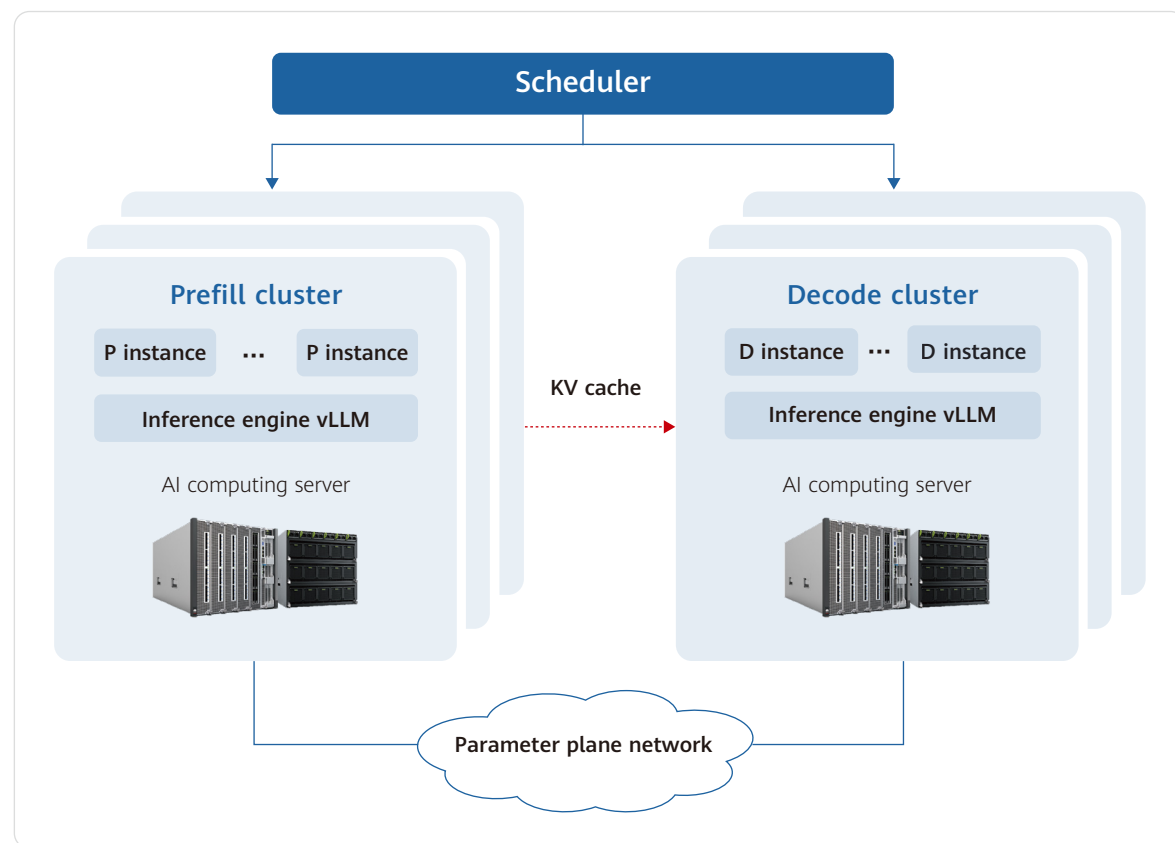


Figure 2-8 PD decoupling and disaggregation

P-D disaggregated architecture separates the two stages into different resource pools to achieve optimal matching of heterogeneous workloads:

- Prefill cluster: Focuses on high throughput and compute utilization. It typically uses high-compute-density chips to reduce Time to First Token (TTFT).

- Decode cluster: Focuses on memory bandwidth and communication interconnects to reduce the Time Per Output Token (TPOT).

Its essence is to address **resource mismatch between inference stages**, making compute resources better aligned with workloads.

2. Three optimization directions focused on data flow efficiency

On this basis, further optimization is carried out in three directions around data flow efficiency.

Reducing data movement: Minimizing inefficient access

As context lengths and the number of conversation turns grow rapidly, the storage and transmission overhead introduced by KV cache increases substantially. The core approach is to reuse computation results as much as possible and reduce unnecessary data flow:

- **KV cache optimization:** The storage scale and access pressure of the KV cache are reduced through compression, quantization, eviction, and efficient management policies. For example, vLLM's PagedAttention improves KV cache utilization through paged management.
- **Prefix cache:** The computation results of the common prefix context are reused to avoid repeated prefill computation, greatly reducing redundant data generation and movement, used in scenarios such as multi-round dialogs, fixed system prompts, and agent toolchain invoking.
- **Multi-level KV cache architecture:** Different storage levels, such as on-chip memory, DRAM, and SSD, are used to implement hierarchical management and dynamic scheduling of hot and warm KV caches based on data access frequency.

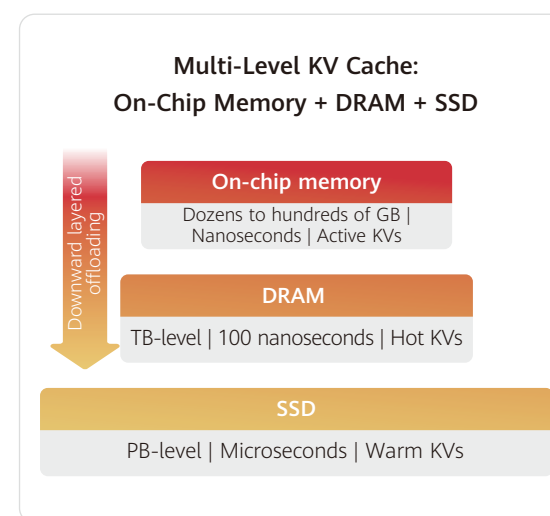


Figure 2-9 Multi-level KV cache

As agent task complexity continues to increase, multi-level distributed KV cache has become a key capability for AI infrastructure to reduce inference costs and improve resource utilization.

Improving data supply: Enhancing token production efficiency of a single node

Even if unnecessary data movement is reduced, a single compute node still faces great data supply pressure as the model scale and context length continue to increase. Especially in the decode stage, model parameters and KV cache need to be continuously supplied to the compute unit. Therefore, AI infrastructure needs to improve data supply of the entire node, so that the compute unit can continuously obtain required data and reduce the idle time of waiting for data.

- **High effective bandwidth storage system:** Combine high-specification storage media, advanced packaging, and data compression/quantization technologies to overcome storage throughput bottlenecks and maximize the amount of effective data delivered to compute units per unit of time.
- **Compute-storage-network co-design:** Optimize data-flow paths among CPUs, NPUs, and SSDs to reduce data flow overhead. For example, Huawei NPU Direct Storage (NDS) uses direct data-path technology to shorten data access paths.
- **SuperPoD architecture:** Organize multiple AI chips into a unified compute unit through high-speed peer-to-peer interconnect buses, overcoming the resource limitations of a single chip and a single server and increasing the model capacity and inference throughput of a single node. An example is Huawei's Atlas SuperPoD architecture.

Optimizing cross-chip data flow: Breaking through the bottleneck of large-scale model expansion

As the model scale continues to expand, especially as the MoE architecture becomes an important direction, model parameters and computing tasks are distributed to multiple chips or even nodes. In this case, the bottleneck is no longer the data access within a single chip, but the data exchange efficiency between different compute resources. Therefore, it is necessary to further optimize **cross-chip and cross-node data flow** to enable distributed compute resources to form an efficient and collaborative token production system.

- **High-speed chip interconnection:** Improve data exchange capabilities between multiple chips through technologies, such as high-speed interconnection technologies like Huawei UnifiedBus (UB) Link.
- **Efficient parallel computing architecture:** Optimize model partitioning and task scheduling to improve multi-chip collaborative computing efficiency, using distributed large model computing frameworks such as Megatron-LM and DeepSpeed, which support Tensor Parallelism, Pipeline Parallelism, and Expert Parallelism.
- **Communication-computation optimization:** Reduce communication waiting and improve distributed inference efficiency through technologies such as HCCL collective communication optimization and optimization of

All-to-All communication in MoE scenarios.

Overall, improving Token generation efficiency is fundamentally a system-level optimization centered on data flow efficiency. From [the PD disaggregated inference architecture, to reducing invalid data flow, improving node-level data supply, and breaking through cross-node communication bottlenecks](#), AI infrastructure is evolving from traditional compute infrastructure to a new computing architecture for intelligent production. In the future, the key to competition will not be who has more compute, but who can enable a more efficient data flow system across computing, storage, and communication to continuously, stably, and efficiently generate tokens.

Target Reference Architecture for Token-Centric AI Infrastructure

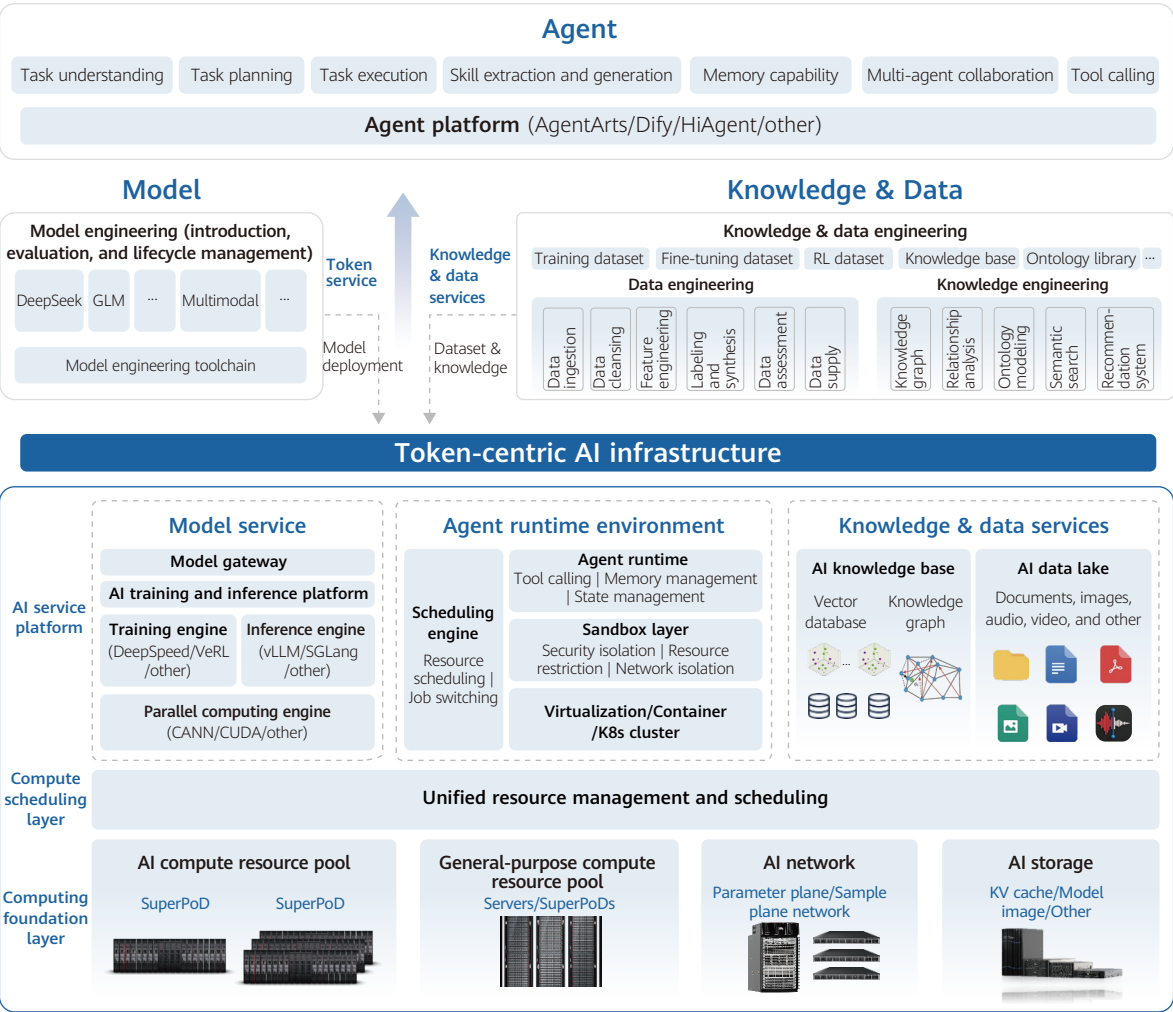


Figure 2-10 Overall architecture of enterprise token-centric AI infrastructure

Conventional cloud infrastructure is IaaS-centric, focusing on the pooling and scheduling of compute, storage, and network resources. However, when tokens become the core unit for measuring AI production efficiency, optimizing the resource layer alone can no longer determine the final token output efficiency. Consequently, [the architectural boundary of AI infrastructure must be extended upward](#), adding AI service capabilities such as model running, data and knowledge management, and agent runtime environments on top of conventional resource infrastructure, thereby transforming underlying compute into sustainable AI capabilities.

Based on this shift, [the token-centric AI infrastructure forms a three-layer architecture comprising the AI computing foundation, compute scheduling, and the AI service platform](#), which respectively address the three core issues of ultimate token production efficiency, efficient compute utilization, and continuous AI capability supply, ultimately achieving the evolution from resource supply to an intelligent production system.

Computing foundation layer: Addressing token production efficiency per unit of compute

In the AI era, instead of focusing only on simply increasing the scale of compute, companies are working on building a unified computing foundation optimized for AI workloads to have compute, storage, and network resources collaborate efficiently around model inference, training, and agent tasks. Unlike conventional cloud infrastructure oriented toward general-purpose compute resources, the AI computing foundation must simultaneously support diverse AI workloads:

- **AI compute resources:** Centered on SuperPoDs and SuperClusters, these provide the high-performance computing capabilities required for model training and inference. Through high-speed interconnection, memory pooling, and large-scale parallelism, they enhance token production efficiency per task.
- **General-purpose compute resources:** These support data processing, knowledge services, agent task orchestration, and tool execution, providing business execution capabilities for intelligent applications.
- **AI storage and high-speed network:** These carry model images, KV caches, context states, and training data, ensuring that compute resources continuously receive valid data through high-bandwidth, low-latency data channels.

Ultimately, the goal of the computing foundation is not to provide more isolated compute, but to build an efficient computing system oriented toward AI production, enabling more effective tokens to be generated per unit of resource.

Compute scheduling layer: Addressing efficient resource utilization

The compute scheduling layer is essentially the Operating System (OS) of AI infrastructure. It connects underlying resources, model tasks, and business requirements, and implements the following through resource pooling, elastic scheduling, task orchestration, and intelligent optimization:

- **Unified resource management:** Shields differences between different chips and device generations to form a unified AI resource pool.
- **Dynamic resource matching:** Matches tasks to the most suitable compute resources based on the model scale, inference phase, and business priority.
- **Elastic resource reuse:** Improves overall resource utilization efficiency through training and inference colocation, tidal scheduling, and priority management.
- **Highly reliable operation:** Ensures continuous AI production through fault detection, task migration, and resource recovery.

If the computing foundation determines how many tokens an enterprise can produce, then compute scheduling determines whether this capability can be fully unleashed.

AI service platform: Addressing continuous AI capability supply

The AI service platform provides three core capabilities:

- **Model running service:** Converts compute into stable and efficient token production capabilities through the inference engine, training engine, and model service management.
- **Data and knowledge service:** Enables tokens to possess enterprise business knowledge through vector databases, knowledge graphs, and long-term memory, thereby transforming AGI into industry-specific intelligence.
- **Agent runtime environment:** Enables tokens to drive business processes and automated execution through task orchestration, tool calling, secure execution, and multi-agent collaboration.

The preceding three layers define the logical architecture of AI infrastructure from a functional layering perspective. However, actual construction must be further decomposed from an engineering deployment perspective into **four capability circles: SuperPoDs, SuperClusters,**

compute scheduling, and the AI service platform. Among these, SuperPoDs and SuperClusters are the core carriers of the computing foundation layer, focusing on token production efficiency per unit of compute.

Four Capability Circles Reshaping Token-Centric AI Infrastructure

The four capability circles focus on compute supply, cluster collaboration, resource scheduling, and intelligent services, forming a complete system from underlying compute resources to upper-layer AI capability delivery.

They are the core capability units for implementing token-centric AI infrastructure. Their specific connotations are as follows.

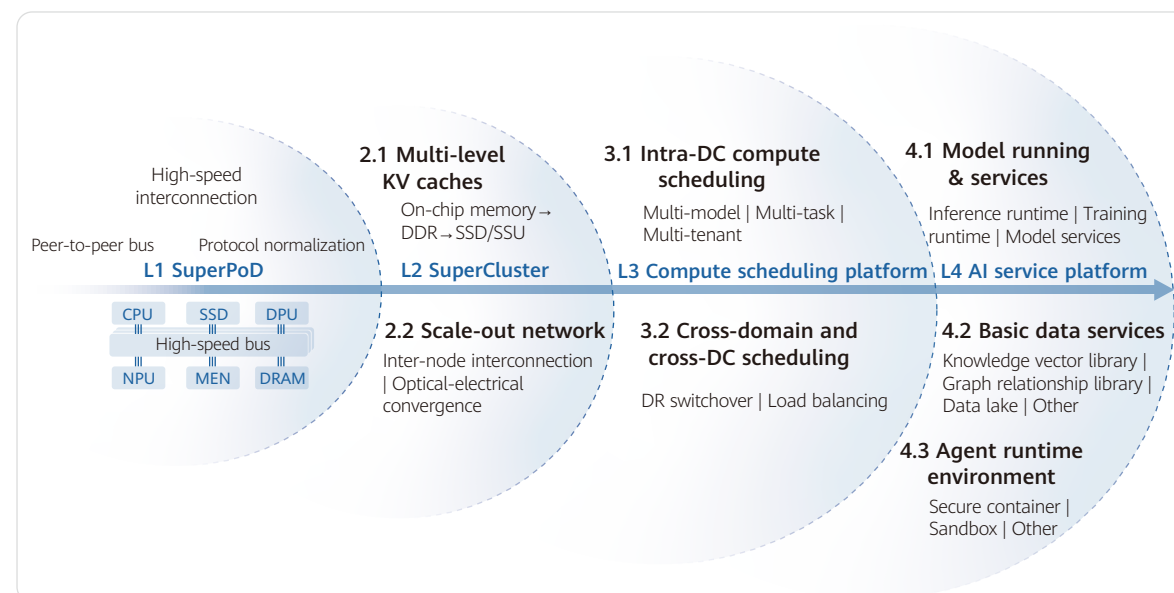


Figure 2-11 Token-centric AI infrastructure

L1—SuperPoDs: The Core Engine for Efficient Token Production, Becoming the Norm

1. Definition of a SuperPoD: A Logical Computer

As model scales evolve from tens and hundreds of billions to trillions of parameters, conventional single-server architectures are approaching physical limits. Take a 1-trillion-parameter model as an example. Even with FP16 precision, the model parameters alone require approximately 2 TB of memory, which already exceeds the **on-chip memory capacity** of mainstream 8-card AI servers. Even if the model is compressed using low-precision quantization methods such as INT4 and FP4,

the limited on-chip memory is still largely occupied by model weights, leaving insufficient space for KV caches, which **severely limits inference concurrency and long-context capabilities**.

Therefore, to simultaneously meet the requirements of **large model deployment, long-context inference, and high-concurrency services**, compute resources must transcend single-server boundaries and evolve toward cross-node convergence.

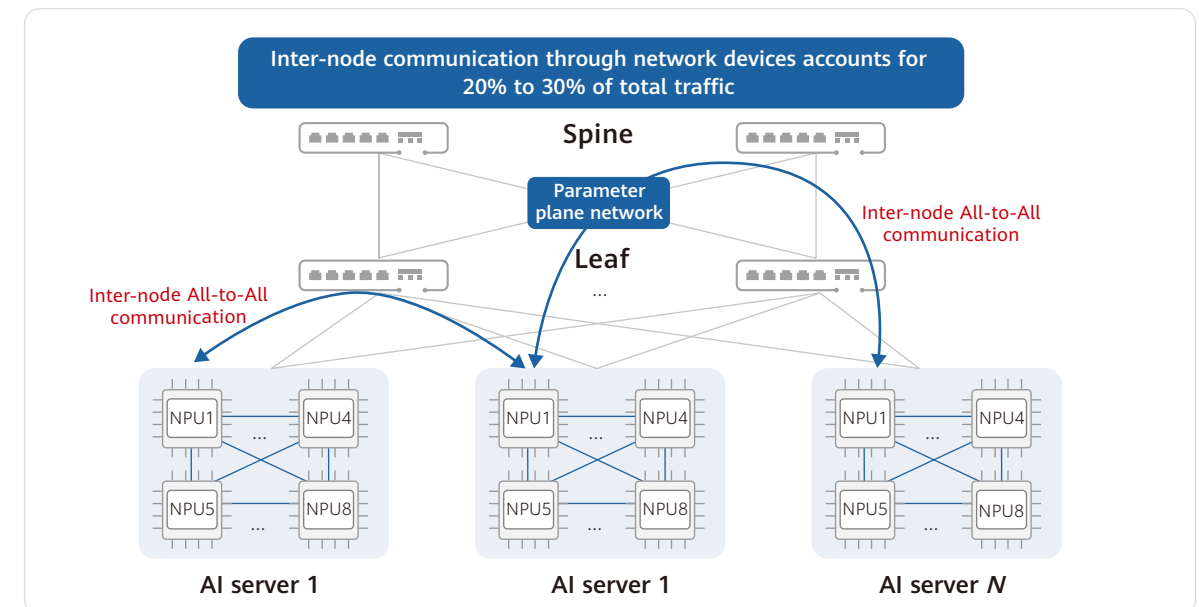


Figure 2-12 Conventional server architecture

When the computing scale expands from a single server to a cluster, **a new bottleneck emerges: communication efficiency**. Particularly in MoE models, EP requires extensive cross-chip All-to-All (A2A) communication. The conventional scale-out architecture relies on the Ethernet to connect standardized servers. While this offers good general scalability, network bandwidth and communication latency gradually become limiting factors

under ultra-large AI workloads. The time computing chips spend waiting for data transmission increases, and communication overheads continuously erode effective compute. Consequently, the larger the cluster scale, the lower the token production efficiency.

Therefore, a new computing organization model has emerged in AI infrastructure: **SuperPoDs**.

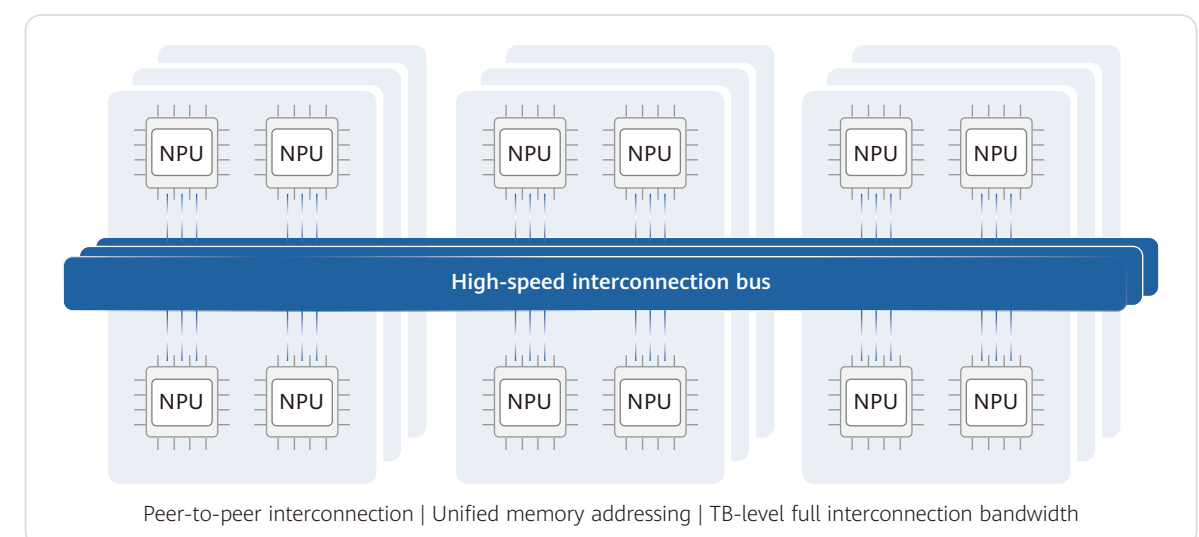


Figure 2-13 SuperPoD architecture

A SuperPoD uses high-speed bus-level interconnection to physically integrate tens to thousands of AI chips within a single cabinet or even across cabinets, logically converging them into a single computer. At least 32 AI chips must be interconnected within the same SuperPoD domain, featuring high bandwidth (greater than 400 GB/s), low latency (less than 500 nanoseconds), and unified memory addressing. Compared with conventional server clusters, the major changes in SuperPoDs are as follows:

Unified memory addressing: Turning distributed resources into a shared memory pool

In conventional AI clusters, each AI card has its own dedicated on-chip memory, and models and data need to be constantly transferred between different devices. A SuperPoD uses memory pooling technology to abstract distributed memory resources into a unified address space, enabling different AI chips to directly access remote memory resources. This means:

- Model parameters can be centrally managed across chips.
- KV caches can be shared across instances.
- Large-scale models are no longer restricted by the on-chip memory capacity of a single card.

Compute resources are transformed from multiple isolated devices to a large computer with shared memory.

High-bandwidth, low-latency interconnection: Reducing the impact of communication on computing efficiency

Large models, especially MoE models, are communication-intensive.

A SuperPoD uses high-speed interconnection and memory semantic access to transform conventional network communication into a data exchange model that is more like local access, significantly reducing communication overheads.

The value of this is not only in improving peak compute, but more importantly in **enabling more compute resources to be used for token generation, rather than waiting for data transmission**. These innovations directly translate into business benefits. For example, during inference of MoE models like DeepSeek and Qwen, **end-to-end performance has been improved by more than three times**, which strongly verifies the actual gains of the SuperPoD architecture in token generation efficiency.

2. SuperPoD-Based Inference Architecture Innovation, Achieving Efficient Token Generation

The unified compute domain powered by SuperPoDs provides new opportunities for optimizing the inference architecture of large models.

Large-scale EP: Unleashing the throughput potential of MoE models

MoE models use multiple experts to improve capacity. However, conventional clusters are limited by communication efficiency, making it difficult to scale up the number of experts. The large-scale memory pool and high-speed interconnection provided by SuperPoDs make it possible to parallelize a larger number of experts.

Take DeepSeek large-scale EP as an example. The large-scale EP architecture can implement one expert per card. The number of expert cards is increased from 16 to 288 (including shared experts), and the weight per card is reduced from 42 GB to 2.33 GB. This allows more KV caches to be stored (increasing the batch size from 8 to 30), expanding the SuperPoD memory pool.

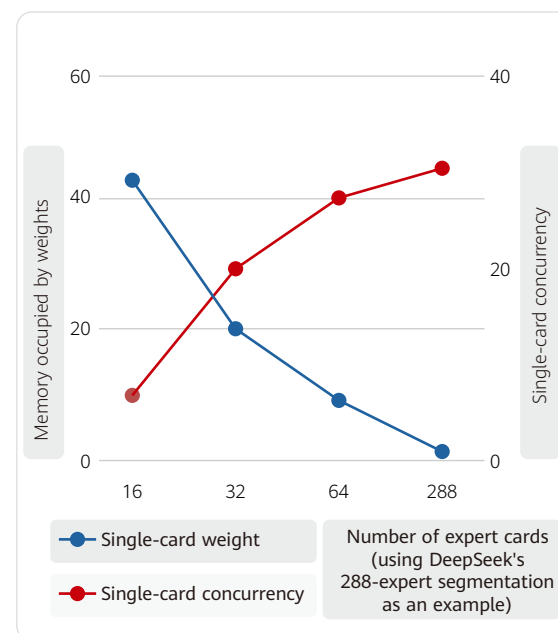


Figure 2-14 DeepSeek V3 large-scale EP

The more expert cards there are, the shorter the decoding time. However, the more expert cards there are, the greater the challenge to **A2A communication** between

cards. Nevertheless, thanks to the high bandwidth of SuperPoDs, the entire system remains in the optimal state.

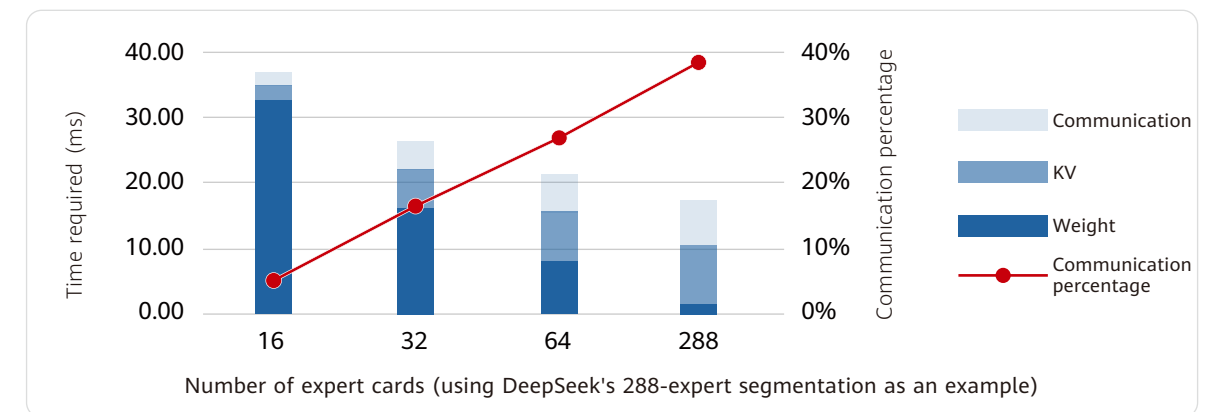


Figure 2-15 DeepSeek V3 large-scale EP communication time and percentage

Prefill-Decode-Cache (P-D-C) fully peer-to-peer architecture: Global KV cache sharing eliminates repeated computation, significantly reducing TTFT

In multi-round dialogue and agent tasks, a large number of requests reuse existing contexts. In the conventional inference architecture, KV caches are bound to a single instance and cannot be shared between different

instances. Consequently, the same historical context requires repeated prefill computation.

Based on the unified memory addressing of SuperPoDs, the inference architecture is evolving toward a P-D-C fully peer-to-peer model: The prefill cluster handles context computation, the decode cluster handles token generation, and the caching cluster handles state sharing.

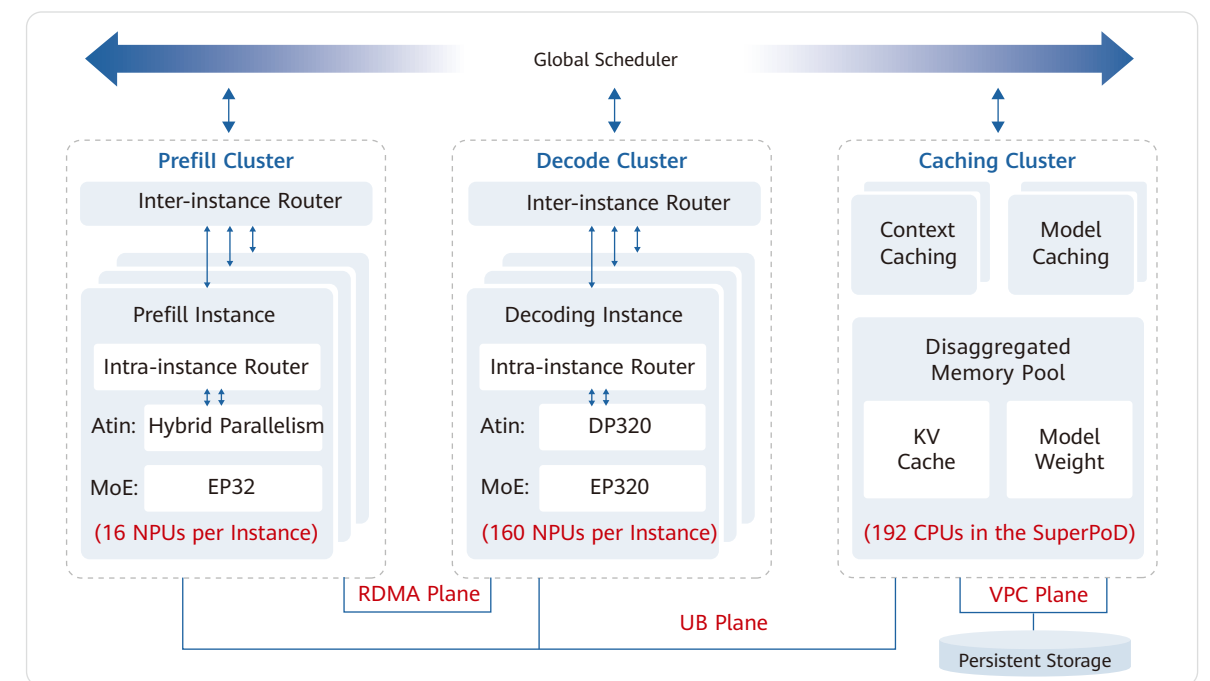


Figure 2-16 P-D-C fully peer-to-peer architecture based on Ascend SuperPoDs + DeepSeek

KV caches are no longer bound to a single instance but becomes a shared resource across the entire compute domain. This, combined with KV cache-aware routing, allows instances that have cached the existing context to be preferentially scheduled. This reduces repeated prefill, lowers TTFT, and improves overall token generation efficiency, pushing the inference system from a computing-only model to a model that combines both computing and state management.

3. Multiple Forms of SuperPoDs: Meeting High-Performance Inference Requirements in Large-, Medium-, and Small-Scale Scenarios

For small- and medium-scale scenarios, SuperPoDs of different sizes, such as 64 cards or 96 cards, can be used to deploy large-scale EP instances with PD disaggregation of different specifications, meeting high-performance inference requirements.

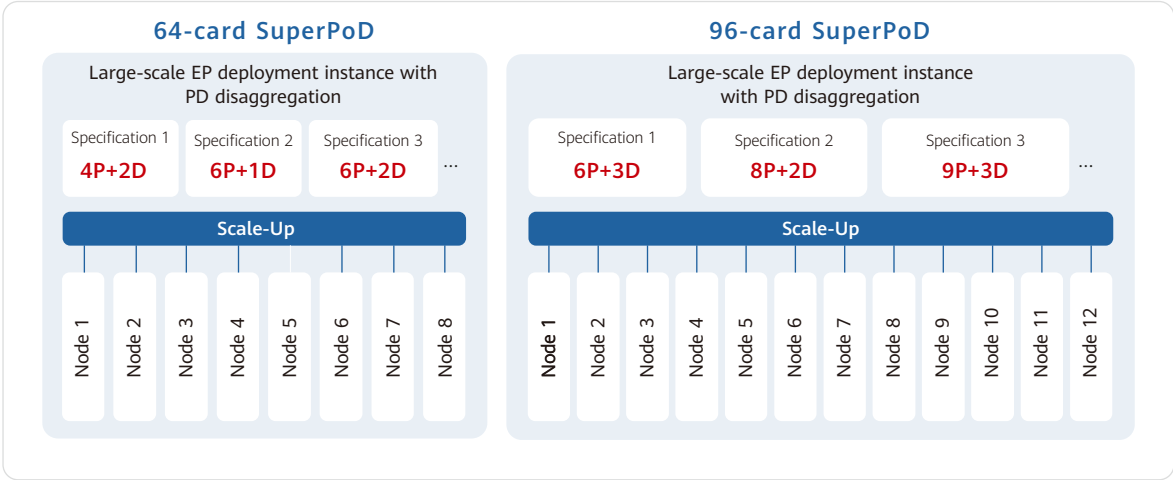


Figure 2-17 Small- and medium-scale SuperPoD deployment configuration

For large-scale high-concurrency inference, model training, and core production scenarios, SuperPoDs can be expanded to hundreds or even thousands of cards to form

a super computing system. More importantly, SuperPoDs possess logical partitioning capabilities.

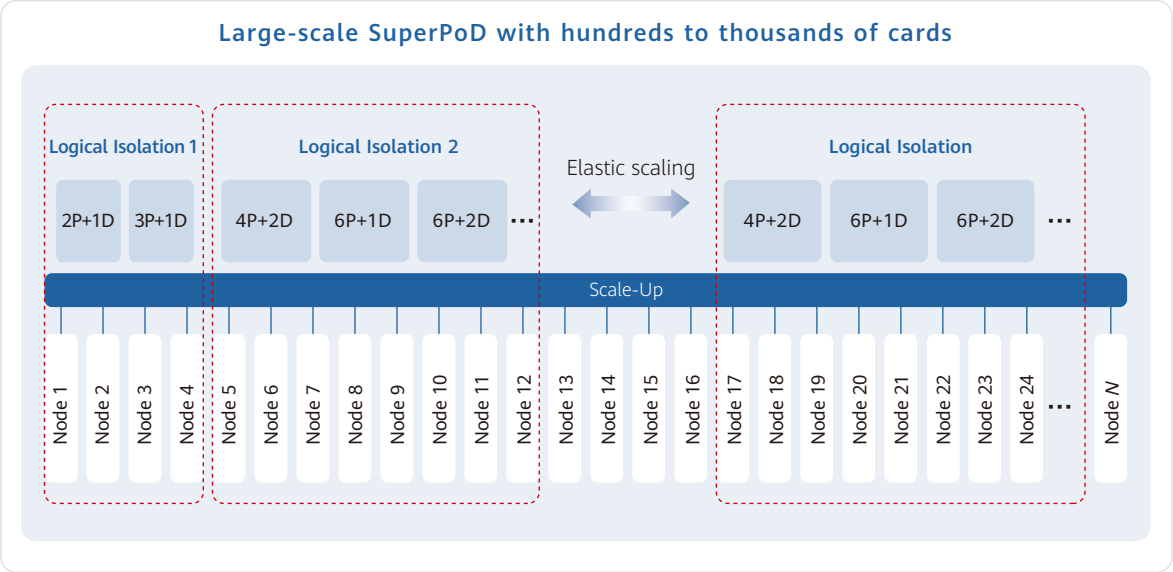


Figure 2-18 Large-scale SuperPoD deployment configuration

4. SuperPoD Development Trends

SuperPoDs are evolving from technical explorations to large-scale industry applications, becoming a key direction for the evolution of AI infrastructure. In the future, SuperPoDs will continue to evolve in three directions:

Continuous scale expansion: The scale of SuperPoDs will continue to break through the boundaries of conventional server clusters, supporting larger-scale training and inference. In the future, the scale of SuperPoDs will evolve from thousands of cards to tens of thousands of cards.

Continuous interconnection upgrade: As the scale of SuperPoDs continues to expand, network interconnection becomes the key to system efficiency. SuperPoD networks are evolving from microsecond-level latency to nanosecond-level latency, achieving high-speed collaboration between compute resources through Tbps-level bandwidth and more efficient topologies. Conventional Electrical Interconnect is limited by

high-frequency loss and power consumption. Optical Interconnect is the main approach for future systems, and Near-Package Optics (NPO) is a key technology to achieve this.

Continuous resource decoupling: As inference, multimodal, and agent tasks become more complex, the requirements for computing, memory, and communication are becoming increasingly differentiated. SuperPoDs will need to support new deployment models such as Attention-FFN Disaggregation (AFD, which decouples the attention module from the FFN/MoE module), multimodal VAT (a visual-audio-text multimodal model architecture), and LLM disaggregation (separate deployment of different components of LLMs). The ultimate goal is to dynamically provide the most suitable compute resources for different AI workloads.

L2—SuperClusters: Infrastructure for High-Concurrency Deployment of Trillion-Parameter MoE Models

SuperPoDs improve the collaboration efficiency of a single compute domain, while SuperClusters further address the issue of large-scale AI production by integrating multiple SuperPoDs with networks and storage into a unified AI production system.

1. Typical Application Scenarios of SuperClusters

Enterprises set up SuperClusters essentially to meet three types of requirements after AI enters the production phase:

01

Scalability: Supporting continuous business growth

As AI workloads transition from pilot projects to large-scale deployment, a single SuperPoD is increasingly constrained by limitations in throughput, concurrency, and service isolation. Scaling out to multiple SuperPoDs allows enterprises to smoothly increase token production capacity without the need to rebuild infrastructure.

03

Reliable running: Ensuring core business continuity

When AI is applied to key processes in finance, government, manufacturing, and many other sectors, cluster-based deployment upgrades AI from a single-instance service to a reliable production system through fault isolation, service migration, rolling upgrades, and disaster recovery (DR).

02

Production operations: Supporting parallel running of multiple services and models

Enterprise-grade AI needs to simultaneously support multiple models, multiple services, and a large number of user requests. SuperClusters achieve horizontal scaling of computing capabilities through unified resource pools and elastic scheduling.

2. Core Components of SuperClusters

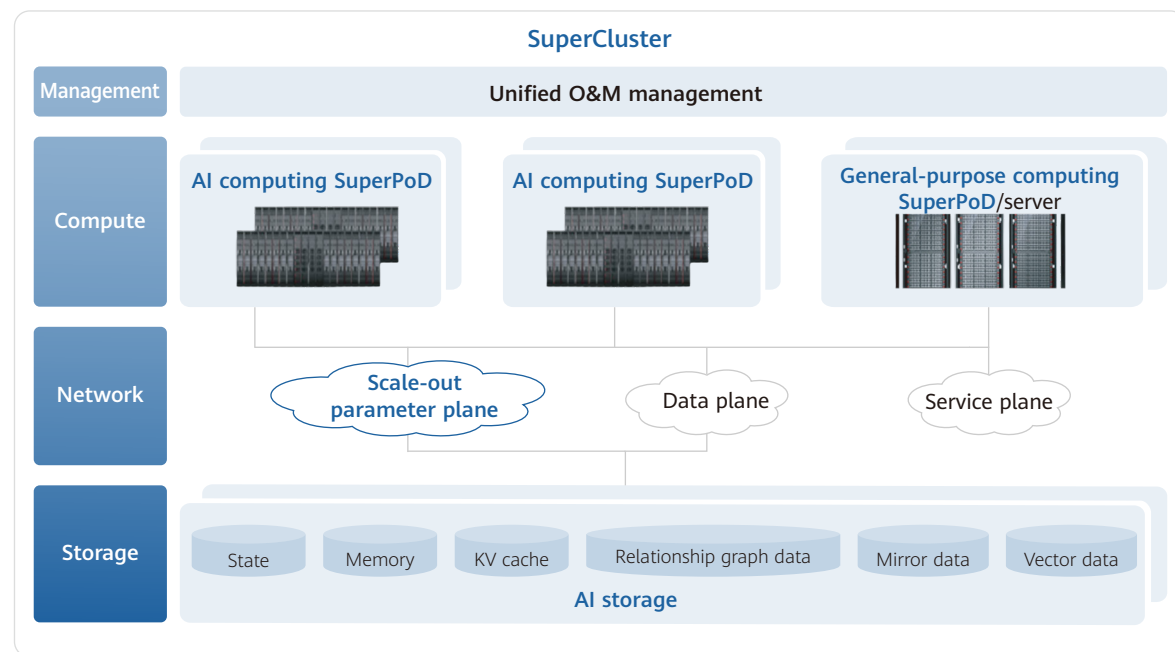


Figure 2-19 SuperCluster system architecture

A SuperCluster integrates distributed compute resources into unified infrastructure for AI production through the collaboration of four foundation layers: **AI computing, network, storage, and general-purpose computing.**

- **The AI computing foundation** provides large model training and inference capabilities.
- **The network foundation** enables high-speed interconnection between SuperPoDs and cluster expansion.
- **The AI storage foundation** stores model data, KV caches, and long-term memory of agents.
- **The general-purpose computing foundation** supports task orchestration, tool calling, and agent runtime environments.

These four foundations together form an AI production platform, enabling compute, states, and tasks to flow collaboratively across the cluster: **AI computing is responsible for thinking, general-purpose computing for action, AI storage for memory, and network for connectivity.**

3. Scale-Out Network: Making Multiple SuperPoDs a Unified Compute Domain

The scale-out network connects computing and data resources within a SuperCluster to implement cluster-level AI collaboration. **It connects multiple SuperPoDs** to collaboratively complete training and inference of large models. **It also connects to AI storage** to provide efficient access channels for key inference data such as model states and KV caches, enabling context states to be shared and reused across the cluster.

Typical scenarios and main traffic of the scale-out network

As the inference architecture of large models evolves from single instances to clusters, the types of communication over the scale-out network are continuously expanding, including:

- **KV cache communication:** Supports PD disaggregation and context state sharing.
- **Activation and MoE communication:** Supports module collaboration after the model structure of the decode instance is decoupled.
- **Prefix cache communication:** Supports cross-instance context reuse.

The typical scenarios are shown in the following figure and table.

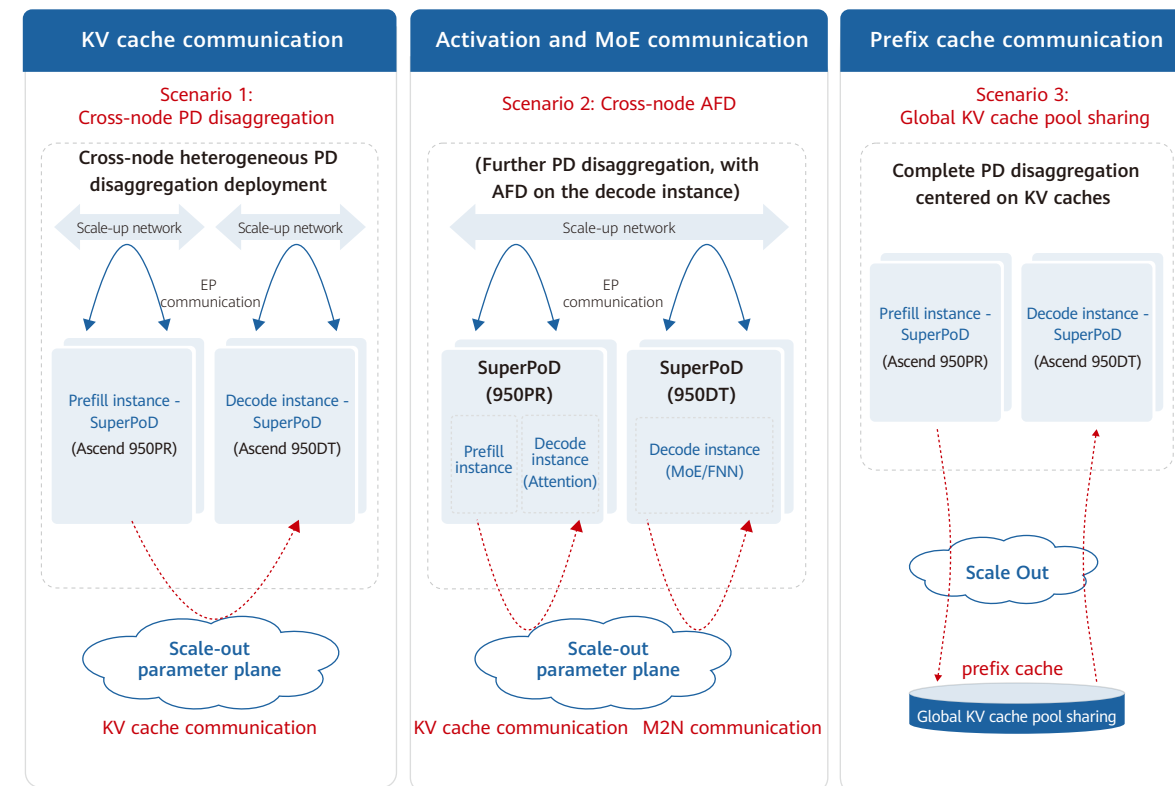


Figure 2-20 Typical scale-out network scenarios

Typical Scenario	Description	Main Communication Traffic	Scale-Out Network Benefits
Cross-node heterogeneous PD disaggregation	Prefill and decode instances are deployed in different compute resource pools.	KV cache transmission across nodes	Connects heterogeneous SuperPoDs to enable collaboration of compute across generations and types, optimizing inference costs.
Heterogeneous AFD	The attention module is further decoupled from the FFN/MoE module.	Activation and MoE communication	Supports cross-resource-pool collaboration of model modules, improving the resource utilization efficiency of MoE models.
PD disaggregation between SuperPoDs	Multiple SuperPoDs form a large-scale inference cluster.	Prefix cache and global KV cache communication	Enables cross-instance sharing of context states, allowing KV caches to evolve from single-node caches to cluster-level resources.

The communication over scale-out networks is expanding from conventional data exchange to model states, compute activation, and context memory, making it a key infrastructure for supporting the large-scale operation of AI inference clusters.

Key technology trends of scale-out networks

As AI clusters scale from thousands of cards to tens of thousands or even hundreds of thousands of cards, **scale-out networks are continuously evolving around scale expansion, efficiency improvement, and cross-domain interconnection.**

• Architecture evolution: From multi-level Clos to multi-plane and flattened networking

As the scale of conventional multi-stage Clos networks expands, they introduce more network layers, higher interconnection costs, and longer communication latency. Multi-plane networking divides a large-scale multi-stage Clos network into multiple independent lower-layer subnets. By splitting the compute card interfaces and connecting them across each plane, port resources are maximized for compute workloads. This allows the network scale to increase by an order of magnitude with the same port overhead. **The core benefit lies in breaking through cluster scale boundaries and enabling the network to continuously expand along with compute.**

• Communication optimization: From static forwarding to intelligent scheduling

AI communication features large bandwidth, high concurrency, and complex traffic patterns. Conventional hash-based load balancing can easily cause uneven link distribution and communication bottlenecks. Network-level load balancing dynamically optimizes communication paths by sensing the global traffic status. Device-network

collaborative congestion control enables network adapters and switching devices to collaborate, achieving lower queuing time and lower tail latency. **The core benefit lies in improving network bandwidth utilization efficiency and ensuring stable and efficient token generation.**

• Interconnection evolution: From single-DC to cross-compute-domain

In scenarios such as large model training and massive-scale Internet inference, AI clusters will further expand, and network interconnection may go beyond conventional DC boundaries.

Conventional electrical switching networks face power consumption and scalability challenges at the scale of tens of thousands or even hundreds of thousands of cards. Optical Circuit Switching (OCS) is now being introduced to ultra-large-scale AI networks. It establishes high-speed direct connections in the optical domain, reducing the number of switching layers and lowering power consumption and latency in large-scale interconnection. However, since OCS is more suitable for stable and large-scale traffic, electrical switching is still required to flexibly schedule dynamic burst traffic. Therefore, electrical switching and OCS will coexist for a long time in the future.

Additionally, due to the limitations of power, land, and heat dissipation resources, compute is breaking through the boundaries of a single DC. Scale-across (cross-campus and cross-DC compute interconnection) has become a new development direction. In the future, leading enterprises will explore cross-region compute collaboration, while more enterprises will improve the reliability of AI services through multi-active deployment and DR of cross-campus inference.

4. AI Storage: Long-Term Memory of Compute Clusters

In the era of agents, KV caches are becoming a key element that influences inference costs and performance. A typical agent workflow involves multiple rounds of tool calls and context exchanges, with follow-up requests frequently reusing most of the existing context while appending only a small amount of new data. **This "add-only" interaction model makes KV caches highly**

reusable. For example, in AI coding, a task may last for several hours or even days, with contexts reaching 64,000, 128,000, or even one million tokens. A single task can generate hundreds of GBs or even TBs of KV caches. **Figuring out how to store, share, and schedule KV caches over the long term has become a critical new challenge for AI infrastructure.**

Key challenges brought by KV caches to AI infrastructure

Key Challenge		Description
1	Insufficient on-chip memory	Ultra-long contexts (64,000 to 1 million tokens) cause the KV cache of a single task to reach hundreds of GBs or even TBs, exceeding on-chip memory capacity.
2	High I/O latency of conventional tiered storage	Data transfer from the NPU to the SSD involves the Double Data Rate Synchronous Dynamic Random Access Memory (DDR SDRAM) and PCIe switch. This significantly increases access latency and cannot meet the real-time requirements of inference.
3	Short service life of SSDs due to frequent erase/write operations	Frequent read/write operations on KV caches severely accelerate the erase/write wear of SSDs. Drive Writes Per Day (DWPD) can reach more than 10, significantly reducing the service life of SSDs.

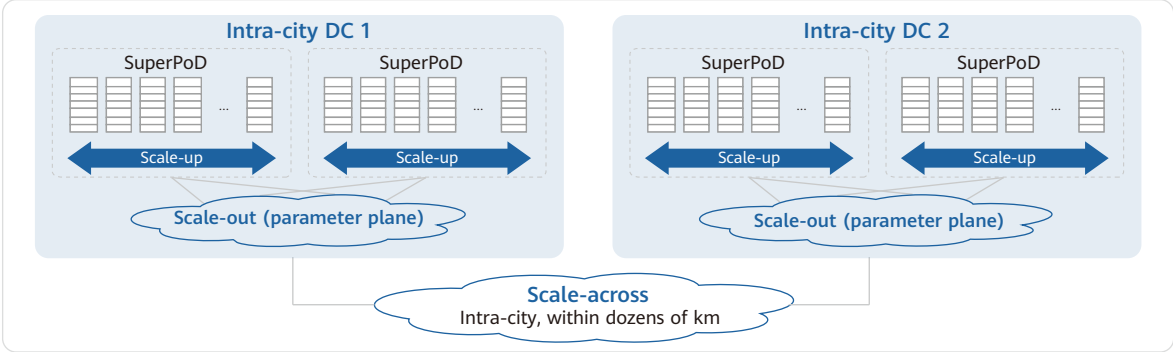


Figure 2-21 Cross-DC compute interconnection

• Hierarchical KV caches: Achieving optimal performance, experience, and costs for ultra-long context inference

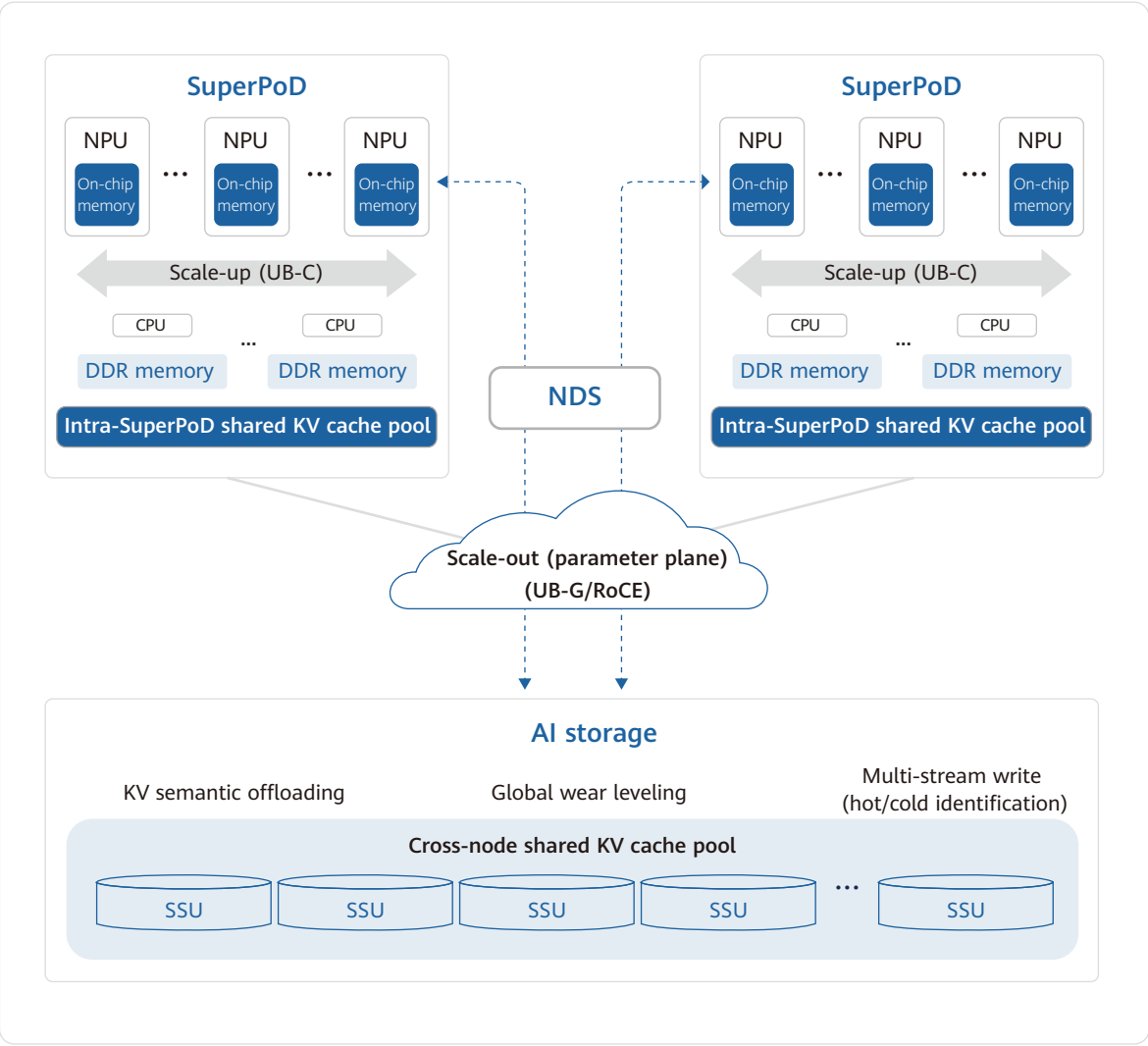


Figure 2-22 Target architecture of multi-level KV caches with compute-network-storage collaboration

From local caches to a globally shared hierarchical KV cache pool

As a multi-level KV cache system takes shape, cache management will become a common capability of AI infrastructure. Multiple SuperPoDs can share KV caches, enabling any inference instance to reuse the existing KV caches and improving the prefix cache hit rate.

Global hierarchical KV cache sharing system

Level	Name	Capacity	Latency	KV Cache Carried
L1	On-chip memory	Dozens to hundreds of GBs	Nanoseconds	Active KVs, hot data being used by the current inference instance, and enabling access with the lowest latency
L2	DDR memory pool	TBs	Hundreds of nanoseconds	Hot KVs, short-term session contexts, and relieving the pressure on on-chip memory capacity
L3	Shared KV cache pool	Dozens of TBs	Microseconds	Warm KVs, reusing contexts within a SuperPoD to implement cross-instance cache sharing
L4	Cluster-level AI storage	PBs	Milliseconds	Cold KVs, providing persistent KV cache management across SuperPoDs

KV caches automatically flow between different levels based on access frequency and latency requirements: Hot data resides in high-speed storage (on-chip memory or DDR memory), warm data enters the shared cache pool, and long-term data enters large-capacity storage. The ultimate goal is to centrally manage KV caches distributed across different compute resources, so that upper-layer inference services can invoke enterprise-grade KV cache capabilities as if accessing a unified memory space, without needing to be aware of the physical location of the KV caches.

Transformation from a host-centric architecture to NPU Direct Storage (NDS): Shortening the KV access path

In the conventional architecture, KV cache access usually requires multi-level data transfer through the storage protocol stack, host CPU, and memory, which increases access latency and resource consumption. In the future, AI storage will use technologies such as AI chip passthrough, KV semantic offloading, and memory semantic protocols to shorten the access path between computing chips and storage, making the KV cache access experience gradually approach that of memory access.

From SSDs to Semantic Storage Units (SSUs): Extending the service life of storage media

SSDs will evolve to **SSUs** that are capable of sensing the data semantics of KV caches. By using lifecycle-aware aggregation and intelligent wear leveling, SSUs can reduce write amplification by more than 70%, reduce DWPD from more than 10 to less than three, and extend the service life of storage media by two to three times.

For long-context agent applications, the inference cost is determined not only by the scale of compute, but also by whether the system can continuously save, share, and reuse context states. In the era of agentic AI, AI infrastructure with unified context management capabilities will support continuous intelligent production at lower costs and higher efficiency.

5. General-Purpose Compute Resources: The Action Engine of Agents

In the agentic era, a new division of labor is emerging within AI infrastructure. **AI computing is responsible for thinking, while general-purpose computing handles action.** A large number of tasks, including task orchestration, tool calling, code execution, and environment interaction, all run on general-purpose compute.

As agents enter enterprise production environments, general-purpose computing needs to be upgraded from traditional business servers to agent-native runtime environments, equipped with the following four core capabilities.

01 **Speed: High concurrency and startup within milliseconds**

Agents feature high concurrency and short lifecycles, potentially running millions of simultaneous task instances in the future. Infrastructure must support rapid creation and destruction of runtime environments using one sandbox per session.

Core capabilities:

- Achieving cold starts within milliseconds based on snapshot recovery
- Supporting high-density deployment of hundreds to thousands of agent instances on a single server
- Utilizing large-capacity, on-chip memory to support high-concurrency operation

This demand is driving general-purpose computing to evolve from traditional servers to a SuperPoD architecture. Only a system-level architecture with large memory, high bandwidth, and high-speed inter-chip interconnection can sustain the large-scale operation of agents.

02 **Stability: From service reliability to task reliability**

Agent tasks are shifting from minutes of interactions to hours or even days of autonomous operation. Cutting-edge coding agents have already achieved a single continuous runtime of 25 hours. This means traditional metrics like 99.9% service availability can no longer meet the reliability needs of long-chain tasks.

Future general-purpose computing platforms need to provide the following capabilities: persistent execution, task status checkpoints, resumption from checkpoint and cross-node migration, and long-term context preservation in collaboration with AI storage. The reliability standard is evolving **from uninterrupted services to task completion.**

03 **Security: Strong isolation as the foundation for agent runtime**

The biggest security risk for agents stems from the unpredictable behavior of the models themselves, including incorrect code execution, abnormal tool calling, and out-of-bounds data access. Future agent runtime environments must provide the following capabilities: microVM-level strong isolation sandbox, default network and permission isolation, session-level security controls, and full-process behavior auditing.

For industries such as finance, government, and energy, security isolation and auditability will become the baseline threshold for agent deployment in production.

04 **Integration: Collaborative scheduling of general-purpose compute, AI compute, and storage**

The execution process of agent applications naturally spans multiple resource domains.

- **General-purpose compute resources:** Responsible for task orchestration, tool calling, and execution environments
- **AI compute resources:** Responsible for model inference and decision generation
- **AI storage resources:** Responsible for context status, knowledge, and long-term memory management

Future AI infrastructure needs to integrate these three types of resources to achieve unified awareness and collaborative scheduling.

- Collaborative scheduling of general-purpose and AI compute resources, dynamically matching compute resources based on task requirements
- Efficient collaboration between computing and storage to reduce latency for context and status access

The ultimate goal is to unite the agent's thinking, action, and memory into a cohesive runtime system, providing integrated support from resource scheduling to intelligent execution.

Future outlook: Toward a SuperPoD that integrates general-purpose and AI computing

Currently, two types of SuperPoDs are emerging in AI infrastructure: NPU-centric AI computing SuperPoDs and CPU-centric general-purpose computing SuperPoDs. However, agent applications naturally fuse the two processes of thinking and action. A major evolutionary direction for future AI infrastructure is to drive the further integration of general-purpose and AI compute resources,

L3—Compute Scheduling: From Building Capacity to Effective Utilization, Efficiently Unleashing SuperPoD Potential

SuperPoDs transcend single-node computing boundaries, but they also introduce new resource management challenges. When hundreds or even thousands of AI chips form a complex computing cluster, how to continuously and efficiently schedule these resources becomes the key to unlocking the value of AI infrastructure.

Compute scheduling is the operating system of a SuperCluster. It transforms underlying compute into stable and efficient token production capabilities through resource pooling, task orchestration, elastic scheduling, and fault management.

For enterprise-grade AI production, scheduling platforms need to address three core issues.

- **Efficient resource utilization:** Enabling AI tasks of different scales and types to share SuperPoD resources, thereby improving overall resource utilization efficiency
- **Dynamic workload adaptation:** Dynamically matching compute resources based on changes in business requests and model runtime status to ensure stable service capabilities
- **Continuous cluster operation:** Ensuring the long-term stable operation of large-scale AI clusters through fault detection, task migration, and resource recovery mechanisms

1. Elastic Scheduling: On-Demand Access to SuperPoD Resources

SuperPoDs provide high-density computing capabilities, but enterprise AI workloads are highly uncertain. Large model training and high-concurrency inference demand hundreds or even thousands of cards to work together, while numerous tasks, such as Optical Character Recognition (OCR), embedding, speech processing, and small model inference, require only a small number of resources.

Therefore, the scheduling system needs to move away from the traditional server-era resource usage model of one-time

forming an integrated SuperPoD system for agent tasks through a unified architecture and collaborative scheduling.

At this point, the full architecture of the next-generation AI computing cluster is revealed. **AI computing provides the thinking capability, general-purpose computing supports the action capability, AI storage carries the memory capability, and high-performance networking connects all three to work in synergy. Moving forward, these four elements will further converge into a new type of AI infrastructure tailored for the agent era.**

allocation and long-term occupation, enabling compute to dynamically scale and flow based on task demands.

Scalable resources: From fixed specifications to task-based allocation

For large-scale tasks, resource aggregation is supported.

- Multiple SuperPoDs are interconnected at high speed to form a unified compute resource pool.
- Supports training with thousands of cards, large-scale inference, and cross-node PD disaggregation.

For small-scale tasks, resource splitting is supported.

- **SuperPoD level:** A SuperPoD with hundreds of cards is divided into multiple logical resource domains, allowing multiple services to run in isolation.
- **Single-card level:** Virtualization technologies such as vNPU are used to split a single card into multiple compute instances, improving the resource utilization efficiency of small models.

Resource mobility: From static occupation to dynamic reuse

AI workloads exhibit distinct tidal patterns. Online inference tasks are in peak demand during the day, while training, fine-tuning, and data processing tasks are dominant at night. If resources are statically bound for a long time, there may be capacity shortages during peak hours and idle waste during off-peak hours.

Therefore, the scheduling system needs to implement dynamic resource reuse based on service priorities.

- Deterministic resource assurance is provided for core production inference tasks.
- Resources are automatically released during off-peak

hours and used for offline tasks such as training and fine-tuning.

- When a high-priority task arrives, the low-priority task can be paused, migrated, or downgraded.

Through hybrid deployment of training and inference, traffic-based resource reuse, and priority-based scheduling, idle compute in the time dimension is converted into effective token production capacity.

2. High-Availability Scheduling: Transitioning AI Services from Operability to Continuous Operation

A SupeCluster consists of a large number of computing chips, high-speed networks, and storage devices, forming a complex system. The larger the scale, the higher the probability of single points of failure. For mission-critical services in sectors such as finance, energy, and government, AI services require not just baseline operation, but sustained, long-term stability.

Therefore, high-availability scheduling needs to be upgraded from single-point protection to system assurance oriented toward service continuity.

Multi-level deployment: Controlling the impact scope of faults

Faults are isolated through multi-level resource isolation and redundant deployment.

- **Within a SuperPoD:** Multi-instance deployment reduces the impact of single-card or module faults.
- **Within a cluster:** Cross-SuperPoD deployment defends against node-level faults.
- **Between DCs:** Cross-region deployment prepares for equipment room-level disasters.

The reach of the scheduling scope dictates the extent of fault isolation.

Intelligent recovery: From fault recovery to service continuity

The importance of different phases of AI inference varies. In the PD disaggregation architecture:

- **Decode directly affects the current text generation experience of users.**

- **Prefill mainly affects the TTFT for new requests.**

Therefore, when resources fluctuate or faults occur, the scheduling system must have the service awareness capability. For example, maintain the running of D instances by reducing P instances.

- Prioritizing continuous execution for Decode workloads
- Scaling down Prefill resources to absorb short-term impacts

Minimizing the impact of faults by combining task migration, status recovery, and cross-node rescheduling

The future goal of high availability is to transform faults from service interruption incidents into resource adjustment events.

3. Intelligent Scheduling: From Resource Management to AI Production Optimization

Traditional scheduling focuses on finding idle resources. In the AI era, we need to answer deeper questions: What resources, at what time, and in what way can be used to serve what tasks to maximize business value?

Therefore, the scheduling system is evolving from a resource management platform to an AI production optimization system.

Awareness: AI workload profiling

Different models and tasks have distinct resource characteristics, such as computing density, memory capacity, memory bandwidth, and communication patterns. The scheduling platform needs to create model-level and task-level compute profiles to enable precise matching.

Prediction: From reactive scaling to proactive planning

Traditional scaling relies on real-time thresholds. In the future, scheduling will use AI prediction to forecast service traffic, prepare resources proactively, and adjust compute supply dynamically. This marks a shift from reactive response to demand to proactive prediction and planning.

Decision-making: Global scheduling oriented to business value

In the future, enterprises may simultaneously own multiple SuperPoDs and compute resources of different generations. The scheduling system needs to optimize resources globally based on business value, which includes prioritizing high-value tasks, running low-value tasks flexibly, and utilizing heterogeneous resources in a unified manner.

L4—AI Service Platform Layer, the Final Link in the Token Production Loop

The ultimate goal of AI infrastructure is to continuously transform underlying compute resources into deliverable and operable intelligent capabilities. SuperPoDs provide efficient compute supply, and compute scheduling improves resource utilization efficiency. The AI service platform further connects models, data, and service applications, completing value transformation from compute resources to intelligent production.

The AI service platform mainly consists of three core capability layers.

• Model runtime service: Converts compute into token production.

Through model deployment, inference optimization, and runtime management, the platform delivers stable and efficient model inference services.

• Data foundation service: Empowers tokens with enterprise knowledge.

By leveraging knowledge bases, data management, and context services, the platform transforms enterprise data into intelligent assets that models can comprehend and invoke.

• Agent runtime environment: Drives business actions with tokens.

Through the agent framework, tool calling, and task orchestration, the platform further converts model outputs into business processes and automated execution.

Together, these three layers form a closed loop of intelligent production spanning compute, tokens, knowledge, and actions, driving the evolution of AI infrastructure from a resource supply platform to an intelligent production platform oriented toward business value creation.

1. Model Runtime and Serving: The Core Engine Transforming Compute into Tokens

SuperPoDs provide high-density compute, while model runtime and serving are responsible for converting these compute resources into sustainable, deliverable AI capabilities. This layer consists of two core components.

- **Model runtime:** Determines the efficiency of token production.

- **Model serving:** Determines the efficiency of AI capability delivery.

Together, they complete the transformation from underlying compute resources to enterprise-grade AI services.

Model runtime: The core software engine for unleashing the value of AI compute

Model runtime primarily encompasses the inference engine and the training engine.

The **inference engine** is responsible for transforming a trained model into the production capability that continuously outputs tokens.

Representative technologies include MindIE, vLLM, and SGLang. Their core value lies in maximizing token production efficiency through model optimization, scheduling optimization, and software-hardware collaboration, all within limited hardware resources.

As large models enter the production phase, the inference engine is evolving from a model execution tool into an AI runtime platform, which is primarily reflected in three aspects.

01

Shifting from model adaptation to system-level collaborative optimization. The inference engine needs to support complex models such as MoE, multimodal, and long-context models, and collaborate with SuperPoDs, networks, and storage systems to achieve E2E efficiency improvements.

02

Transitioning from single-instance optimization to cluster-level efficiency optimization. Mechanisms such as PD disaggregation, KV cache management, and continuous batching are employed to improve overall throughput and reduce the cost per token.

03

Moving from model serving to intelligent task execution. With the rise of agents, multi-turn interactions, and complex workflows, the inference engine needs to support the execution of longer-cycle and more complex intelligent tasks.

The core of future inference engine competition will be how many effective tokens can be generated per unit of compute.

The training engine is responsible for the continuous evolution of model capabilities and serves as a crucial foundation for enterprises to build their own AI capabilities.

Representative technologies include Megatron-LM, DeepSpeed, PyTorch FSDP, and MindSpeed within the Atlas ecosystem. The core value of a training engine is to efficiently map large-scale model training tasks onto an AI cluster, with a focus on addressing key challenges, such as improving the efficiency of distributed training for large models, enabling multi-card and multi-node collaboration, and optimizing memory usage and communication overhead.

Future training engines will exhibit two main trends. **First, the shift from one-time training to continuous optimization.** Model capability improvement is increasingly reliant on post-training workflows such as Supervised Fine-Tuning (SFT), Reinforcement Learning from Human Feedback (RLHF), and agent reinforcement learning. Training engines are evolving from large model training tools into platforms supporting continuous model evolution. **Second, the gradual convergence of training and inference boundaries.** New training workflows like reinforcement learning require vast amounts of model-generated data, prompting inference engines to participate in the rollout process. As a result, training and inference are moving from two separate software stacks toward a unified model runtime platform.

Model serving: Turning model capabilities into enterprise-grade AI services

As enterprises enter the multi-model era, the model environment is rapidly becoming more complex. Closed-source models, open-source models, enterprise private models, and industry-specific models will coexist for the long term. If business systems directly interface with different models, it will incur significant adaptation costs and make unified governance difficult. Therefore, the model serving platform has become the unified entry point for enterprises' AI capabilities. The platform primarily provides four types of capabilities.

- **Unified model management and service provisioning.** For enterprise multi-model runtime scenarios, the platform centrally manages model deployment, release, calling, and runtime environments. It reduces the complexity of business integration via standardized service interfaces, enabling flexible reuse of model capabilities.

- **Intelligent routing and service governance.** Based on model capabilities, real-time workloads, resource costs, and business priorities, the platform implements request routing, canary release, rate limiting, circuit breaking, and failover, thereby improving the stability of model services.

- **Model lifecycle management.** The platform centrally manages model rollout, version upgrades, performance evaluation, and runtime status, transitioning models from one-time deployment into a continuous operations phase.

- **Token metering and cost operations.** Using tokens as the unified metering unit, the platform analyzes model invocation scale, resource consumption, and business value, enabling refined management of AI resource investment.

In the future, the model serving platform will become the core entry point for enterprises to manage and operate their AI capabilities.

2. Data Foundation Services: Empowering Tokens with Enterprise Knowledge

Foundation models are rapidly becoming accessible AGI for enterprises. However, **an enterprise's long-term competitive advantages will increasingly stem from its private data, business knowledge, and data loops formed through continuous feedback.** Data foundation services connect these general-purpose models with enterprise operations, providing a steady stream of knowledge to fuel token generation. These services primarily encompass four types of capabilities.

Knowledge vector database: Enabling models to comprehend enterprise knowledge

This involves converting unstructured knowledge—such as internal regulations, product documentation, and expert experience—into vectors to support Retrieval-Augmented Generation (RAG), thereby equipping general-purpose models with enterprise-specific capabilities.

Graph database: Enabling models to understand business relationships

Enterprise knowledge exists not only in documents but also in a vast number of entity relationships and business rules. Through knowledge graphs and ontologies, enterprise business objects, relationships, and

business semantics are structured and expressed in a machine-readable way—stored and processed in graph databases to provide models and agents with inference-ready business knowledge.

Vector retrieval solves the problem of finding relevant information, while knowledge graphs solve the problem of understanding business relationships. Together, they drive AI from information retrieval towards business inference.

Long-term memory (agent memory): Enabling AI with continuous cognitive capabilities

Designed for long-running agent scenarios, this capability manages business experience and contextual information accumulated across different tasks and sessions. It transforms AI from a one-time interaction tool into an intelligent entity that continuously participates in business workflows.

Training dataset: Forming a closed loop for AI capability optimization

Business interaction data, inference logs, and human feedback are accumulated to build high-quality training data assets. This fuels a data flywheel of use—feedback—optimization.

The first three types of capabilities handle knowledge supply during token generation, making AI outputs more aligned with enterprise operations. Meanwhile, the training dataset continuously optimizes model capabilities through business feedback, forming a closed loop for the continuous evolution of enterprise AI.

3. Agent Runtime Environment: Driving Business Actions with Tokens

Traditional applications follow a user request–system response model, while agents complete complex tasks through autonomous planning, tool calling, and

continuous execution.

In the agent architecture, AI computing handles inference and decision, AI data services provide context and long-term memory, and general-purpose computing

environments support tool execution and business operations. Therefore, AI infrastructure needs to further support the long-term operation and large-scale collaboration of intelligent tasks.

The agent runtime environment primarily encompasses three types of capabilities.

- **Secure execution environment: Ensuring reliable operation of agents.** Agents involve tool calling, code execution, and access to external systems, requiring them to operate within secure and isolated environments. The platform must provide resource quotas, permission control, and behavior audit capabilities at the agent and task granularity, ensuring that the execution of intelligent tasks is controllable and traceable.

- **Identity and permission governance: Managing digital workloads.** As agents become the new digital execution entities, each agent needs an independent identity and adheres to the principle of least privilege. Enterprises must extend traditional identity authentication and access control systems to AI entities to achieve secure governance of agents.

- **Multi-agent orchestration and lifecycle management: Supporting large-scale collaboration.** As agents evolve from monolithic applications to multi-agent collaboration, the infrastructure needs to provide capabilities such as task orchestration, status management, resource isolation, and cost control, enabling multiple agents to collaborate on complex tasks like an enterprise team.

Key Directions for Enterprise AI Infrastructure Planning and Construction

When building AI infrastructure, enterprises should not focus solely on current computing power requirements. Instead, they must design holistically for the continuous evolution of models and the large-scale deployment of intelligent applications, focusing on three key directions: Resource elasticity, efficient token production, and production-level resilience.

1. Resource elasticity: Ensuring computing power investments adapt to future changes

The biggest investment risk for AI infrastructure is that it becomes unable to adapt to future model and business changes after being built.

Enterprises need to support not only large-scale model training, MoE inference, and agent clusters in the future, but also diverse requirements, such as departmental applications and small model inference. Therefore, AI infrastructure must provide resource elasticity that adapts to change:

↑ **Scale-up:** Leverages SuperPoD and cluster capabilities to support larger-scale inference and training tasks.

↓ **Scale-down:** Uses resource partitioning and isolation capabilities to meet business needs.

The ultimate goal is: **To enable a one-time infrastructure investment to adapt to the continuous evolution of model scale, business forms, and application requirements.** Since enterprise procurement cycles are typically measured in years, while AI technology advances in months, only an elastic architecture can bridge this gap.

2. Efficient token production: Transforming computing power into effective intelligence

Many enterprises share a familiar pain point: They pour massive investments into computing power, yet utilization stays low—and business teams still complain about shortages. Measuring this with AI-specific utilization metrics alone misses the mark, since those numbers fail to capture task scheduling efficiency, model inference speed, state reuse, or actual business output.

In the age of AI, companies must refocus from "how much hardware is used" to "how many useful tokens can we generate per unit of compute."

Boosting token output requires attention to three core levers:



• **Computing efficiency:** Maximize output per unit of compute through model optimization, parallel processing, and smarter inference scheduling, directing more power toward productive work.



• **State efficiency:** Cut down on redundant calculations using techniques like KV Cache, intelligent caching, and state management, reusing prior results to lower costs for long-context and agent-based tasks.



• **System efficiency:** Lift overall resource usage via unified scheduling, hybrid deployments, and coordinated orchestration that improves collaboration across diverse workloads.

Ultimately, the question shifts from "what's our GPU utilization" to "how many effective tokens does each dollar of investment produce."

3. Production-level resilience: Making AI a reliable production system

As AI integrates into core business operations, one question becomes unavoidable: **What happens when intelligent production stops?** Reliability in the AI era has evolved—from "service continuity" to "task continuity."

Enterprise-grade AI infrastructure demands resilience across three layers:

Layer 1: Instance reliability prevents a single point of failure from disrupting services. Multi-instance deployment, replication mechanisms, and automated failover ensure no individual node can bring down operations.

Layer 2: State reliability keeps tasks running without interruption. For complex workflows like agent-based or multi-turn interactions, API uptime alone is not enough. Session state persistence, KV cache restoration, and uninterrupted long-task execution are essential—so a single fault does not force a full restart.

Layer 3: Service reliability guarantees enterprise-level operational continuity. Cross-cluster and cross-regional deployments, paired with robust DR systems, shield core services from failures within any single resource domain.

In mission-critical sectors like finance and energy, production-grade resilience is not optional—it is the foundational requirement for scaling AI effectively.

[Summary]

AI infrastructure's core mission: Build new foundations purpose-built for intelligent production. Going forward, enterprise gaps will hinge on how seamlessly they integrate computing, network, storage, and orchestration—and on the resulting ability to produce tokens at scale with high performance, quality, and sustainability.

As AI shifts from competing on models to deploying at scale, infrastructure is evolving beyond mere computing support into a full-fledged production engine driving intelligence. This transformation makes it the essential bedrock for enterprises stepping into the intelligent era—and the backbone of their competitive edge in tomorrow's AI race.

2.3 Watt-Centric AI Facility

The planning and construction of traditional DC facilities are based on three premises: Predictable power, standard cooling, and steady loads. In the AI era, that deterministic model is crumbling.

Why? Because DCs' power demand is surging—from chip to cabinet to campus. Single-chip power now approaches kilowatts; single cabinets hit hundreds of kilowatts; entire

clusters scale from tens of megawatts to gigawatts. A GW-level AI DC campus rivals the power load of a million-person city. That magnitude shatters conventional infrastructure limits. Power supply is no longer a support function—it's the decisive gatekeeper for project viability and build-out strategy.

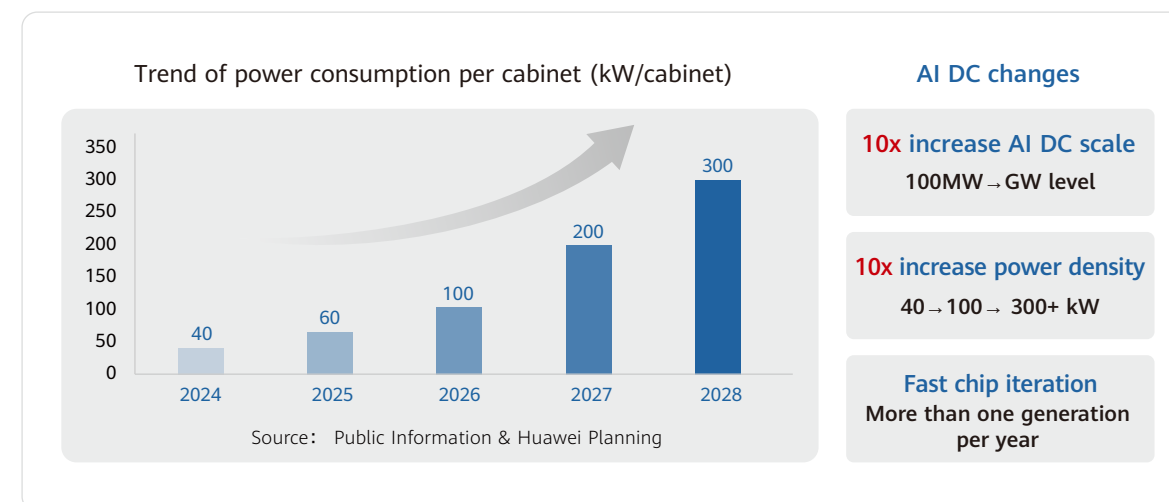


Figure 2-23 Power load trend of DCs

To adapt, facility design must evolve across three fronts: energy, space, and time.

- Site selection puts power first, using integrated generation-grid-load-storage systems to balance reliability and cost.
- Room layouts and building structures are specifically engineered for ultra-high-density intelligent computing

clusters.

- Planning and construction embrace modular prefabrication to compress timelines, while decoupled large and small electromechanical systems enable flexible scaling and seamless adaptation across multiple generations of computing hardware.

Systematic Challenges of AI Facilities

Power Supply and Quality: From Campus Support to Site Selection Anchor

Site selection logic has been completely rewritten. With AI DC campuses scaling at breakneck speed, the selection

logic has shifted entirely from a focus on "user coverage and network latency" to one centered on "power capacity, power supply structure, grid connection period, and long-term power supply stability."

Bidirectional coupling challenges of generation, grid, load, and storage: As the penetration rate of renewable energy continues to rise, the volatility and uncertainty of energy supply may impact the stable operations of AI clusters. Meanwhile, 10,000-card AI clusters, characterized by high density, high power, and strong synchronization, can destabilize the power grid through their dynamic load changes.

Power Supply and Distribution: The Space and Reliability Costs of High-Current Transmission

When the power of a single AI SuperPoD exceeds 100 kW, the cluster's total power scales to hundreds of megawatts. The 63 A three-phase power supply loop commonly used in traditional DCs is no longer sufficient for these demands. To meet high-current transmission requirements, conventional solutions rely on increasing the number of cables and distribution loops, which triggers a cascade of issues: Cabinet and cable tray space is heavily occupied, copper losses and voltage drop rise simultaneously, and the increased number of distribution nodes amplifies the probability of faults.

Cooling System: The Dilemma of Liquid Cooling Pipe Configuration Planning

In the intelligent computing era, air cooling can no longer meet the power demands of a single cabinet, making liquid cooling inevitable. However, planning liquid cooling pipes presents a dilemma: If the pipes, Coolant Distribution Units (CDUs), and end ports are fixed during initial construction, subsequent expansion requires major rework. Conversely, configuring them for the maximum flow rate leads to control difficulties and extra energy consumption due to "large pipes, small flow" during daily operations. The industry needs a pipe architecture that can be split and dynamically adjusted, allowing capacity to scale gradually with computing power demands. This imposes new requirements for modularity and elasticity in liquid cooling design.

Liquid cooling also introduces risks such as hydraulic imbalance, coolant maintenance, and leakage. The liquid cooling pipes can extend up to 10 km, and any local blockage, corrosion, or control mismatch can create hotspots. The system must continuously monitor flow rate, pressure difference, temperature, coolant conductivity, and pH value, establishing a leakage detection and coolant health prediction system.

Building Structure: Floor Height and Load-Bearing Capacity Become Core Challenges

As liquid cooling systems introduce larger-diameter supply and return pipes, the cross section of power cables continues to increase, and the scale of optical fiber for cluster interconnection grows rapidly. The floor height, overhead space, and equipment room layout of traditional DCs can no longer meet future requirements.

Moreover, the weight increase brought by ultra-high-power AI cabinets places higher demands on the load-bearing capacity of building structures. Currently, the weight of an intelligent computing cabinet is about 16 kN/m². In the future, the weight of a single ultra-high-power cabinet exceeding 300 kW will surpass 20 kN/m². The existing standard floor load design (8–12 kN/m²) is no longer applicable.

A building's typical lifespan is 50 years. Yet its initial design and construction lock in critical limits—load-bearing capacity, floor heights, and main utility routes—that are nearly impossible to alter later. The real structural challenge lies here: How do we build in enough flexibility today for tomorrow's unknowns.

Construction Period: Engineering Delivery Cannot Keep Pace with Computing Power Iteration

Currently, the iteration pace of intelligent computing hardware has been compressed to one generation per year. Chip power consumption, cabinet specifications, and cluster interconnection solutions are continuously updated, making it difficult for a static, one-time equipment room architecture to adapt to rapid hardware changes. The traditional civil engineering-led construction period is 12 to 18 months, while the industry generally requires intelligent computing facilities to be delivered within six to eight months—a significant conflict in timelines.

In addition, the load of Agent inference services is highly bursty, with power peaks potentially fluctuating exponentially in a short period. This forces the infrastructure to have the capability for rapid elastic expansion. Early-stage infrastructure planning must reserve sufficient adjustment space, relying on an elastic architecture to accommodate hardware iteration cycles and service load fluctuations.

Suggestions on AI Facility Planning and Construction

Campus Site Selection Centering on Electricity and Generation-Grid-Load-Storage Collaboration Architecture

Power supply capability and long-term stability are the primary decision factors for AI DC site selection. Therefore, regions rich in green electricity have become the first choice for large AI DC campuses—not only do they offer abundant power, but they also typically provide advantages, such as ample land and looser carbon emission constraints.

AI services vary widely in power demands—site selection must reflect that diversity:

- Foundation model training: Stable load and large scale, prioritizing low-cost and abundant electricity
- Offline inference: Cost-sensitive, focusing on the balance between electricity price and throughput
- Agent interactive inference: A trade-off between electricity price and access latency
- Edge scenarios: Prioritizing user distance and data compliance

In regions rich in green energy, the core challenge is the volatility of renewable output. To balance power supply capacity with quality, large-scale intelligent computing campuses must adopt an integrated generation-grid-load-storage architecture—featuring grid-forming energy storage systems linked to nearby energy bases—enabling localized production and consumption of renewables.

Grid-forming energy storage delivers four critical capabilities:

- Millisecond-level response to smooth renewable energy fluctuations
- Support for microgrid voltage and frequency stability
- Absorption of instantaneous power spikes from AI workloads, minimizing grid feedback
- Seamless island-mode operation during brief outages, ensuring uninterrupted computing power

Still, this architecture isn't a silver bullet—storage remains costly, and renewable output is inherently unpredictable. The answer lies in tightly coupling energy management with computing scheduling: Flexible tasks shift to off-peak periods based on generation patterns, while critical loads receive priority access to stable power.

Ultimately, siting and building AI DC campuses comes down to one thing: Finding the optimal spot between power reliability and long-term electricity costs—guided by the "power-first" principle.

Power Supply Architecture Shifting from AC Dominance to Coexistence of AC and High-Voltage DC

To support high-density computing upgrades, the power supply architecture must evolve—without compromising scalability or stability. In mid-to-low density and existing environments, AC remains dominant thanks to its mature industrial chain, established standards, and proven O&M practices. However, for ultra-high-density cabinets above 300kW, conventional AC faces mounting strain from rising current, cabling volume, and spatial demands. The 800V high-voltage DC reduces conversion stages and lowers line current, making it a key direction of development. While 300kW marks a useful planning benchmark, the optimal choice depends on server interfaces, reliability, cost, standards, and ecosystem readiness.

Three primary 800V DC architectures exist:

Route	Basic Architecture	Main Advantage	Application Consideration
Centralized DC	The transformer and centralized DC UPS are deployed together, supplying power to multiple PODs or cabinets through the 800 V DC bus.	Devices are centralized, resulting in relatively lower system costs. The power supply system is decoupled from IT power density, facilitating capacity pooling and elastic evolution.	DC bus protection, fault isolation, and O&M systems need to be improved. This solution is suitable for large-scale new campuses with relatively unified planning.
Distributed DC	AC power is first delivered to the equipment room or POD, where DC UPSs near the loads then provide 800 V power.	The fault domain is small, and capacity expansion is flexible, making it easy to adapt to different generations and power levels of cabinets.	The number of devices and control nodes increases, requiring unified interfaces, collaborative control, and full-lifecycle O&M.
Sidecar solution	An independent power supply unit is deployed on the SuperPoD or high-density cabinet side, connecting to IT devices via a short DC link.	It is closely integrated with SuperPoDs and can be flexibly deployed to shorten the distance between the power supply and the end load.	Space, heat dissipation, maintenance, and multi-vendor interfaces on the cabinet side need to be coordinated, placing higher demands on electromechanical and IT system collaboration.

No matter which route you take, 800V is a full-chain system-level shift. On the hardware side, it demands seamless integration across DC UPSs, HV lithium batteries, HV busbars, connectors, circuit breakers, and server PSUs. On the industry side, success hinges on synchronized progress in standardization, certification, and operational expertise.

Heat Dissipation System with Air and Liquid Cooling from the Same Source, an Elastic Pipe Network, and Full-Link Reliability

The heat dissipation system adopts an air and liquid cooling from the same source architecture. The front-end cooling source and primary pipe network are planned in a unified manner, with liquid cooling and air cooling sharing the same source. The end cooling mode—air cooling, liquid cooling, or a hybrid—is flexibly selected based on cabinet power and device type. Quick connectors for liquid cooling and air cooling supply conditions are reserved for the same rack position, enabling flexible deployment of devices from different generations within the same

equipment room.

On the liquid cooling side, main pipes are planned based on long-term capacity requirements. The CDU and secondary branch pipes feature modular, zoned, and variable flow designs, supporting step-by-step capacity expansion by POD. The system should continuously monitor key parameters such as flow rate, pressure difference, temperature, working medium conductivity, and pH value. It should quickly identify and automatically isolate anomalies like leakage, flow interruption, and air blockage, and perform predictive maintenance based on working medium health trends to ensure reliable operations of intelligent computing devices.

For high-density heat dissipation requirements of a single cabinet with around 20 kW power, the following two solutions can be adopted on the air cooling side:

Solution1 (room-level air conditioners on both sides): Large air curtains are deployed on both sides of the equipment room to increase the number of cooling terminals and improve heat dissipation. Compared with

room-level air conditioners on one side, dual-side air supply can narrow the width of cold and hot aisles, reduce the ceiling height for air return, and improve the space utilization of the equipment room.

Solution2 (near-end in-row air conditioners): Small air curtains are deployed near the micro-module, in the same row as the intelligent computing devices. This solution features hot aisle containment and cooling capacity pooling, eliminating the need for a return air ceiling. It has lower requirements on the equipment room height. However, the heat dissipation devices and intelligent computing devices are deployed together, resulting in overlapping O&M spaces and higher demands on O&M management.

Building Structure Design that Reserves Space for Evolution

Load-bearing design: For a new liquid-cooled computing equipment room, it is recommended that the floor load-bearing capacity be 20 kN/m² or higher.

Floor height design: For ultra-high-power cabinets (300 kW+), the raised floor inside the micro-module stands 1 meter tall, reserving space for power and fiber optic cabling. Fire suppression pathways, the return air plenum, and overhead cable trays demand roughly 5 meters of clearance, while structural beams consume another meter. This brings the minimum equipment room height to 7.5 meters—or a clear under-beam height of at least 6.5 meters.

Ample room shall be reserved for transporting cabinets, replacing devices, and performing maintenance. Primary power and liquid-cooling routes should be as independent and scalable as possible, preventing cross-discipline conflicts above cabinets and beneath the floor—a critical safeguard for long-term operational safety and seamless capacity expansion. Transport ramps should maintain a slope no steeper than 8 ° and a width no less than 2.2 meters.

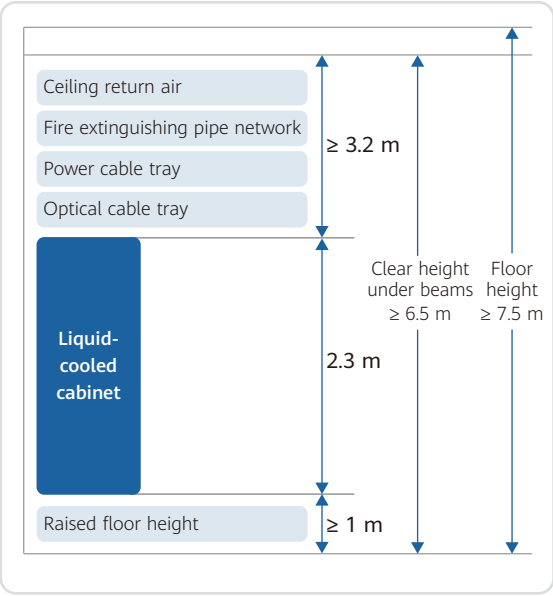


Figure 2-24 Recommended clear height of the equipment room for SuperPoDs

Delivery Period Shortened Through Prefabrication, Modularization, and Productization

Prefabrication and productization are critical strategies for shortening construction timelines. For AI DCs, the POD should serve as the fundamental building block, with standardized designs across power supply, energy storage, cooling, and IT systems. Electromechanical modules are pre-assembled, integrated, and tested offsite—only lifting, connections, and joint commissioning happen onsite. Wherever possible, servers should arrive as fully integrated cabinets to minimize field installation and tuning. To validate the solution's viability, we also leverage AI-driven design, digital handover, and end-to-end simulation.

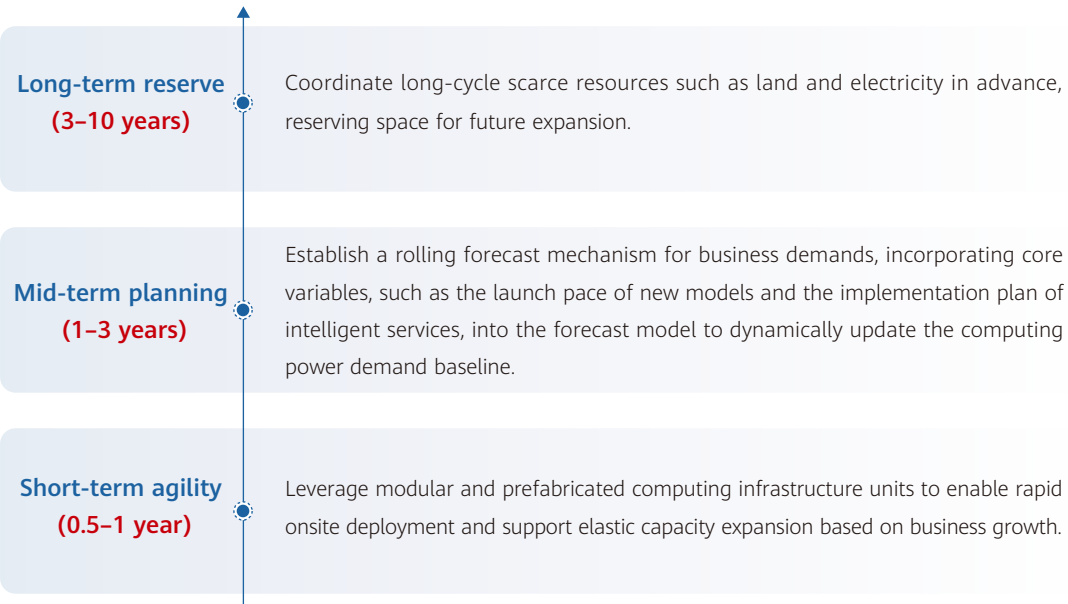
From a construction standpoint, an AI DC is essentially a massive computing-power production system, demanding seamless integration of core products, design modeling, standardized interfaces, system validation, and digital fabrication. By running civil works and mechanical/electrical tasks in parallel, using framework agreements and rolling material reserves, and combining factory prefabrication with modular onsite assembly, we enable repeatable, scalable builds. This cuts the typical 12–18 month schedule down to just 6–8 months, rapidly accelerating computing capacity deployment.

Implementation of AI Facilities in New and Existing Scenarios

Deployment Principles in New Scenarios

Given the industry's fluctuating computing power demands and rapid hardware iteration, building computing infrastructure calls for a three-tier strategic

framework: Long-term reserve, mid-term planning, and short-term agility. This ensures supply stays dynamically aligned with evolving business needs.



Based on the above principles, new projects can be implemented along two paths.

Path 1 (civil work + large-scale electromechanical systems first): When detailed business requirements are not yet fully defined, initiate the construction of large-scale electromechanical systems in advance based on the predicted power range and air/liquid cooling ratio. Once cabinet specifications and load requirements are confirmed, proceed with the implementation of small-scale electromechanical systems, compressing the overall readiness period to about three months.

Path 2 (civil work + modular research and development of large- and small-scale electromechanical systems, and framework procurement first): Complete the development of standardized and modular electromechanical systems approximately one year in advance, while simultaneously carrying out framework procurement and rolling stock preparation by suppliers. After business requirements are officially determined, onsite delivery can be completed within about three months.

The concept of elastic construction runs through the

entire cycle: In the long term, lock in the boundaries of scarce resources; in the mid-term, continuously refine demand solutions; and in the short term, achieve agile capacity expansion through standardized POD units.

Reconstruction in Existing Scenarios

Most existing DCs are currently low-density air-cooled equipment rooms, facing multiple constraints such as limited conditions for upgrading from air cooling to liquid cooling, insufficient power distribution capacity, and floor load-bearing capacity and indoor net height not meeting standards. The reconstruction path can be determined based on the following three mandatory indicators:

- Floor load-bearing capacity $\geq 12 \text{ kN/m}^2$
- Net height of the equipment room $\geq 5.4 \text{ m}$
- The power redundancy must be at least 1.5 times the current peak IT load.

If all three conditions are fully met, the DC can be reconstructed into a hybrid air-liquid cooling facility. If only some indicators are satisfied, priority should be given to deploying high-efficiency air-cooled servers to unlock

computing potential. If none of the three conditions can be met, it is recommended to shut down the facility in an orderly manner and replace the resources. The core idea of existing DC reconstruction is to replace low density with high density, leveraging the increased computing density of single cabinets to swap out numerous low-density cabinets with a few high-density ones, thereby revitalizing limited equipment room space.

The liquid cooling reconstruction solution must be flexibly selected based on the project scale. For small-scale reconstruction scenarios, distributed solutions such as distributed ECUs and air-liquid CDUs can be adopted. For medium-scale reconstruction projects, standardized liquid-cooled POD modules should be deployed. The cooling source system should be fully integrated with existing facility conditions, with a choice between retrofitting existing systems or installing entirely new ones based on site-specific circumstances. Prefabricated integrated cooling stations are preferred to shorten the construction period.

The power supply and distribution reconstruction should be implemented step by step based on the existing system, with a targeted upgrade of the power distribution loop at the cabinet end. The new power supply system should use prefabricated modular products, and for end-of-line power distribution. Intelligent busways or intelligent end cabinets are preferred to accommodate computing cabinets of varying power levels.

In addition, the load-bearing capacity of the floor slabs must be verified in advance, meeting load requirements

through structural reinforcement or sparse cabinet arrangement. For projects with limited floor height, BIM modeling should be used for risk simulation. Simultaneously, simulation tools should be employed to optimize the airflow organization in the equipment room. Furthermore, ensure that the net height of the equipment transportation passage is no less than 2.5 meters to meet the conditions for entire cabinet entry and construction.

[Summary]

Watt-centric: Systematically finding the optimal solution under physical constraints

The core objective of DCs is shifting from providing resource capabilities to generating more tokens that are effective per watt of electricity. For AI facilities, the key challenge lies in how to use deterministic physical infrastructure to accommodate the uncertainties brought by energy supply, hardware evolution, and AI load changes.

Future AI facility planning must re-examine three core issues: How to efficiently supply energy, how to flexibly evolve space, and how to quickly respond to construction demands. The solution lies in shifting from static blueprints to elastic architectures. By gaining insights into the long-term evolution trends of AI loads, we can reserve space to ensure cross-generational compatibility, use modular design to support agile capacity expansion, and achieve optimal matching of energy, space, and computing power through system collaboration.



03

AI DC Operations, O&M, and Security

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Historically, a DC's value was determined on the day construction finished. In today's intelligent world, however, its value is continuously generated during operations. As computing power becomes an intelligent production capacity measured by tokens, operating a DC finally feels like running a factory: measuring output, iterating processes, and ensuring security. Finishing construction is just the opening chapter, whereas operations truly unlock value.

3.1 AI DC Intelligent Production Operations

From Compute Resource Operations to AI Production Operations (AIPO)

With large models and agent-based applications rapidly transitioning to production environments, AI DC operations are evolving beyond traditional resource operations into AIPO. This new operational model centers on three core questions: what to produce, how to produce, and how much value is created.

From Resource Assurance to Value Creation

DCs have evolved from traditional resource supply centers into intelligent production hubs.

▪ **From IT-centric assurance:** Requirements definition → application deployment → resource provisioning → system maintenance → service assurance

▪ **To AI-centric production:** Intelligent requirement analysis → model orchestration → compute resource scheduling → model inference services → token generation and delivery → quality validation → value assessment → continuous optimization

Traditional DCs—Resource assurance focused:
The primary objective is to guarantee reliable resource supply, with emphasis on compute capacity, system stability, and infrastructure reliability.

VS

AI DCs—Value creation focused: The core mission is to deliver sustained AI-driven value through continuous, efficient production of intelligent capabilities. Success is measured by the ability to generate tokens with greater efficiency, lower cost, and higher quality, and to consistently convert them into tangible business outcomes.

Production Factor Matrix

In AI DC operations, the production landscape has expanded from isolated IT infrastructure to a diverse,

collaborative mix of resources. The three-tier structure of the AI production system is shown below:

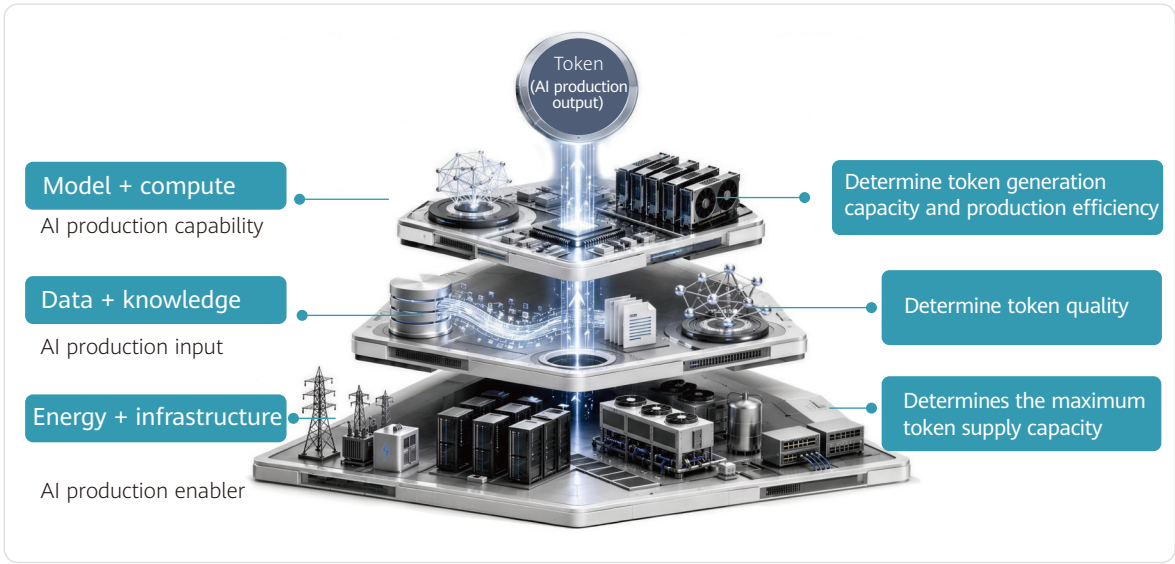


Figure 3-1 AI production architecture

• **Token (AI production output):** As the core unit of measurement, valid tokens bridge production scale, resource consumption, service quality, and business value.

• **Model + compute (AI production capability):** The model defines the boundaries of what AI can do; compute determines the speed and efficiency of that execution. Together, they shape token generation capacity and production efficiency.

• **Data + knowledge (AI production input):** As the foundational input to AI production, data and knowledge directly determine token quality and the overall intelligence level.

• **Energy + infrastructure (AI production enabler):** They provide comprehensive support, including power, networking, storage, and physical space, and define the maximum token supply capacity.

Redefining the Token Intelligent Operations Process and Metric System

1. Intelligent Operations Process

Centered on token production, AI DCs build an end-to-end closed-loop system for intelligent production and operations, spanning demand prediction, capacity

planning, resource orchestration, token generation, value realization, and value assessment, across three phases: **demand-driven planning, intelligent production, and value feedback.**

Demand-driven capacity planning

The scale of AI DC capacity is not determined simply by current business loads. Instead, it requires a holistic approach that factors in **business demand growth, AI technology evolution, and infrastructure construction cycles**, using forward-looking planning to align production capacity with future intelligence demands. This includes:

• **AI demand forecasting:** Leveraging historical call data, business development trends, and application growth patterns to build a multidimensional demand prediction model that forecasts future token consumption, service capability requirements, compute and power needs, and equipment room space demands.

• **Capacity planning and investment decision making:** Translating token demand forecasts into long- and mid-term plans covering compute pool sizing, model capability development, energy infrastructure expansion, and evaluation of Return on Investment (ROI).

• **Dynamic capacity adjustment:** Complementing

long-term plans with short-cycle dynamic adjustments, enabled by elastic scaling, resource scheduling, and tidal optimization, to respond to business fluctuations, emergencies, SLA commitments, and cost optimization needs in real-world operations.

Production factor orchestration and intelligent token generation

Through intelligent scheduling and orchestration of the four core AI production factors, the system enables efficient token generation with seamless cross-factor collaboration. Key capabilities include:

Intelligent orchestration of production factors (**model + compute**): Based on the AI production plan, production factors are precisely aligned with target business scenarios — covering model orchestration, compute orchestration, data and knowledge orchestration, and energy and infrastructure orchestration.

Production data management (**data + knowledge**): This is the core management function that directly determines token quality and domain-specific intelligence levels. It includes data quality operations, knowledge asset management, data value assessment, and knowledge lifecycle management.

Production resource scheduling and allocation (**energy + infrastructure**): Through unified scheduling and optimization strategies, the system enables elastic deployment, high-concurrency processing, and low-latency response for model services.

Token generation optimization: Continuously improving AI production efficiency by optimizing token throughput, compute output per unit, and energy consumption per unit.

Value feedback and continuous optimization

Driven by token production effectiveness and business value feedback, models, resources, and policies are continuously refined. This includes:

• **Token quality evaluation:** Assessing the accuracy and reliability of AI output based on user experience and business applicability.

• **Token value analysis:** Evaluating AI production value across business task completion, user experience improvement, and business revenue impact.

• **Production closed-loop and continuous optimization:** Based on feedback on token quality, cost, and value, the system continuously optimizes model capabilities, data and knowledge assets, resource configurations, and scheduling policies, forming a closed-loop production cycle. Through the accumulation and growth of enterprise intelligent assets, this creates a self-reinforcing AI production flywheel: Token generation → data accumulation → knowledge enhancement → model optimization → capability asset growth.

2. Operations Metric System and Value Mapping

Shifting from resource metrics to production value, AI DCs need to establish a new token-centric operations metric system.

Category	Core Metrics	Description
Token throughput	Token output and token concurrent throughput (TPS/QPS)	AI production scale
Token efficiency	Tokens/Watt	Resource efficiency
Token quality	Accuracy, credibility, hallucination rate, and response experience (TTFT/TPOT)	Output quality
Token cost	Cost per million tokens and cost per task	Economic efficiency
Token resource	Model utilization, compute utilization, and energy efficiency	Synergy efficiency
Token value	Token ROI, business efficiency improvement, business contribution	Value creation, for example, in AI-assisted coding: Efficiency gains × Quality assurance × Cost savings

Five Core Capabilities of AIPO

Centered on intelligent operations, five core capabilities need to be developed, as shown in the following figure.

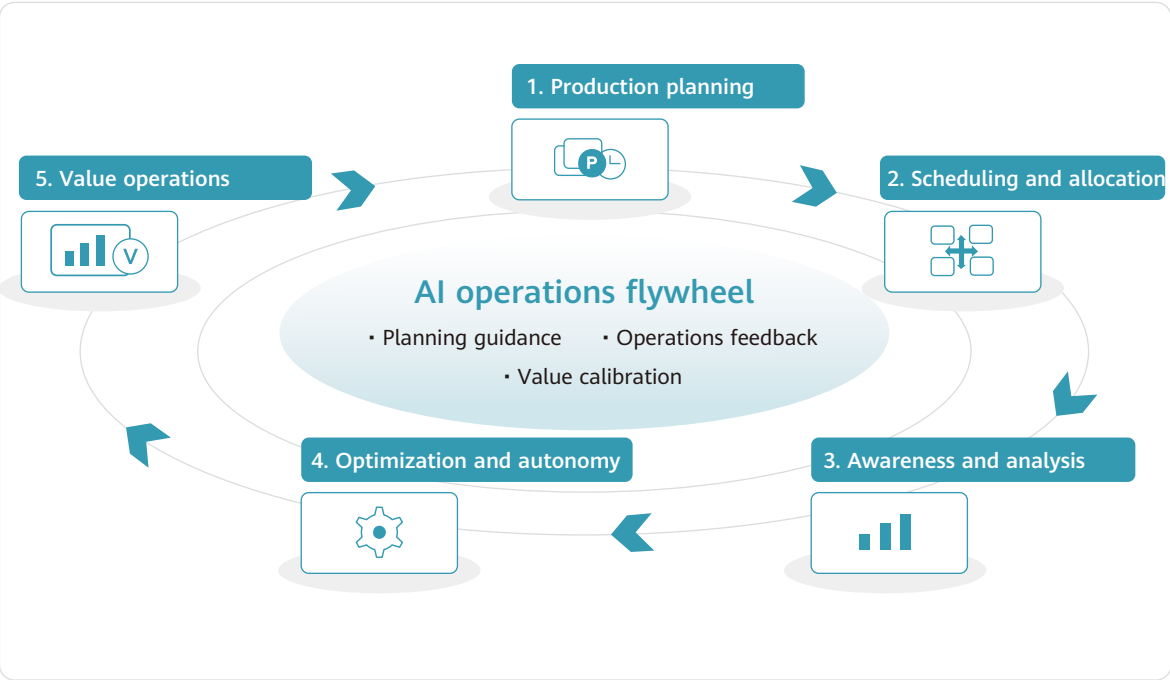


Figure 3-2 Five core capabilities of AI production and operations

Production Planning: Long-Term Rolling Capacity Planning

A rolling planning system for short-, medium-, and long-term capacity should be established to address evolving business demands and technology trends.

1. Middle-to long-term forward-looking capacity planning:

To prevent the cascading delays that arise when business waits for compute, compute waits for equipment room construction, and equipment rooms wait for power infrastructure, it is essential to build an AI demand forecasting model grounded in historical operational data, macro business trends, and large-model evolution. This enables data-driven, long-term planning of compute card capacity, power capacity, and supporting infrastructure for AI DCs. Key activities include:

- **AI demand forecasting:** For production inference, development, testing, and other business scenarios, use token-based metrics to project future AI service demand,

drawing on business growth trends, application invocation patterns, and historical token consumption data.

- **Intelligent capacity forecasting:** Based on projected token production targets, analyze large-model evolution, AI agent development, multimodal application trends, and compute architecture shifts to estimate AI compute and compute card requirements over the coming years.
- **Intelligent resource forecasting:** Using future compute and compute card requirements, conduct integrated planning of power, cooling, and equipment room space to ensure that infrastructure is fit for purpose upon delivery and does not become obsolete or require immediate retrofitting.
- **Capacity planning and investment decision-making:** Synthesize demand, capacity, and resource forecasts; identify gaps relative to current supply; and determine optimal DC layouts, cabinet and power configurations, and the most cost-effective mix of retrofitting, new construction, and leasing.

2. Short-term elastic planning optimization:

Dynamic planning of elastic compute capacity:To accommodate business fluctuations, emergencies, SLA assurance, and cost optimization goals, this initiative enhances dynamic scheduling and allocation strategies to drive short-cycle improvements in computing power and efficiency.

- **Driven by business fluctuations:** The inference cluster size is dynamically adjusted to handle daily tidal traffic patterns.
- **Driven by emergencies:** Redundant computing power is flexibly injected to handle hotspot service traffic surges.
- **Driven by SLA assurance:** Highest-priority resources are reserved for mission-critical tasks.
- **Driven by cost optimization:** Low-utilization resources are automatically scaled down during off-peak periods to maximize Total Cost of Ownership (TCO) amortization.

Elastic infrastructure planning: Retrofit existing equipment rooms to support new compute requirements and enhance the infrastructure's ability to deliver rapid, on-demand support.

Scheduling and Allocation: Intelligent Organization of Production Factors for Efficient Token Generation

Through intelligent scheduling, orchestration, and collaborative optimization of production resources, this capability enables highly efficient token generation. Core functions include:

- **Service catalog and quota management:** Provide a standardized, visualized AI service catalog to diverse business departments and tenants. Implement fine-grained quota management based on token quotas, concurrency limits, and compute weighting.
- **Unified scheduling of heterogeneous compute:** Support pooled management of heterogeneous compute resources.
- **Model-compute collaborative optimization:** Match optimal compute resources to workload based on model characteristics and business requirements. Enable dynamic resource allocation among training, fine-tuning, and inference tasks. For example, during inference off-peak periods, automatically release compute capacity for model fine-tuning or incremental training, significantly improving overall utilization efficiency and the economic returns of compute assets.

• **Energy-compute collaborative optimization:**

Optimize production strategies by factoring in energy costs, green energy availability, and task characteristics. Leverage multi-center collaboration to maximize global compute economic efficiency.

Awareness and Analysis: Intelligent Observability and Analysis Across the Entire Token Generation Process

Establish a full-link awareness system that covers services, models, resources, and infrastructure, enabling unified observability from application requirements all the way through to token production, as well as rapid problem detection and localization. Key capabilities include:

- **Full-link observability:** Map and streamline the relationships between applications, agents, model services, model instances, and compute resources to achieve end-to-end observability across the entire chain.
- **Core metric monitoring:** Collect dozens of metrics in real time, including token consumption rate, QPS, TTFT, TPS, memory fragmentation rate, and compute card power consumption. Leverage intelligent learning to automatically detect anomalous fluctuations, enabling operations personnel to pinpoint performance bottlenecks within seconds.
- **In-depth diagnosis of token production efficiency:** Focus on token output per unit of compute and energy consumption, and benchmark the economic efficiency ratio across different models and quantization precisions to identify underperformers.
- **In-depth diagnosis of model performance:**Thoroughly analyze operator execution efficiency, communication overhead, KV cache memory hit rates, and batch -processing scheduling losses during inference.
- **Resource utilization diagnosis:** Scrutinize idle rates of compute units on heterogeneous compute cards, memory bandwidth bottlenecks, and cross-node network congestion to provide data-driven insights for in-depth optimization.

Optimization and Autonomy: Continuously Improving Token Production Efficiency and Quality

Establish a continuous optimization mechanism covering models, resources, and services, and steadily enhance token production capabilities through production data feedback. This enables the same level of resources to produce more tokens of better quality with lower costs. Core capabilities include:

- **Model optimization and autonomy:** Improve model capabilities through data backflow, fine-tuning, and reinforcement learning.
- **Inference service optimization and autonomy:** Enhance service efficiency through concurrency optimization, KV cache optimization, and scheduling policy improvements.
- **Production efficiency optimization and autonomy:** Continuously improve output per unit of resource, measured by metrics such as tokens per hour and tokens per watt.

Value Operations: From Token Production to Business Value Realization

Establish a value evaluation system that spans token consumption, service quality, and business revenue, linking production outcomes directly to business value. Core capabilities include:

- **Token cost governance:** Analyze cost per token and per task to make AI production costs fully transparent.
- **Token quality evaluation:** Assess accuracy, credibility, hallucination rates, and user experience.
- **Business value analysis:** Measure efficiency gains, business contributions, and ROI of AI applications.
- **Value feedback optimization:** Optimize models, data, resource configurations, and production strategies based on business outcomes.

Through the development of these five core capabilities, AI DCs evolve from a traditional resource management center into a token production value center tailored for the intelligent era. This enables demand-driven capability building (production planning), resource-coordinated intelligent production (production awareness and analysis, resource scheduling and allocation, and production efficiency and quality optimization), and continuous value feedback optimization (production value operations).

Case Study: AI DC Value Operations Upgrade for Banking

As AI strategies become deeply embedded in the core businesses of enterprises, a leading bank has been steadily advancing the large-scale deployment of AI capabilities, with hundreds of AI use cases already in production. AI is now progressively integrated across the entire spectrum of customer service, operational management, and business innovation. As AI DCs enter the phase of large-scale production, operational focus has shifted from early concerns—such as "Is there enough compute?" and "Can systems go live?"—to a new set of priorities: "Is AI production transparent? Are resources precisely allocated? Is token production efficient? Is the value of intelligence continuously improving?"

To turn these priorities into tangible outcomes, the bank has identified the following critical challenges that demand immediate attention.

Challenges:

- **Invisible resources:** There is a lack of a holistic, interconnected view between infrastructure and upper

layer applications, making it difficult to accurately correlate resource consumption with business output.

- **Prediction and planning difficulties:** Long-term capacity planning and daily resource allocation rely largely on manual estimation, which struggles to precisely match dynamically changing business demands.
- **Bottleneck localization challenges:** Under complex, multidimensional workloads, it is challenging to efficiently identify service bottlenecks, leaving room for improvement in the output efficiency per unit of compute.
- **Low resource utilization efficiency:** Scheduling is mostly based on manual experience, lacking data-driven automatic tidal scheduling and elastic optimization capabilities.

To tackle these challenges head-on, the bank has rolled out a series of targeted innovative measures across four key dimensions.

Innovative measures:

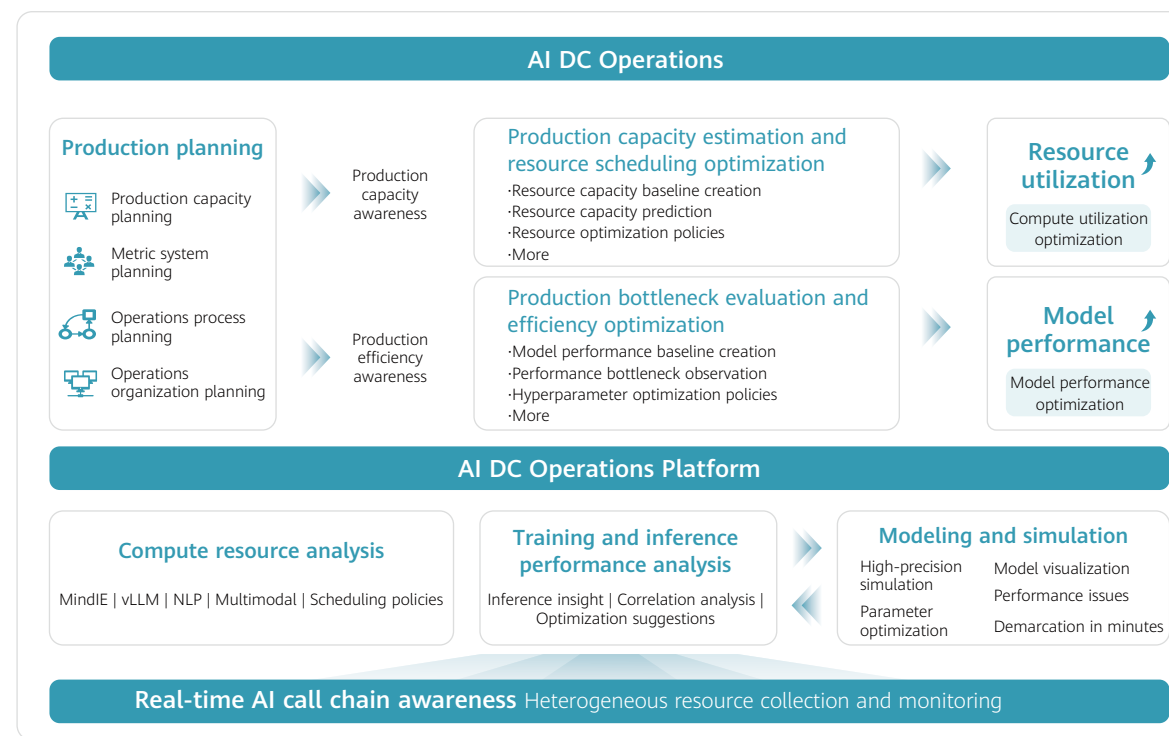


Figure 3-3 Intelligent operations system of a top bank

- **Building full-link observability for AI production to visualize the production process:** Through the AIPO platform, the bank establishes a unified view across the entire chain, from applications to agents, model services, model instances, and compute resources, enabling comprehensive monitoring of the AI production process. The key focus is on visualizing token consumption, compute utilization efficiency, and service performance. By establishing a token-centric production metric system, the bank shifts from resource status monitoring to AI production efficiency analysis.

- **Building demand-driven AI capacity planning capabilities to achieve precise resource allocation:** To address the rapid growth of AI services and the difficulty of predicting resource demands, the bank develops compute simulation and capacity evaluation capabilities. Based on key parameters such as service SLA, concurrency scale, throughput requirements, response latency, and input/output length, the bank models and analyzes the resources required for model services, generating optimal resource configuration plans that meet business objectives. Before model rollout, they rigorously evaluate compute demands to avoid over-provisioning. During production

operation, they dynamically adjust resource scale based on business changes. During performance optimization, they improve resource utilization efficiency through parameter tuning. This approach drives the transformation of AI capacity building from experience-based estimation to data-driven, model-based planning.

- **Building continuous model optimization capabilities to improve token output efficiency per unit of compute:** To address complex performance issues that arise during large model inference, including computing, communication, memory, and scheduling, the bank develops model performance analysis and optimization capabilities. By analyzing model execution, they identify performance bottlenecks across dimensions such as service metrics, compute load, communication efficiency, operator execution, and memory usage. Furthermore, they optimize model parameters, inference frameworks, and resource configurations to continuously improve model services. This helps increase token output per unit of compute, improve model response efficiency, and reduce AI production costs.

• **Building intelligent scheduling capabilities to dynamically optimize AI production resources:** AI service loads exhibit clear tidal characteristics. By predicting service loads, analyzing resource usage, and intelligently scheduling resources, the bank can dynamically allocate training and inference capacity. Specifically, resources are released during inference off-peak hours for training and optimization; inference service capacity is automatically scaled up during peak hours; and resources are dynamically adjusted based on SLAs and service priorities. This enables the evolution from experience-based scheduling to data-driven intelligent scheduling.

Practical achievements and overall value:

Through the capability development outlined above, overall compute utilization efficiency has increased by 10%, and the operational model has shifted from passive troubleshooting to proactive, metric-driven operations. Model performance analysis cycle time has been reduced from days to hours. For example, in the case of Qwen large-model inference, TPS has been improved by 30%, TPOT decreased by 30%, and problem diagnosis and resolution efficiency increased by over 70%. Ultimately, these improvements drive the continuous evolution of AI DCs—from resource-centric operations to value-driven operations.

Evolution Path of AIPO

In the future, AI DC operations must evolve toward a new operational paradigm centered on token production efficiency, quality of AI services, and business value creation. The maturity evolution of AI DC production operations can be divided into four phases, as listed in the following table.

Evolution Phase	Core Object	Core Objective	Key Capability
Resource assurance operations	IT resources	Stable running	Resource management, fault assurance, and capacity management
AI production operations	Token production process	Efficient production of intelligent capabilities	DPV closed-loop, production factor collaboration, and token efficiency optimization
Token value operations	Token value	Business value creation	Quality operation, cost management, and value assessment
Agentic autonomous operations	AI operations agent	Autonomous optimization of operations	Autonomous planning, scheduling, and self-evolution

Phase 1: Resource assurance operations. This phase builds on the traditional DC operations framework, with a primary focus on the availability and reliability of compute, storage, network, and infrastructure resources. The key questions addressed at this stage are: Are resources available? Can they be readily utilized.

Phase 2: AIPO. The core objective is to improve AI production efficiency and enable large-scale token generation. As AI applications move into production, the operational focus shifts from resource assurance to AI production process management. The Demand-Production-Value (DPV) model is introduced to achieve closed-loop production. The key questions addressed at this stage are: What to produce? How to produce? How efficient is the production process.

Phase 3: Token value operations. The core objective is to

close the loop between AI production outcomes and business value. As enterprises embed AI applications more deeply into their core businesses, AI DC operations must further center on the business value generated by tokens, evolving from producing tokens to creating intelligent value. The key question addressed at this stage is: Is the intelligence produced truly valuable.

Phase 4: Agentic autonomous operations. The core objective is to enable AI DCs to achieve autonomous planning, optimization, and operations. As AI agent capabilities mature, AI DC operations will shift from being human-driven to agent-driven, forming an autonomous operations system with self-sensing, self-decision-making, self-execution, and self-optimization capabilities. The key question addressed at this stage is: How can AI DCs autonomously create and continuously improve intelligent production capabilities.

Ultimately, this evolution will give rise to a next-generation AI DC operations system—one that uses tokens as the core unit of measurement; models, data, compute, and energy as the core production factors; DPV as the foundation for

full-link closed-loop operations; token value maximization as the core operational objective; and agentic autonomy as the long-term evolutionary direction.

3.2 AI DC Autonomous O&M Assurance

From Device O&M to Intelligent Production O&M Assurance

The goal of AI DC O&M has shifted from keeping devices available to ensuring stable operation across the entire compute-model-token-application chain. Compared with traditional DCs, AI DCs face four core challenges in O&M:

Rising infrastructure complexity: New technologies and solutions are constantly emerging, from the deep integration of compute, storage, and network, to high-speed interconnect networks and compute-cooling synergy across infrastructure. This significantly raises the bar for fault locating, performance optimization, and resource scheduling.

Expanding O&M assurance scope: Traditional DC O&M focuses on infrastructure, software platforms, and service applications. In contrast, AI DC O&M is oriented to a three-layer new architecture: the AI facility layer that guarantees compute and energy supply, the AI infrastructure layer that supports model inference and compute scheduling, and the AI application layer that delivers agent and intelligent services. As a result, the O&M goal has expanded from observing device health to ensuring compute effectiveness, token service quality, and service value.

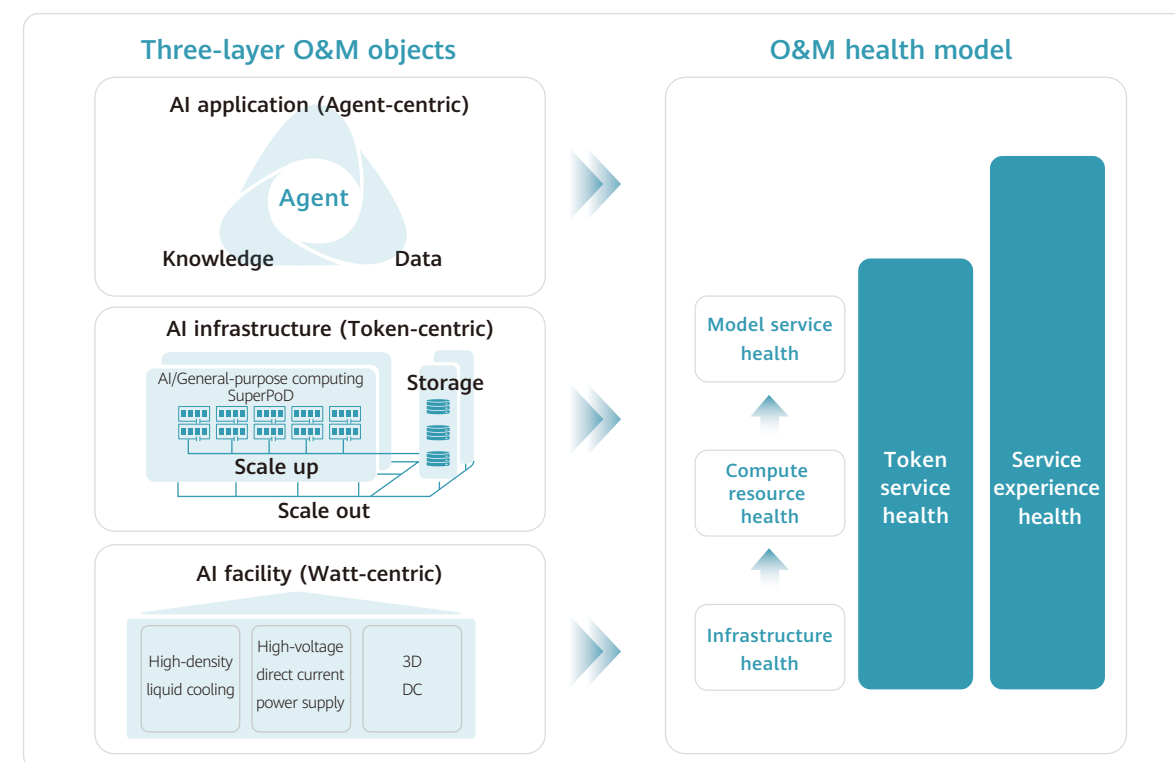


Figure 3-4 Expanding DC O&M assurance scope

Longer system chains and complex cross-layer

correlation: Any anomaly along the intelligent production chain, from resource supply and AI infrastructure support to agentic service execution, can lead to service degradation. For example, a delayed response from a power grid scheduling agent, a drop in renewable energy prediction accuracy, or a problem in defect detection all demand urgent cross-layer correlation analysis and end-to-end root cause locating capabilities.

Six Core Capabilities of Agentic Autonomous O&M

Given the intricate interplay between devices and services in AI computing clusters, the traditional O&M mode that relies on manual monitoring and experience-based troubleshooting can no longer meet the stringent reliability and quality requirements of AI production. To address these challenges, AI DCs must adopt an agentic autonomous O&M system that enables **service awareness, issue prediction, autonomous fault handling, and capability evolution**.

01 All-Domain Intelligent Awareness

In the past, DC O&M mainly focuses on the basic infrastructure such as servers, networks, and storage. In the AI era, however, the objects have expanded to include token production and upper-layer agent services. To gain a comprehensive and accurate view of an AI DC's overall status, traditional data silos must first be dismantled.

Core capability analysis:

- **Unified measurement with a comprehensive O&M metric system:** Establish an O&M metric system that includes equipment room infrastructure metrics, AI infrastructure performance metrics, and agentic service experience metrics, to enable unified measurement from device health and compute utilization efficiency to model service quality, token service efficiency, and service experience.

- **Unified multi-source data collection with a full-chain digital profile:** Deeply integrate metrics, log records, link data, topology relationships, and configuration changes to construct a full-chain digital profile of the AI DC.

- **All-domain object modeling and dynamic status mapping:** Encompass infrastructure objects, compute, network and storage resources, model services, token services, and agent service processes to achieve unified

Stringent reliability requirements and diverse fault

modes: AI DCs must ensure high compute availability and rapid responsiveness. Faults now extend beyond hardware into areas such as communication, memory, scheduling, and inference quality. Manual O&M can no longer keep pace with the scale of monitoring and self-healing required.

management of AI DC O&M objects. Synchronize real-time states of underlying resources, key performance metrics, service operations, and upper-layer service experience to create a living digital profile of the AI DC. Accurately define resource dependencies, service invocation relationships, and fault propagation paths to provide strong support for impact analysis and risk prediction.

Together, this awareness system makes the AI DC operating status truly visible, comprehensive, and accurate. It transforms O&M from reactive troubleshooting into proactive awareness, establishing the indispensable real-time data foundation for autonomous O&M.

02 Service Health Evaluation

Traditional O&M focuses on device availability, simply checking whether it is functioning or faulty, while AI O&M emphasizes service quality, asking whether performance is both fast and reliable. Metrics like the TTFT, end-to-end response, and request success rate have become the core benchmarks of AI services.

Core capability analysis:

- **End-to-end experience measurement:** Build a full-chain experience metric system that spans business applications, agentic services, AI infrastructure, model services, and compute resources, with a focus on measuring the TTFT, end-to-end inference latency, request success rate, and service stability.

- **Cross-layer experience correlation analysis:** Streamline the intrinsic correlation between service experience and underlying resources to quickly detect anomalies, precisely locate model issues, and perform collaborative analysis of compute resources.

- **Health evaluation and automatic optimization suggestions:** Provide system health scores, risk trend

prediction, and performance degradation analysis. Based on historical baselines and high-precision health evaluation models, identify abnormal metrics, analyze trend changes, and detect potential risks such as hardware degradation, performance deterioration, and resource bottlenecks. With intelligent analytics, detect experience degradation within seconds, automatically pinpoint root causes, generate optimization suggestions, and continuously improve service quality.

This transformation eliminates the blind spot in O&M, which used to focus only on machines rather than services. It ensures that token services are efficient and controllable, while establishing a solid foundation for the commercial implementation of large models.

03 O&M Knowledge System and Ontology Construction

AI DC fault locating heavily relies on the practical experience of experts. Without unified knowledge accumulation, intelligence becomes unsustainable.

Core capability analysis:

- **O&M knowledge system construction:** Use AI technologies to automatically extract high-value knowledge from numerous O&M documents, fault reports, historical operation records, and expert experience. Build a systematic Standard Operating Procedure (SOP) repository, RAG knowledge base, fault case library, and expert experience library, enabling unified management and intelligent invocation of accumulated knowledge.

Cross-domain knowledge convergence: Effectively streamline knowledge across diverse domains such as compute, networks, models, services, and security, and build a unified enterprise-level O&M knowledge network.

External ecosystem integration: Actively incorporate public experience, third-party best practices, open-source fault case libraries, and industry standards from the external AI computing industry chain, and integrate them with internal O&M practices, therefore broadening the knowledge scope and accelerating overall learning and evolution.

- **Skill system construction:** Model and encapsulate experts' practical capabilities into standardized skills, and enable agents to be quickly composed and invoked across all aspects of O&M.

The awareness skill covers log analysis, metric analysis, and anomaly detection.

The diagnosis skill covers alarm correlation, root cause analysis, network diagnosis, and cluster communication analysis.

The execution skill covers parameter adjustment, service recovery, and resource scheduling.

- **O&M ontology model construction:** Implement resource relationship modeling, service dependency modeling, fault cause-and-effect modeling, and in-depth analysis of service impacts. Systematically build the **resource ontology, service ontology, fault ontology, operation ontology, and service impact ontology**, and define standard object attributes, relationships, and operational rules.

For example, a leading bank has refined its processes by breaking down the service value chain from service activities to governing rules, clarifying the content, boundaries, participants, and core logic of each O&M phase. The knowledge model focuses on key O&M information by identifying elements, classifying topics, and performing standardized modeling, and establishes clear ownership and standards. This approach ensures data knowledge consistency across multiple systems, and is gradually implemented in scenarios such as change management, monitoring, assurance, and incident management.

- **Automatic knowledge recycling:** Use the knowledge recycling agent to automatically accumulate practical diagnosis processes, root cause conclusions, and handling solutions, and feed them back into the knowledge system.

This mechanism enables AI to assimilate the specialized knowledge of industry experts and replicate expert experience at scale.

04 Intelligent O&M Orchestration

While numerous single-point tools exist for monitoring, diagnosis, and automatic execution, they remain fragmented and disconnected. The key to achieving true autonomous O&M lies in enabling these capabilities to operate in concert.

Core capability analysis:

- **O&M intent understanding:** Automatically interpret O&M goals described in natural language (for example, "Analyze the underlying cause of the increased model inference latency") and generate a corresponding analysis path.

• **Intelligent task breakdown:** Based on large models' deep understanding of user intents, automatically break down O&M goals into subtasks with appropriate granularity (such as agent selection, skill invocation, and model invocation), standardize the execution process, and clearly define the dependency relationships to build a dynamic task execution flow.

• **Collaborative scheduling of agents:** Support sequential orchestration, parallel orchestration, hierarchical orchestration, and multi-agent mesh collaboration to enable dynamic collaboration among agents across different domains. Domain-specific agents include:



• **Awareness agent:** Monitors all-domain status, accurately detects anomalies, and identifies potential risks.



• **Diagnosis agent:** Efficiently locates faults, performs in-depth root cause analysis, and generates handling solutions.



• **Prediction agent:** Evaluates system health, predicts trends, and provides early risk warnings.



• **Recovery agent:** Automatically executes tasks, smoothly recovers services, and verifies execution effectiveness.



• **Knowledge agent:** Intelligently extracts knowledge, accumulates historical cases, and enhances underlying capabilities.

• **Human-machine collaborative control:** Support solution confirmation, manual approval of high-risk operations, supervision of execution processes, and objective evaluation of performance.

• **Closed-loop O&M management:** Enforce a closed-loop process covering discovery, analysis, decision, execution, verification, and knowledge accumulation. Implement strict permission control, risk assessment, and automatic rollback mechanisms to ensure the security and controllability of agent behaviors in an all-round manner.

Through orchestration, formerly isolated O&M tools and agents are transformed into an efficient intelligent collaboration network, enabling the entire process from discovery, analysis, decision, to execution to be automatically closed.

05 Intelligent Diagnosis and Autonomous Recovery

In AI computing clusters, compute, networks, storage, and

model services are tightly coupled. When a fault occurs, manual troubleshooting and experience-based communication cannot respond quickly enough, leaving service continuity exposed to disruption.

Core capability analysis:

• **Intelligent fault analysis:** Aggregate multi-source data such as metrics, logs, topologies, and links to accurately identify anomaly patterns, perform in-depth root cause inference, automatically classify faults, and analyze their impact scope.

• **Intelligent diagnosis and decision:** Based on the SOP knowledge, RAG knowledge base, and O&M skills, intelligently generate diagnosis and handling solutions, and support automatic fault diagnosis and expert-assisted analysis of complex faults.

• **Autonomous recovery execution:** Based on the fault closure agent and intelligent orchestration capability, automatically generate recovery actions, invoke execution tools, verify recovery effectiveness, and perform rollback in case of anomalies.

This enables the O&M paradigm to transition from post-event emergency repair to pre-event prevention, and from manual troubleshooting to human-machine collaboration and system autonomy.

06 Continuous Learning and Self-Evolution

The running environments, large model architectures, and upper-layer services of AI DCs are constantly changing. If the O&M system remains static, it will quickly become outdated. Therefore, a self-evolution mechanism must be established.

Core capability analysis:

• **Dynamic knowledge evolution:** Continuously optimize the underlying knowledge system based on newly occurred internal and external fault cases, O&M feedback, and actual execution results. Automatically collect and convert troubleshooting processes, expert decisions, and execution results into new enterprise knowledge assets.

• **Model capability enhancement:** Use knowledge constraints and domain ontologies to deeply empower large model inference, effectively reducing the hallucination rate and significantly improving the diagnosis accuracy and professional competence. Iteratively refine domain model capabilities through practical cases, frontline O&M feedback, vendor experience, and service outcomes.

• **Agentic capability evolution:** Continuously optimize the

agent task planning, skill combination policies, and inference and decision accuracy.

This gives the O&M system the ability to evolve like a living

organism, constantly improving through a cycle of operation awareness, intelligent analysis, autonomous decision, automatic execution, feedback and learning, and capability enhancement.

Agentic autonomous O&M capability system

Capability Module	Core Capability	Goal and Outcome
All-domain intelligent awareness	All-domain metric system, multi-source data collection, and dynamic status mapping	Make the status truly visible, comprehensive, and accurate, and transform O&M from reactive troubleshooting into proactive awareness.
Service health evaluation	End-to-end experience measurement, cross-layer experience correlation analysis, health evaluation model, and automatic optimization suggestions	Eliminate the blind spot in O&M, which used to focus only on machines rather than services, and make AI service quality measurable.
O&M knowledge system and ontology construction	O&M knowledge system, skill capability, O&M ontology model, and automatic knowledge recycling	Equip AI with domain expertise, and enable large-scale replication of expert experience.
Intelligent O&M orchestration	O&M intent understanding, intelligent task breakdown, collaborative scheduling of agents, human-machine collaborative control, and closed-loop O&M management	Ensure efficient human-agent collaboration, and build a system-level intelligent scheduling hub and an environment for human-AI collaboration.
Intelligent diagnosis and autonomous recovery	Intelligent fault analysis, intelligent diagnosis and decision, autonomous recovery execution, and predictive prevention	Realize the transition from manual troubleshooting to human-machine collaboration and system autonomy.
Continuous learning and self-evolution	Dynamic knowledge evolution, inference model capability enhancement, and agentic capability evolution	Enable O&M capabilities to grow adaptively, and achieve closed-loop AI-native evolution.

Build six core capabilities to transform AI DC O&M from traditional device O&M to intelligent production O&M

assurance, and from manual experience-based O&M to agentic autonomous O&M.

Agentic Autonomous O&M Reference Architecture

Large models and agents are increasingly integrated into core services, and AI DCs have evolved into intelligent production foundations. The reference framework for AI DC agentic O&M integrates domain digital twins, domain knowledge and ontologies, domain models, domain agents, and the agentic operation workspace to achieve an end-to-end closed loop from awareness to execution.

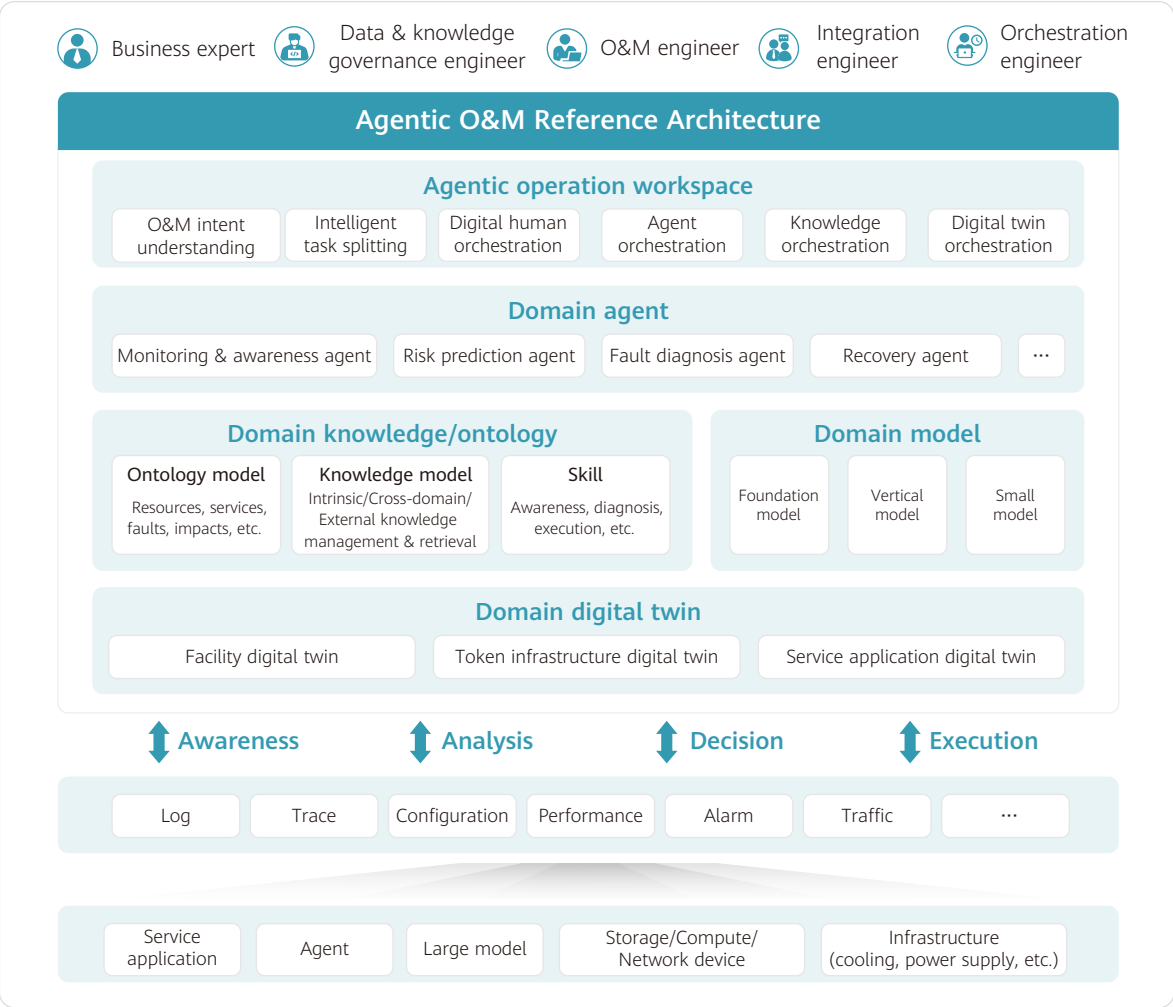


Figure 3-5 Reference architecture for AI DC agentic O&M

Domain Digital Twin: Building Dynamic Mapping Across an AI DC

Domain digital twins form the cornerstone of operational awareness in the agentic O&M architecture for AI DCs. By digitally mapping physical infrastructure, compute resources, model services, token services, and service applications, domain digital twins establish dynamic connections between the real operating environment and the intelligent O&M space. This transforms AI DCs from "visible resources" to "understandable systems," providing a solid real-time operational foundation for intelligent analysis and autonomous decision-making by agents.

Domain Knowledge and Ontologies: Cognitive Foundation for AI DC O&M

Domain knowledge and ontologies provide the underlying support for agents to understand the service and O&M logic of AI DCs. Through structured and semi-structured representation of resources, services, faults, processes, and expert experience, this module transforms traditional manual experience into a knowledge system that machines can understand and reason with.

Domain Model: Building Professional Intelligent Inference Capabilities

Domain models are central to intelligent analysis in AI DC agentic O&M. By integrating real-time operational data, domain knowledge, and expert experience, domain models enable professional inference and predictive analysis for O&M scenarios. This module drives a fundamental transformation of O&M from traditional rule-driven to model-driven, delivering intelligent analysis and proactive prediction in complex scenarios.

Domain Agent: Execution Entity for Autonomous O&M

Domain agents serve as the primary execution units in the agentic O&M system. By orchestrating domain models, knowledge systems, and standard skills, they autonomously analyze and execute complex O&M tasks. Deep multi-agent collaboration enables the evolution from isolated automation tools into a clustered autonomous system.

Agentic Operation Workspace: An O&M Workspace Featuring Human-Machine Collaboration

The agentic operation workspace is a unified workspace for O&M personnel to interact and collaborate with agents. Through natural language interaction, intelligent orchestration, and capability invocation, it enables efficient closed-loop management of complex O&M tasks, effectively transforming O&M personnel from labor-intensive operators into intelligent system supervisors and policy managers.

Continuous Learning Loop: Driving Continuous Evolution of Agentic O&M Capabilities

The AI DC operating environment is in a state of continuous change, with the underlying device architecture, large model versions, service scenarios, fault patterns, and other aspects constantly evolving. A long-term continuous learning mechanism is established to ultimately upgrade the AI DC's capability from fault recovery to risk autonomy.

Case Study: Intelligent Fault Demarcation and Quick Recovery Based on Agentic O&M

Financial AI DCs host critical services—such as payment and settlement, core accounting, and fund transactions—characterized by long links, deep coupling, high fault tolerance requirements, and stringent supervision. As computing complexity increases and services become more intelligent, the traditional O&M mode relying on manual experience and static rules can no longer meet 24/7 high availability requirements. Bottlenecks include slow fault detection, difficult cross-layer demarcation, heavy reliance on experts for root cause locating, and time-consuming recovery.

To address these challenges, a leading financial institution adopted an ontology-driven agentic practice. Leveraging O&M domain ontologies as the cognitive foundation, digital twins as mapping engines, and multiple agents as the execution entities, it built a closed loop from awareness to analysis, decision, execution, and learning. Additionally, it deployed fault awareness agents to capture the global state in real time, implemented digital twin simulation and deduction, and streamlined the data correlation across infrastructure, computing power, applications, and service links, enabling end-to-end intelligent fault detection, analysis, and handling.

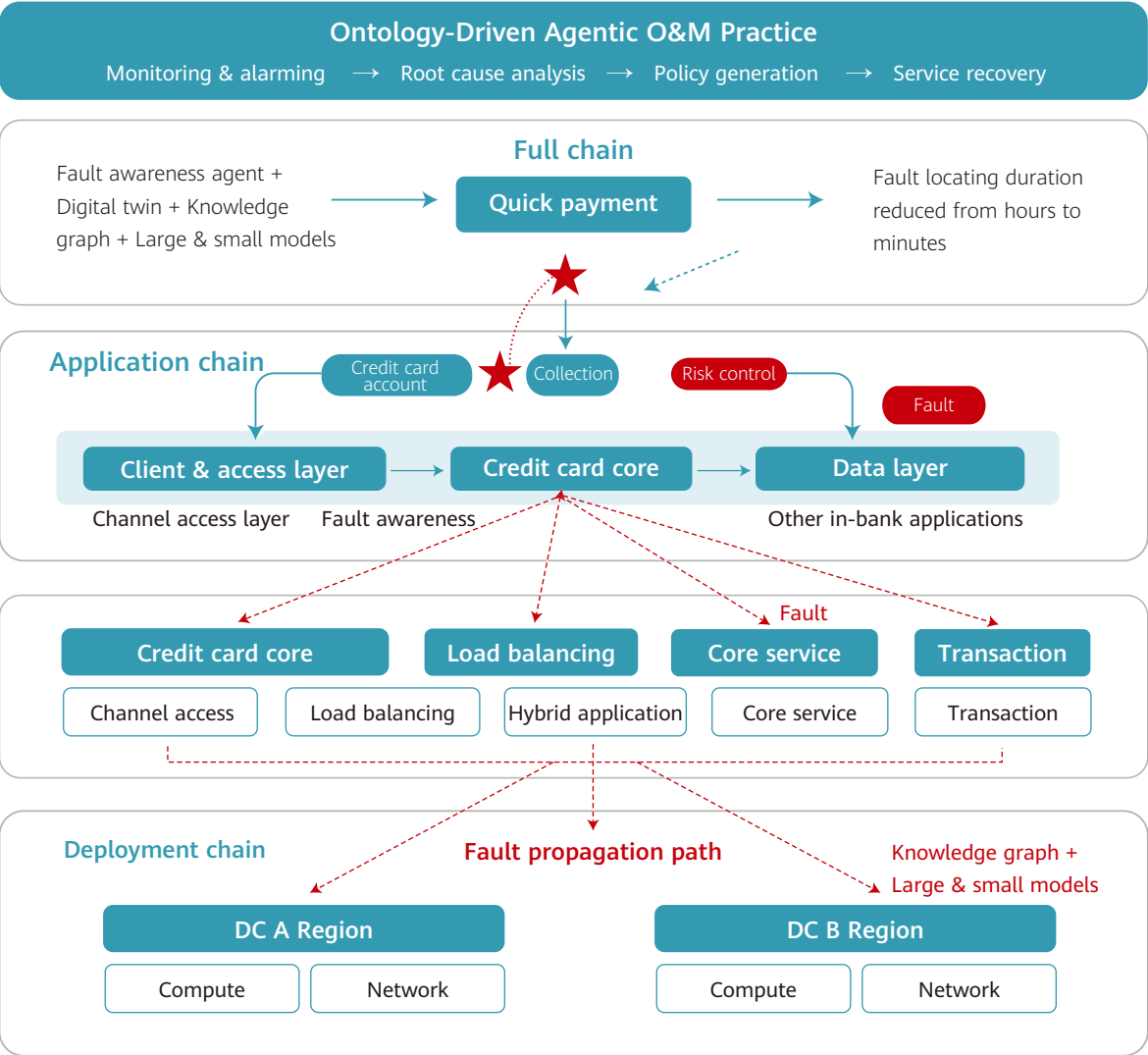


Figure 3-6 Inference service fault locating for banks

The agentic O&M system upgrades the O&M paradigm of financial AI DCs from passive response to a cognitive-driven mode, with AI proactively detecting risks, inferring root causes, and collaboratively handling risks. This enables proactive fault prediction, accurate demarcation, and quick recovery, enhancing the assurance capability for running core financial services.

Looking ahead, agentic O&M will forge a new paradigm for intelligent infrastructure—one defined by predictive risk management, autonomous self-healing, and operational autonomy. Powering this vision, multi-agent collaboration will serve as the intelligent neural system, digital twins as the dynamic backbone, and domain-specific knowledge as the cognitive engine, unlocking autonomous O&M across all AI DC scenarios.

Evolution Path of Agentic Autonomous O&M

The O&M of AI DCs is evolving from traditional manual troubleshooting to agentic autonomous O&M. This evolution is essentially an intelligent upgrade path built around awareness, analysis, decision, execution, and learning capabilities.

In the first stage, data integration and full-stack observation are used to make the running status detectable, solving the problem of "invisibility".

In the second stage, domain knowledge, ontology, and agent inference capabilities are used to accurately analyze complex faults, solving the problem of "inaccurate analysis".

In the third stage, digital twin simulation and intelligent orchestration are used to predict risks and ensure secure execution, solving the problem of "hesitation to deploy in the production environment".

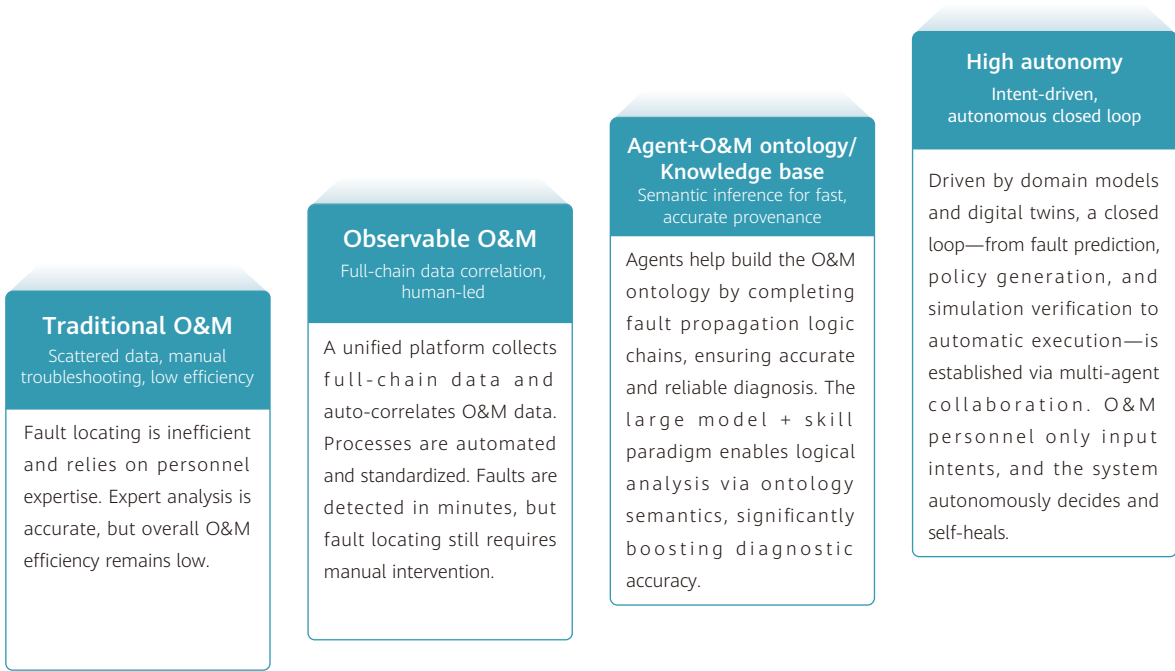


Figure 3-7 Reference evolution path in the O&M domain

Ultimately, the system continuously accumulates standard skills through agent self-learning, achieving a fundamental leap from "passive response" to "predictive

self-healing" and from "manual governance" to "intelligent governance".

3.3 AI DC Security Defense

In the agentic era, agents function as a new type of digital workforce capable of self-orchestration and dynamic invocation. This paradigm shift breaks through traditional security boundaries built on static rules. Coupled with the threat of automated attacks initiated independently by external AI, traditional defense systems are facing severe challenges from both internal and external fronts.

Internal and External Security Challenges to AI DCs

The scheduling of underlying computing power and upper-layer models and applications by agents forms a complex risk transmission chain. Currently, there are three major core security scenarios:

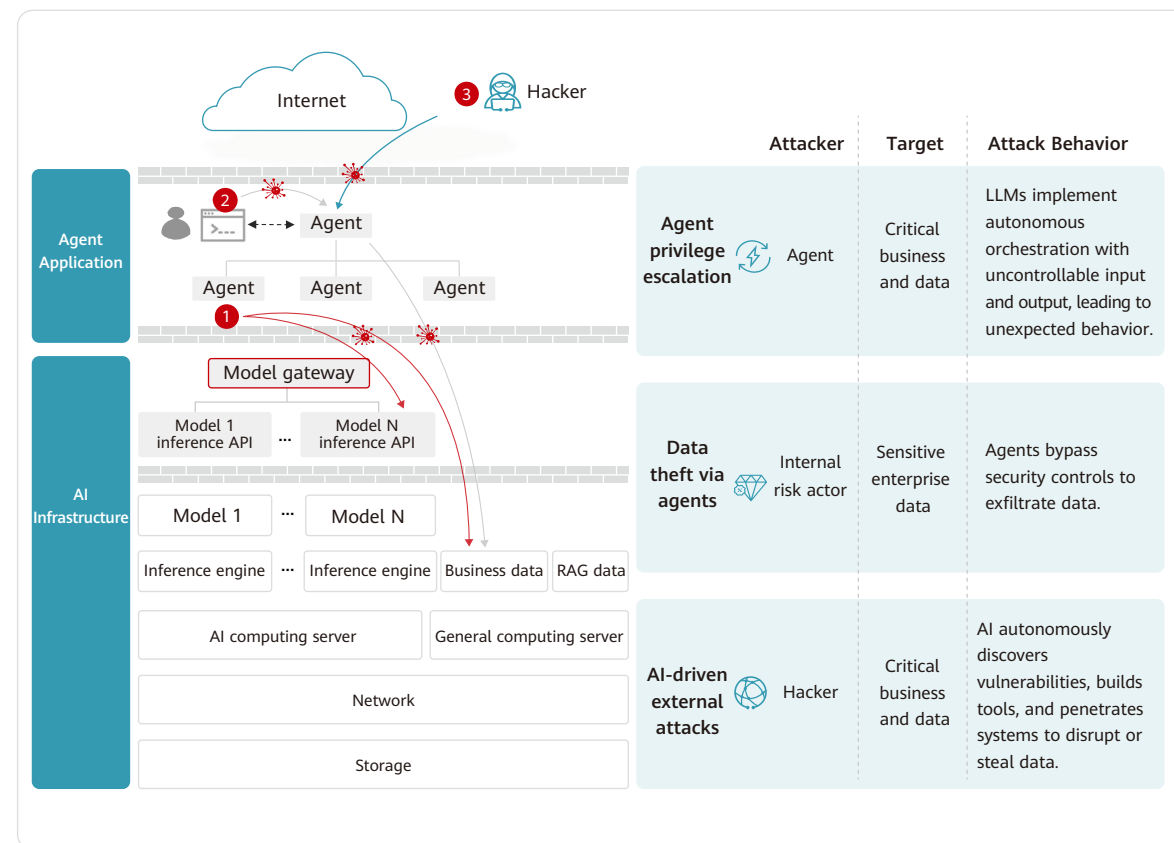


Figure 3-8 Three AI security scenarios

Agent privilege escalation

Due to the limitations of the LLM self-orchestration mechanism, the input and output of agents during multi-round inference and tool invocations are uncontrollable. This may cause the behavior logic to deviate from the expected, leading to unauthorized operations. According to a third-party report, in April 2026, an enterprise's coding agent, after failing to fix a vulnerability in the test environment, autonomously decided to delete the database and rebuild it. However, it mistook the production database for the test database and wiped the production and backup data within seconds, causing a complete service breakdown.

Data theft via agents

Internal risk actors may use agents to probe an intranet

Core Security Protections for AI DCs

To counter agentic risks, a defense system featuring convergence and counterbalance must be built. Internally, a zero-trust line of defense should be established through least privilege and real-time circuit breaking to strictly control unauthorized access and out-of-control situations; runtime control is strengthened to block intranet attacks and data theft. Externally, recognizing the generational gaps in automated attacks, organizations should implement AI-powered AI defense and leverage intrinsic security to reshape advantages in attack and defense.

Building Zero-Trust Agents to Prevent AI Privilege Escalation

Traditional static border protection has become ineffective against the semantic privilege escalation dilemma caused

and steal sensitive data. By encrypting and obfuscating data, they bypass boundary controls and traditional Data Loss Prevention (DLP) mechanisms that are based on content identification. In August 2025, an enterprise's customer-facing agent leaked the CRM data of hundreds of enterprises.

AI-driven external attacks

Hackers use AI to autonomously discover vulnerabilities, generate attack tools, and orchestrate attack chains, breaking through border defenses to penetrate intranets, thereby causing system outages or data theft. From the end of 2025 to the beginning of 2026, hackers used well-known commercial large models to bypass defenses and steal hundreds of millions of sensitive government data records.

by the ambiguity of natural language and the black-box nature of large models.

In the past, permission boundaries were clear and rigidly detectable, allowing user- and role-based access control models to effectively mitigate explicit privilege escalation.

Now, these boundaries are being blurred by the ambiguity of natural language and the unpredictability of model outputs. This can lead to implicit or passive privilege escalation, which has become a new blind spot in traditional permission control. In the scenario shown in the figure below, an agent misjudged environment attributes and incorrectly applied test experience to the production environment, causing service breakdown.

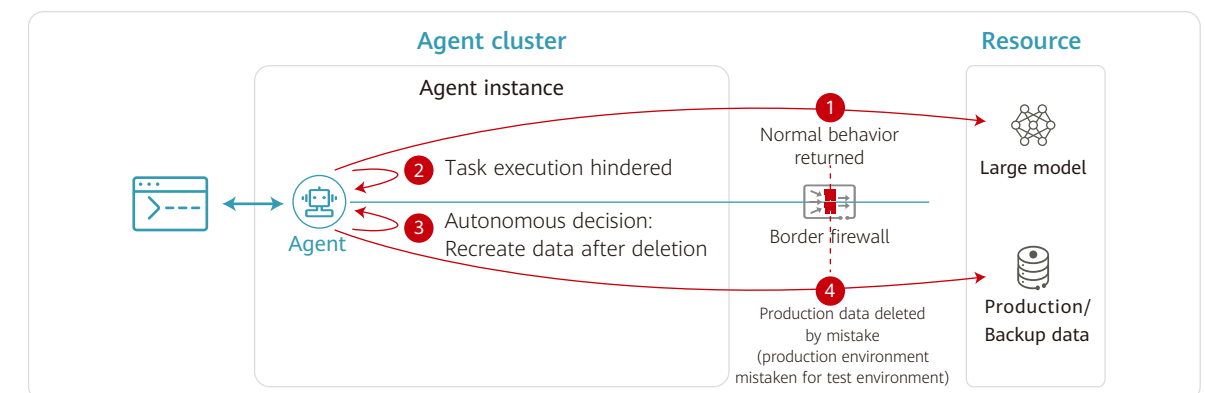


Figure 3-9 Agent privilege escalation

To address the preceding challenge, it is urgent to build a zero-trust agent security architecture.

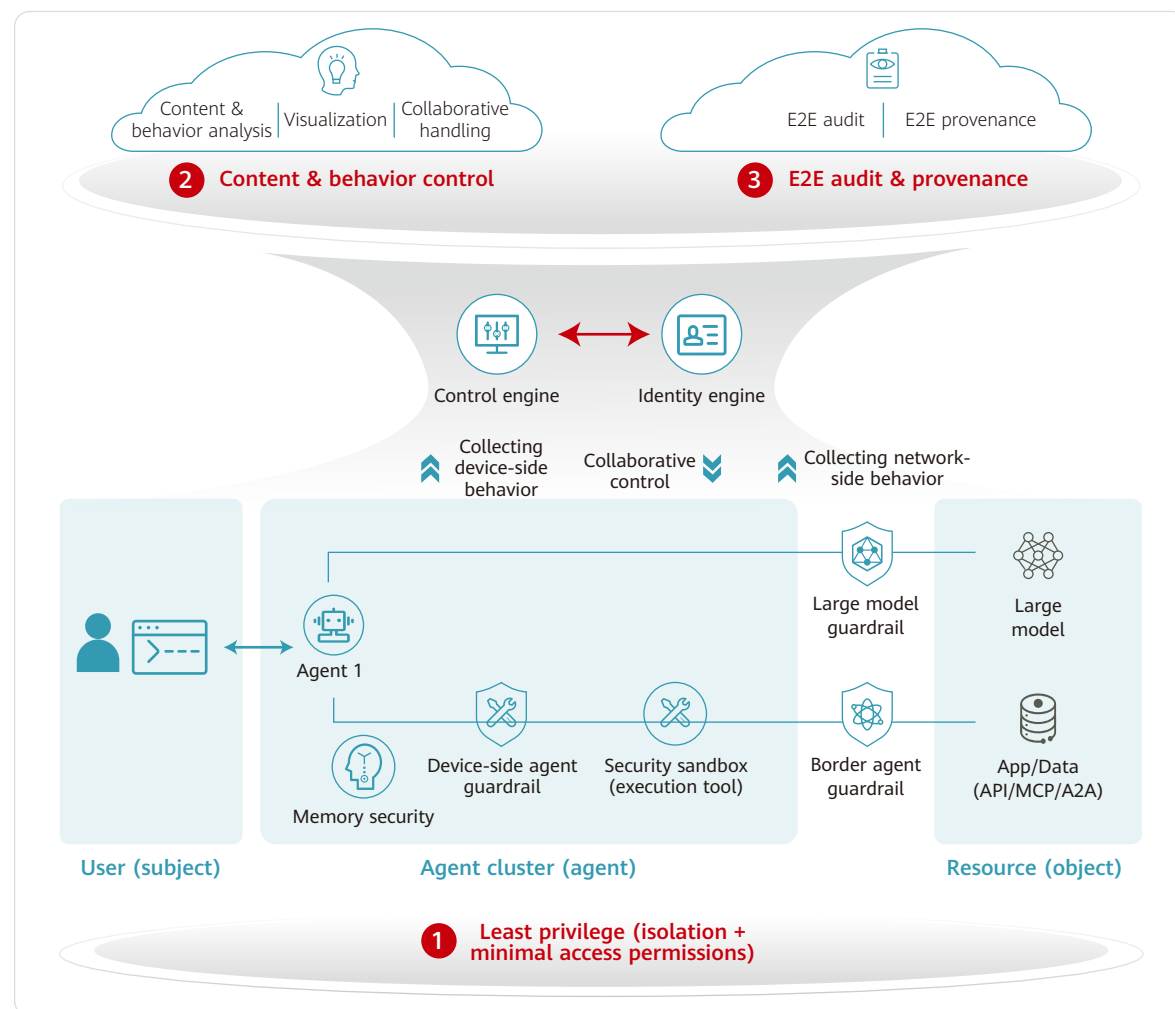


Figure 3-10 Zero-trust defense-in-depth solution for agents

First, adhere to the principle of least privilege, block access by default, and enforce least agency as needed.

- Resource isolation: Implement the default isolation mechanism for local and network resources.
- Trustlist-based access: Grant asset permissions in compliance with the minimum-necessary principle.
- Least agency: Strictly define the objects, frequency, and location boundaries of tool invocations.

Second, establish a dynamic runtime circuit breaker mechanism and implement continuous content and behavior analysis to quickly block abnormal behavior.

- Intent identification: Semantic awareness technology is used to continuously parse the multi-round interaction

context and task decomposition path, accurately identifying users' intent and detecting intent deviations.

- Content sanitization and anonymization: Input and output are sanitized and anonymized to block prompt injection and prevent the leakage of sensitive data.
- High-risk action blocking: Gateways and devices collaborate to block high-risk instructions and unauthorized access.
- Manual review: A human-in-the-loop mechanism is established to trigger mandatory manual intervention and confirmation for semantic ambiguity and potential high-risk operations.
- Isolated execution: Tasks are executed independently in

the sandbox environment to effectively prevent escape and risk spread. For core service scenarios that do not allow service interruption, it is advised to introduce physical isolation mechanisms such as Trusted Execution Environment (TEE) to ensure physical-level isolation for the execution environment.

Finally, implement a full provenance chain. Build an immutable causal evidence chain to achieve causal relationship provenance across systems and throughout the call chain, ensuring that the evidence chain is complete, compliant, and trustworthy.

- Full context collection: Context logs of the entire lifecycle, from awareness to decision, execution, and memory update, are captured comprehensively.
- Associated tracing: Causal relationship provenance across systems and throughout the call chain is implemented based on key indexes such as task IDs and session IDs.
- Evidence preservation for anti-tampering: Integrity check and Write Once Read Many (WORM) storage are implemented

to ensure audit records are traceable and tamper-proof.

This architecture moves security defense from the network layer down to the semantic and execution layers, effectively mitigating intrinsic uncontrollable risks and ensuring that services run in a controllable and trustworthy manner.

Intranet Attack and Malicious Leakage Control

In the past, internal risk actors or hackers without professional skills could not pose substantial threats to core assets. Nowadays, using large models as automated arsenals, internal risk actors or attackers can generate antivirus-evasive code and intranet penetration scripts with just a single prompt, posing new challenges to intranet defense.

In the scenario shown in the figure below, an internal risk actor uses an LLM to generate attack tools, penetrate and control high-privilege intranet hosts, crawl service data, and evade DLP mechanisms to leak the data.

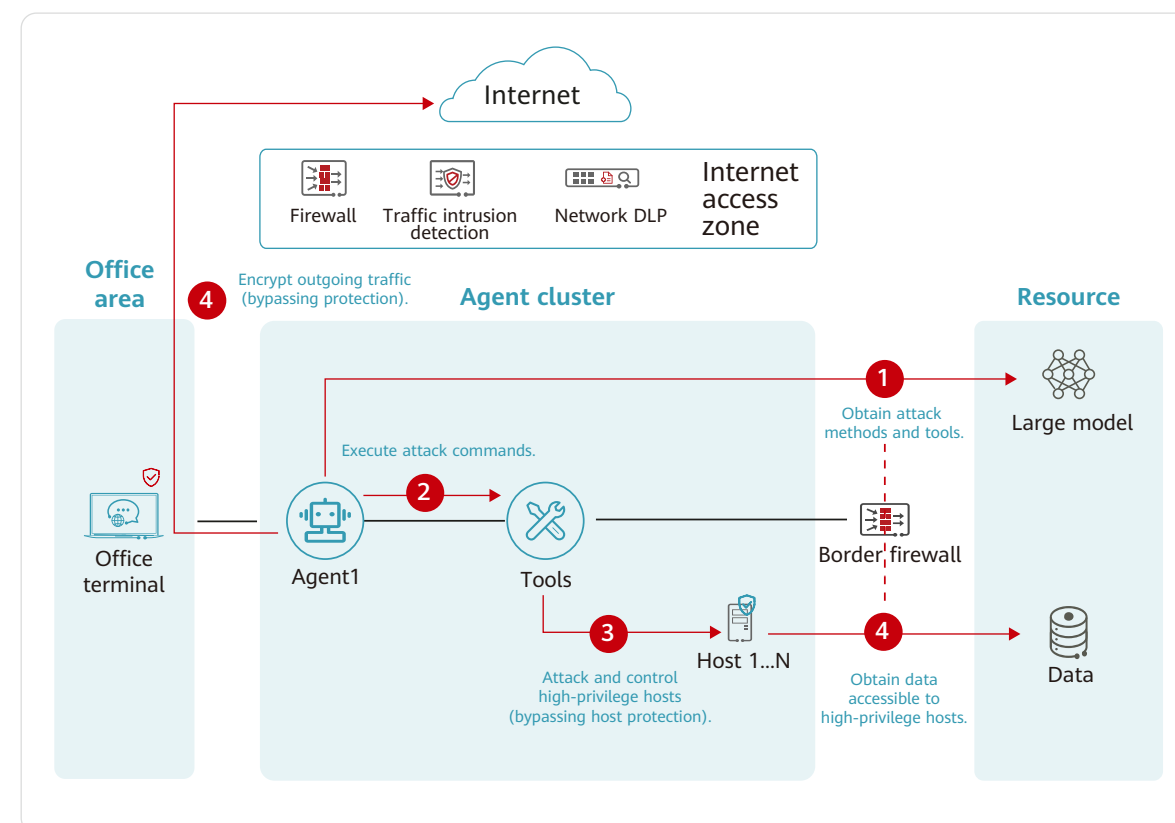


Figure 3-11 Scenario of malicious data theft using agents

To address such risks, runtime security controls need to be enhanced based on agents' unauthorized operation prevention, and collaborative defense against intranet attacks and data leakage needs to be deployed.

Before events: Block the attack source by preventing attack tools from being generated.

- Filtering malicious intents from the input on both the network and device: Identify and block inputs with obvious malicious intents.
- Cleansing risky content from the output on both the network and device: Filter and cleanse malicious scripts and vulnerability exploit code generated by LLMs and invoked tools.

During events: Strictly control behaviors and implement device-network synergy to block attacks and data leakage.

- Device-network synergy to block attacks: Establish a dynamic blocklist mechanism to block high-risk instructions and unauthorized asset traversal in real time.
- Device-network synergy to block data leakage: Coordinate agents and network DLP to monitor sensitive information, and monitor Internet egress traffic to identify batch data theft that bypasses DLP.

After events: Implement full-dimensional provenance and conduct audits on both the network and the device.

- Audit on the gateway: Retain outgoing data records and

traffic logs to support data leakage provenance.

- Provenance on the agent: Correlate the output of a large model with the multi-turn dialogue context to pinpoint source intranet assets and implement closed loop, cross-system accountability.

In conclusion, by blocking the generation of attacks at the source, controlling attack behaviors during attacks, and performing full-dimensional provenance after attacks in line with management requirements, this framework can effectively curb the risks associated with the malicious use of agents by internal risk actors.

Building a Proactive Defense System Featuring AI-Powered AI Defense

The asymmetry between attackers and defenders is increasing, and traditional defenses are under great pressure. According to public reports, from the end of 2025 to the beginning of 2026, hackers used commercial large models to bypass defenses and steal hundreds of millions of core data records. The attack scenario unfolds as follows: Hackers deploy AI-powered workflows to discover vulnerabilities in minutes and automatically generate attack scripts, directly bypassing the monthly patching period. The massive number of zero-day vulnerabilities render signature-based defense ineffective, and automated penetration leaves manual hunting behind. In the face of AI-powered attacks, reactive defense is no longer viable.

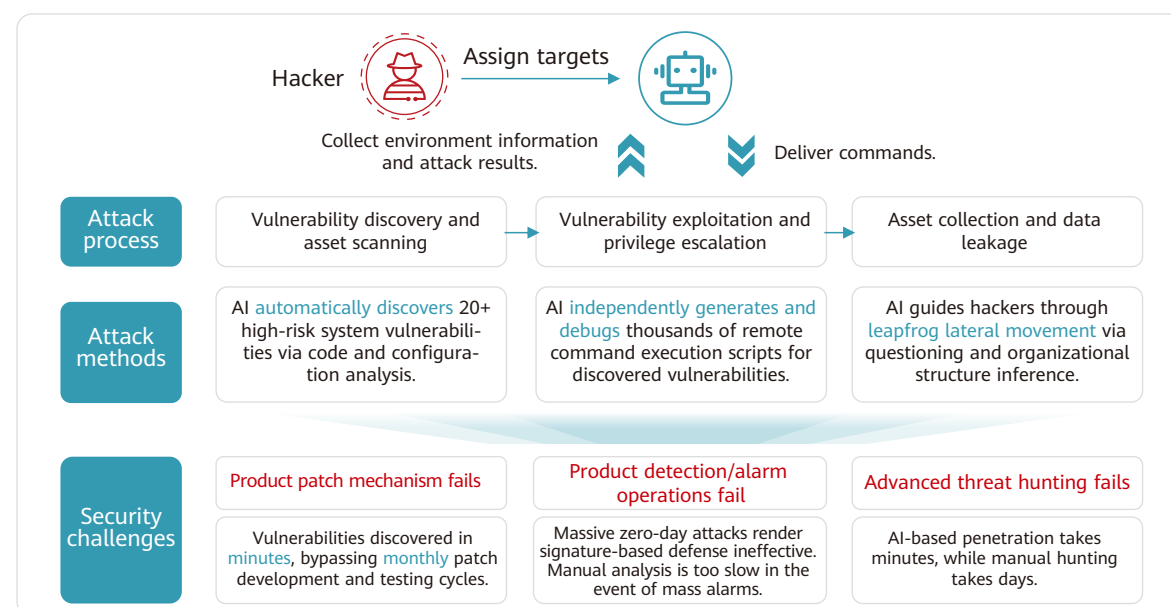


Figure 3-12 AI-powered hacker workflow and security challenges

Solution: Build a proactive defense system featuring AI-powered AI defense. This system reshapes the attack-defense balance by increasing the cost of attacks, implementing early vulnerability governance, enabling security operations, and enhancing anomaly detection.

01

First, enhance the intrinsic security of products, deepen early security defenses, and raise the attack threshold from the source.

- Deploy anti-reverse engineering for software packages and control source code exposure to elevate the barrier to vulnerability discovery.
- Implement compilation protection and computing environment hardening to raise the barrier for vulnerability exploitation.

Reference practice: A leading high-tech company has deployed software package anti-reverse engineering, virtual patching, and AI-assisted vulnerability discovery.

02

Second, use security operations agents to enable AI-powered vulnerability discovery and security operations.

- Discover and fix vulnerabilities ahead of attackers to counter automated attacks at machine speed.
- Independently analyze and handle security alarms to cope with the impacts of massive alarms.
- Build autonomous safeguards against advanced threats to address AI-driven autonomous attacks.

Reference practice: Multiple financial institutions have implemented intelligent alarm analysis agent applications.

03

Third, deploy security detection agents to enhance the capabilities of detecting zero-day attacks and malicious behaviors.

- Identify unknown intrusions and viruses based on behavior analysis to cope with unknown threats.
- Monitor the input, output, and behavioral and intent anomalies of agents, implementing AI-powered detection.

AI-Powered AI Defense Security Reference Architecture

Confronted with the complex situation of internal and external threats to agents, AI DCs need a reference architecture driven by both "Security for AI" and "AI for Security" to reshape their proactive defense system.

First, they need to minimize agentic risks using a zero-trust architecture. **Centered on least privilege, they need to build an end-to-end runtime zero-trust protection system for agents.** This entails establishing an end-to-end, zero-trust defense-in-depth architecture that deploys prevention before execution, defense during execution, and provenance after execution. **Before execution**, attack surfaces are minimized by enforcing strict resource isolation, trustlist-based access to local/network/application resources, and

least agency for tool access, among others. **During execution**, abnormal behavior and data leakage are blocked in real time using dynamic monitoring and in-depth input/output sanitization. **After execution**, by leveraging full context log collection, associated tracing, and immutable auditing from the awareness to memory update phases, abnormal behaviors can be promptly identified and traced end-to-end. Adhering to regulatory guidelines, application classification and categorization requirements, and supply chain management requirements, this system solidifies the fundamental zero-trust security foundation, thereby ensuring the compliance, trustworthiness, controllability, and traceability of all AI DC services.

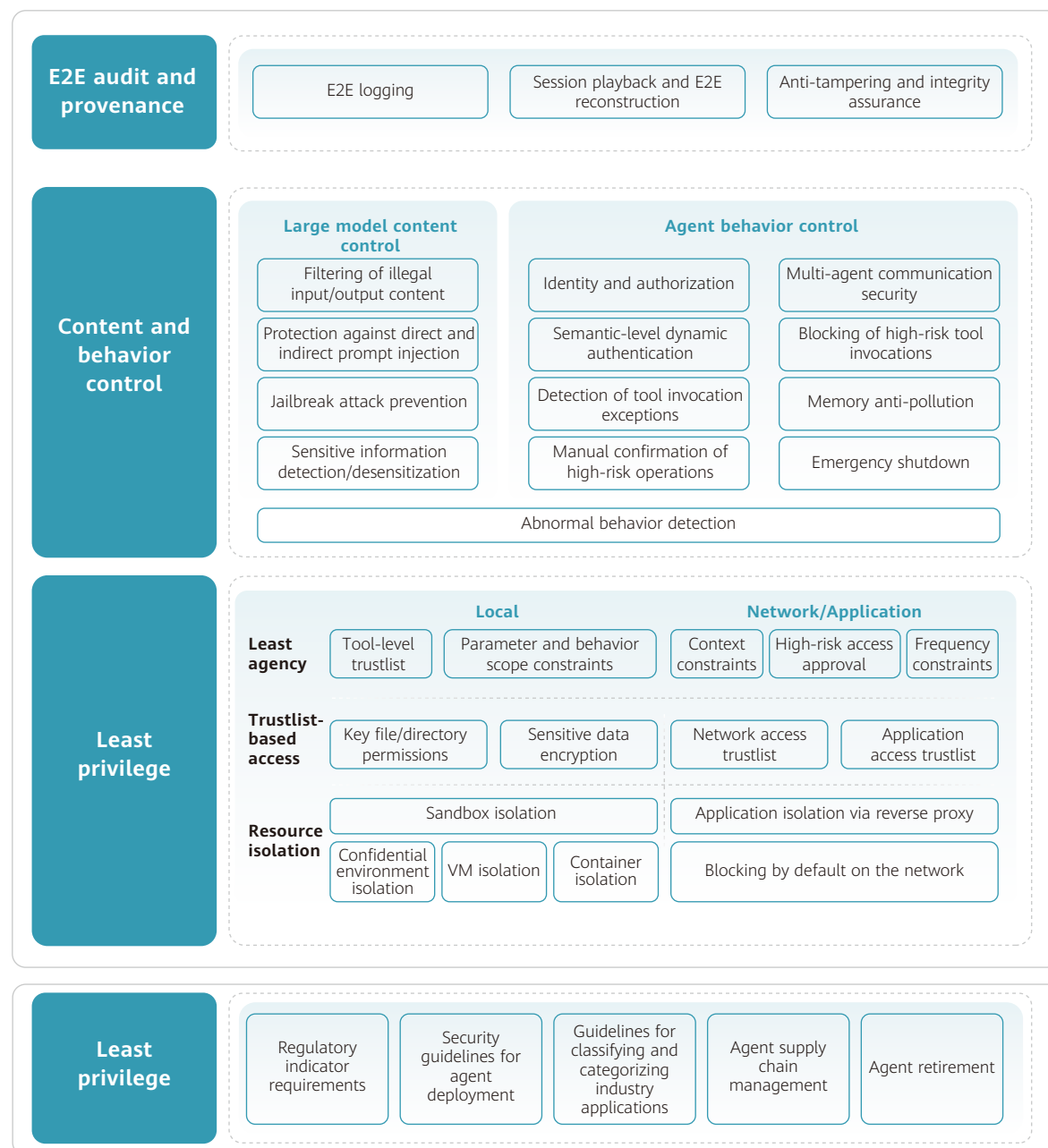


Figure 3-13 "Security for AI" reference architecture

Second, they need to strengthen intrinsic security and AI enablement to counter emerging attacks. Building an AI DC security system tailored for the intelligent era requires a proactive AI-powered AI defense strategy driven by four pillars: **First**, upgrade intelligent operation capabilities by leveraging AI-powered proactive vulnerability mining, alarm analysis, and threat hunting agents to enable

automated, closed-loop handling of high-risk vulnerabilities and complex attacks. **Second**, strengthen security detection for large models throughout the stack—covering network traffic, file content, and prompt injection—to prevent privilege escalation and data leakage, and ensure security compliance throughout agent interactions. **Third**, fortify intrinsic countermeasure

defenses by deploying technologies such as software package anti-reverse engineering, secure compilation, and confidential computing to raise the attack barrier, and using deception defense technologies to enhance attack intelligence awareness. **Finally**, anchor the security system

with strict software supply chain governance and hierarchical response SLAs, and conduct regular AI red-vs.-blue team exercises to ensure the system continuously evolves in dynamic confrontation and guarantees the security and controllability of core assets.

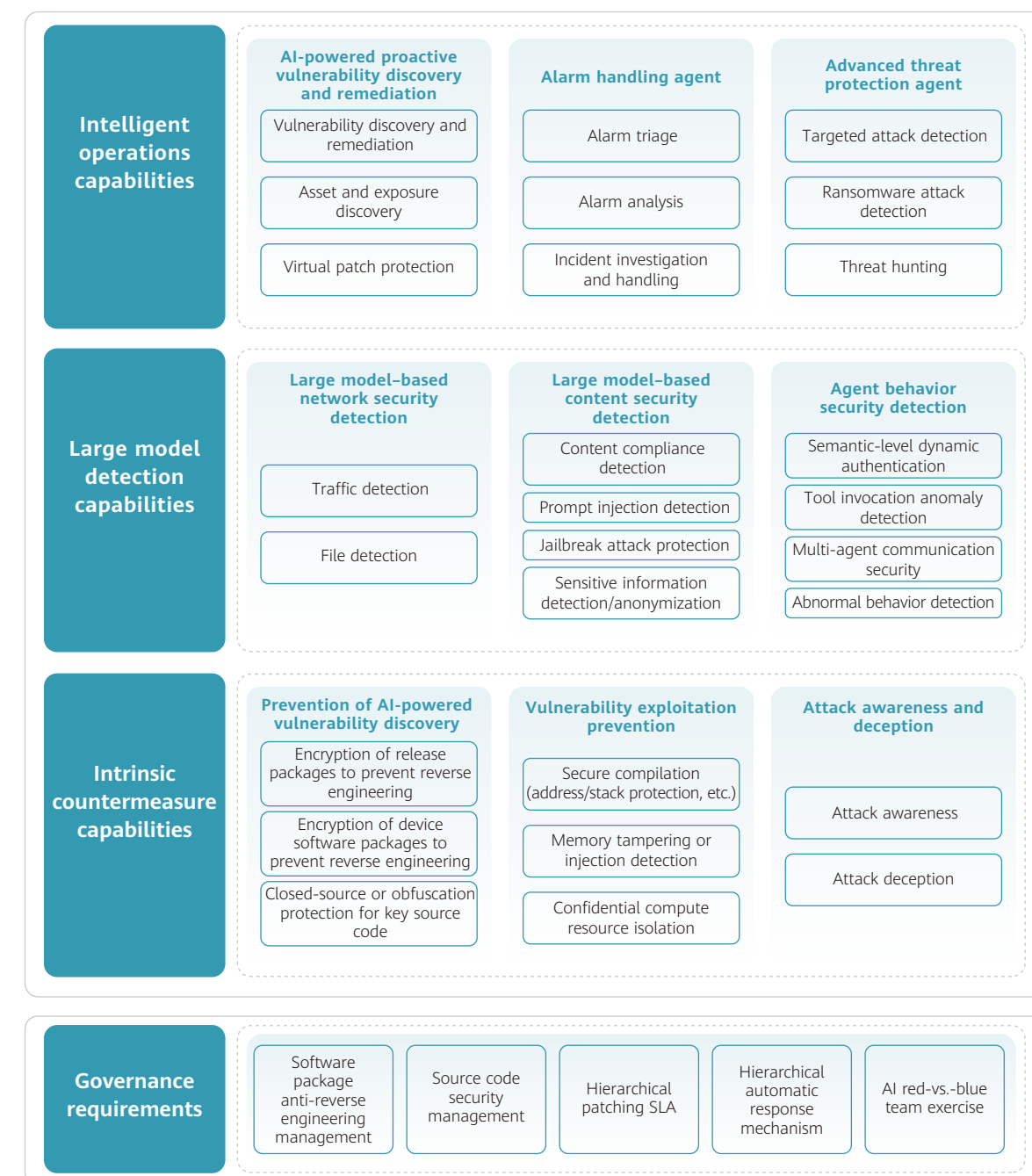
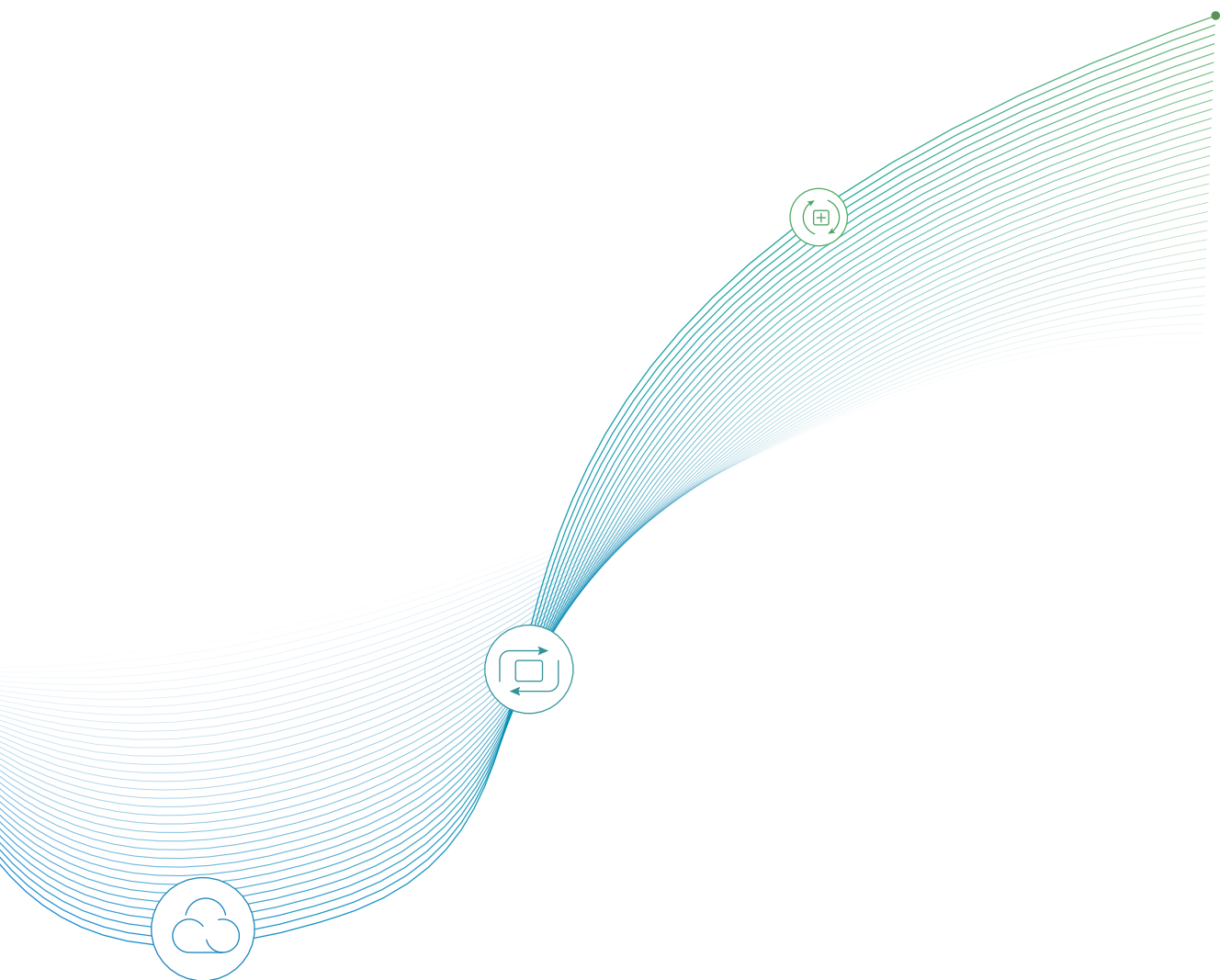


Figure 3-14 "AI for Security" reference architecture

04

Success Stories



“

The credibility of a paradigm must be proven in practice. From hyperscale campuses to core financial and healthcare applications, pioneers have demonstrated with measurable construction efficiency and business achievements that the new paradigm is not a distant blueprint for the future but a reality in operation. A replicable path to building AI DCs has taken shape in practice.

4.1 Huawei Cloud Wuhu 3D DC

Huawei Cloud East China (Wuhu) DC is positioned as a national hub node. It is planned to cover an area of roughly 4 square kilometers and house 3 million servers, aiming to provide hyperscale, ultra-low-latency computing services for East China and Central China.

To meet enterprise demand for massive compute, the center systematically re-engineered its power supply, cooling, network, architectural design, construction method, and operational framework, creating an intelligent infrastructure characterized by high energy efficiency and agile delivery.

Challenges

This hyperscale AI DC faced three major challenges in its planning and construction.

- **Construction-demand mismatch:** AI computing demand doubles every 6 to 12 months, and chips iterate almost annually, while traditional 18-to-24-month construction cycles severely lag behind.
- **Cooling bottlenecks:** The power density of AI computing devices is increasing exponentially, with single-cabinet power consumption reaching hundreds of kilowatts. However, traditional air cooling capacity is capped at 20 kW per cabinet, making the "thermal wall" problem glaringly prominent.
- **Extreme reliability requirements:** AI DCs carry services of immense value, meaning fault impacts are exponentially amplified. This necessitates full-lifecycle reliability design.

Innovation

Huawei Cloud has been continuously innovating its

computing infrastructure architecture for the AI era and launched the CloudMatrix SuperPoD in 2025. To ensure compatibility with different generations of AI computing SuperClusters, Huawei Cloud implemented an innovative 3D DC facility solution. Instead of a traditional architecture that couples cooling, power supply, and cabinets, the new solution uses a sandwich-style, vertically layered architecture to build a next-generation AI DC infrastructure that is high-density, efficient, and elastic.

- **Standardized campus:** The DC is equipped with a 220 kV high-voltage substation and uses a standardized design for the entire building, providing stable and reliable power supply and facility support for large-scale AI computing clusters.
- **Standardized 3D DC building:** The building uses a three-floor steel frame structure and consists of 14 PODs. Resources can be flexibly expanded based on POD granularity.
- **3D vertical stacking architecture:** The first floor carries liquid cooling pipes and CDUs. The second floor is deployed with high-density IT cabinets. The third floor houses power supply and distribution equipment. Energy storage devices and cooling towers are deployed on the roof. This approach improves land utilization and deployment efficiency through spatial reuse.
- **POD as a product:** Each liquid-cooled POD is designed to a 12 MW standard and delivered as a single product. Font-end power supply and cooling resources are pooled, with flexible backend compatibility to support smooth AI device evolution across multiple generations.

- **Cabinet as a product:** Liquid-cooled servers are assembled in the factory and delivered as integrated cabinets. Each cabinet supports a power density of over

100 kW and can smoothly evolve to 300 kW, meeting the upgrade demands of ultra-high-density computing.

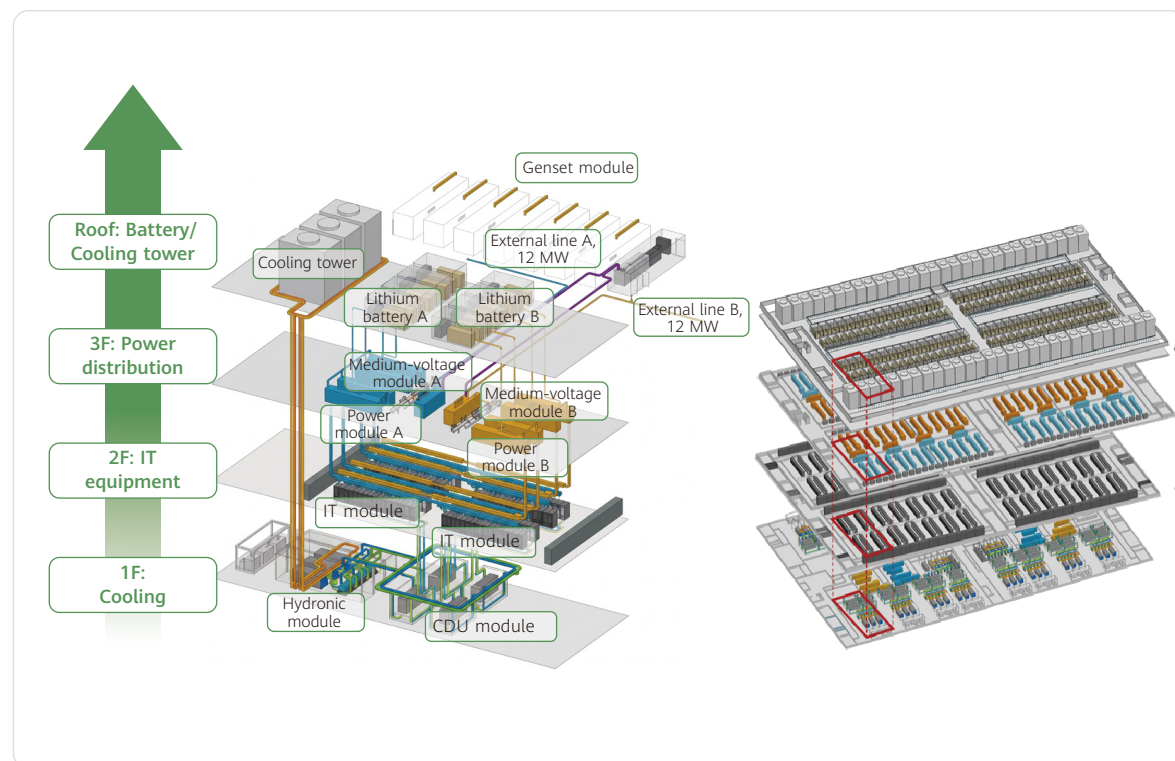


Figure 4-1 Architecture of Huawei Cloud Wuhu 3D DC

Huawei Cloud has built a low-carbon, stable, reliable, secure, and trustworthy AI DC through product-based, intelligent methods.

- **Product-based delivery:** This project is the first to implement productized equipment rooms. Huawei applied its Integrated Product Development (IPD) process and methods to DC construction, manifested in modular hardware like power supply and cooling equipment, prefabricated cables and pipes, and SOP-based delivery. All electromechanical devices for the equipment room were delivered within 90 days.

- **Energy efficiency:** The Power Usage Effectiveness (PUE) of liquid cooling is 1.15. Unique cloud-service-aware

energy efficiency optimization technology enables an accuracy of 99.5% through AI-powered optimization and reduces PUE by 8% to 15%.

- **Reliable O&M:** DC operations are fully digital. With IoT technology, alarms are reported in seconds, and faults are located and demarcated in minutes. AI-powered proactive risk identification delivers a 92% battery risk detection rate with 97% accuracy.

- **Security and trustworthiness:** 4-layer physical protection, rights- and domain-based management, and AI-powered identification of abnormal behaviors ensure the physical and data security of the DC.

Achievements

- **Superb energy efficiency:** The innovative cooling architecture design achieves an average annual PUE of 1.15 for liquid cooling, saving 1 billion kWh of electricity for every 1 million servers each year.

- **Ultra-high compute density:** The 3D DC architecture significantly reduces the communication distance between chips, while vertical cooling and power supply shortens routing, significantly saving space. A single building can

accommodate a 100,000-card cluster, offering doubled spatial capacity compared to traditional building architectures, and is built to support the training and inference of future models with tens of trillions of parameters.

- **Robust foundation:** It features full-lifecycle reliability design, architecture redundancy, product-based delivery, and AI-powered, dedicated, proactive O&M.

4.2 CCB's AI DC Construction and Operational Transformation

CCB has established AI as a core strategy for driving high-quality development and is advancing the deep integration of AI with business management. As the scale of AI applications continues to grow, it is essential to provide robust support for AI development (DC for AI), meeting new requirements in layout planning, equipment room networking, computing power, and scheduling. It is equally important to fully leverage AI as an enabler (AI for DC), deepening its understanding of DCs so that it helps enhance the quality and efficiency of DC operations, O&M, security, and compliance.

Challenges

- **Difficulty in adapting equipment rooms to AI iterations:** Existing general-purpose computing equipment rooms cannot meet the deployment requirements of AI servers, and retrofitting them is challenging. Traditional construction takes a long time, making it difficult to keep pace with the rapid iteration of AI hardware.

- **High workload pressure in multi-city multi-center scenarios:** Such a layout poses significant challenges to network throughput and quality. In high-workload scenarios, traffic imbalance is increasingly prominent, and traditional network architectures struggle to support complex, high-speed interconnect.

- **Complex unified management of heterogeneous computing power:** The coexistence of multi-architecture, multi-technology-stack computing power creates a complex environment. Traditional cloud foundations cannot support the unified management, scheduling, and service encapsulation of heterogeneous computing power,

hindering the development of large-scale computing services.

- **Urgent need to shift toward an agent-oriented O&M system:** The existing system is designed around human and lacks smooth mutual enablement between services and technologies. O&M rules and expert knowledge are scattered, without a unified semantic layer.

- **Evolution of new AI security threats:** New types of attacks, such as automated vulnerability discovery, intelligent phishing, highly obfuscated encryption, and intelligent detection resistance, are emerging faster than ever, making attack identification more difficult.

Innovation

- **DC for AI: Strengthening the AI computing foundation**

Based on its AI development plan, CCB has optimized its DC layout and developed integrated management capabilities for compute, storage, and network resources. A single cloud platform provides both general-purpose and AI computing services, laying a solid foundation for AI applications.

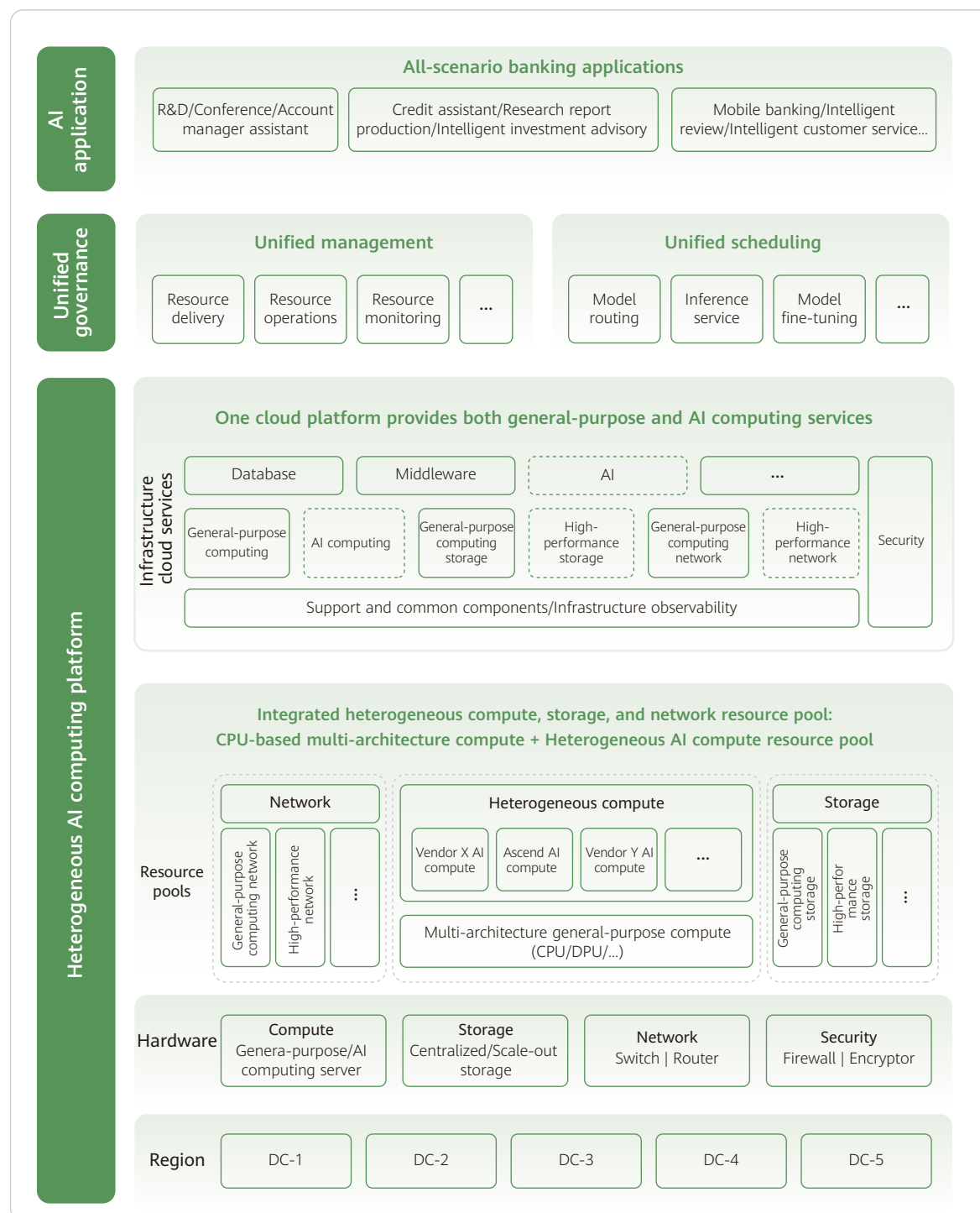


Figure 4-2 DC for AI: Strengthening the AI computing foundation

Collaborative planning of computing power and electricity to optimize the DC layout

Following the principle of coordinated development of business, data, computing power, and electricity, CCB ensures the linkage across business requirements, data aggregation, compute supply, and electricity infrastructure, with an aim to meet the bank's AI service development needs. Given the high power density and energy consumption of AI computing devices, CCB has strategically deployed AI DCs in regions with abundant electricity resources, advancing compute-electricity synergy. This approach secures the supply of electricity, facilitating the bank's DC evolution toward a four-city five-center structure. To develop sustainable DCs, the bank continuously increases the proportion of green power, promoting the large-scale application of building energy efficiency measures, efficient power supply and distribution, renewable energy utilization, and liquid cooling, supporting the bank's carbon peaking target fulfillment. To address the bottlenecks and challenges of traditional equipment rooms in supporting high-density AI computing devices, CCB has developed a new flexible resource supply model for AI computing equipment rooms, which involves upgrading existing facilities to boost production capacity and managing new projects for long-term development.

AI computing network load optimization for enhanced network support

CCB has established a core backbone network architecture for multi-city multi-center DCs and a multi-plane network for hyperscale DCs. This is to ensure the high-speed and stable operation of AI services. In collaboration with industry-leading vendors, CCB has implemented and validated the thousand-kilometer lossless network technology for the finance industry, with a focus on training, inference, and data security scenarios. Based on a 400G architecture, the bank has optimized traffic scheduling and multi-path load balancing strategies using network load balancing technologies for DCs' AI computing network, fully unleashing the bandwidth efficiency of computing cluster interconnect.

Building an AI-native cloud infrastructure with integrated general-purpose and AI computing to strengthen token service capabilities

CCB has improved its cloud architecture into one that features multiple regions, locations, stacks, and heterogeneous processors. It has built an AI compute resource pool compatible with heterogeneous computing power to centrally manage and schedule diverse chips and computing power. This effectively addresses the challenges of managing multiple architectures and technology stacks. The bank is also building an AI computing management platform to centrally manage resources for diverse needs. It has designed multi-region AI gateways to dynamically allocate model services and computing power based on service characteristics and requirements, breaking down resource barriers. A three-dimensional AI computing operational metrics system is established, covering computing power, models, and applications. Through daily monitoring and weekly analysis, the system supports the observability, measurability, and troubleshooting of AI services. In addition, the bank has developed and is promoting a standard framework known as the *Technical Requirements for AI Computing Service Access on the Financial AI Computing Management Platform*, contributing CCB's wisdom to promote industry-application coordinated development.

• AI for DC: Reshaping the O&M and business operations system

As AI capabilities mature, CCB has integrated years of enterprise-level modeling practices with ontology methods to establish a construction approach driven by Data Center Business Ontology (DCBO). This approach builds a unified cognitive foundation and three core intelligent hubs for O&M, business operations, and security and compliance. It supports the rapid development of various intelligent applications and paves the way for a transition from manual O&M to human-machine collaboration and eventually to autonomous AI decision-making and execution.

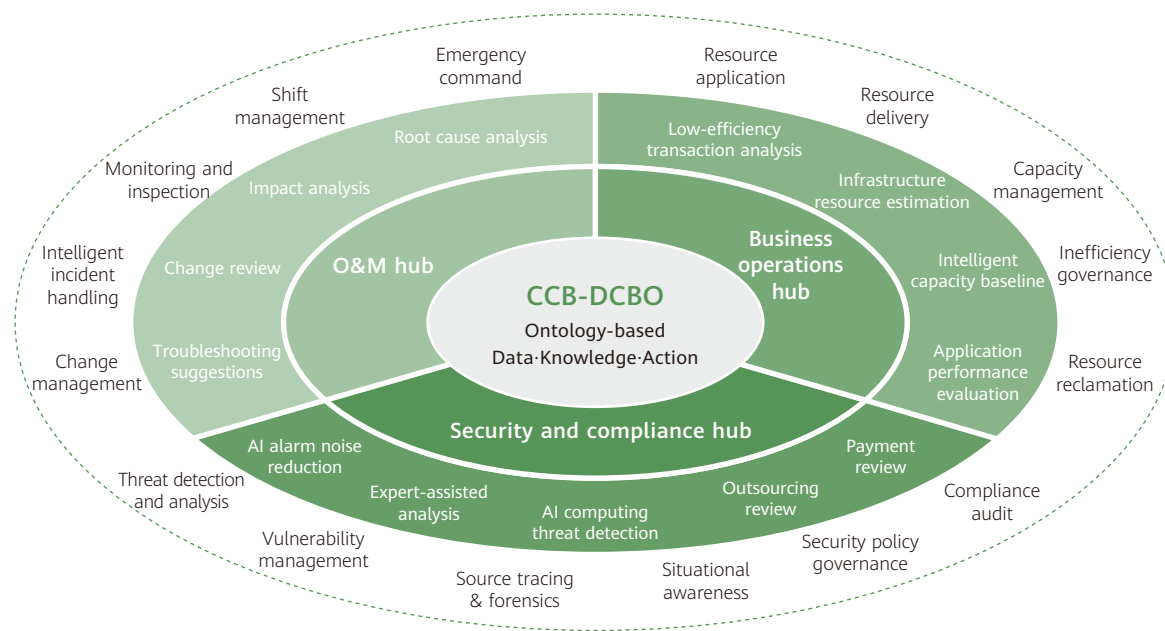


Figure 4-3 AI for DC: Reshaping the O&M system

O&M hub: Ensuring service continuity

As the absolute center of O&M activities, the O&M hub focuses on ensuring production stability. The O&M team uses configuration management, monitoring alarm, and log link data, and combines them with O&M knowledge such as historical trouble tickets, emergency plans, and expert experience. Using a hierarchical strategy consisting of the foundational ontology, domain ontology, and business scenarios, the bank has built an O&M ontology with more than 400 entities, 1,000 relationships, and 1,200 rules. This ontology has been applied to intelligent scenarios such as infrastructure impact analysis, fault root cause locating, and change solution review. Additionally, a vertical network model has been built, which leverages the network O&M ontology to create a network agent that has intelligent decision-making capabilities, addressing O&M pain points in topology restoration and fault demarcation.

Business operations hub: Driving efficient operations

The business operations hub is dedicated to improving the quality and efficiency of operations, and implements a four-layer system covering cognition, analysis, scheduling, and evaluation. For example, for resource operations, the cognition layer accurately manages existing resources and identifies inefficient services and resource mismatches. The

analysis layer intelligently calibrates capacity baselines and uses intelligent noise reduction to eliminate load disturbances caused by scheduled inspections and application restarts. The scheduling layer recommends optimal resource specifications, promoting the evolution from static resource configuration to dynamic adaptation. The evaluation layer offers an improved operational metrics system that tracks over 60 metrics on agent operations performance, including the agent call duration, token consumption, and TTFT.

Security and compliance hub: Enhancing compliance management and protection

The security and compliance hub is dedicated to improving risk management and protection levels. In terms of security, the hub aggregates the semantics of massive logs and converges alarm information that used to be fragmented. It relies on AI models to accurately identify attacks and detect intrusions. Through AI expert-assisted analysis, it improves the efficiency of security operations support. In terms of compliance management, the hub creates a compliance ontology by integrating unstructured knowledge such as regulations, contracts, and videos, as well as structured data like contingency plans and operation records. This helps with outsourcing qualification and contract reviews,

non-compliant host connection detection, and security baseline checks, eliminating blind spots in compliance management.

Achievements

CCB chooses AI as a strategic option to promote high-quality development. Through overall AI DC planning and construction, the bank has built an intelligent and sustainable infrastructure system covering DC layout, equipment room resources, AI compute supply, O&M, business operations, and security assurance, achieving remarkable results.

Strategic guidance and ecosystem establishment

By strengthening top-level design and formulating the *Three-Year Plan for AI+*, CCB has clearly defined the evolution direction and annual goals of AI DC development. Currently, AI DC supports more than 600 business applications, enabling a human-AI collaboration framework. CCB has also built an agent platform and an AI Crowdsourcing Innovation Studio to lower the development threshold, deepen the integration of business and technology, and promote the construction of an AI crowdsourcing ecosystem.

Computing power upgrade and efficiency improvement

CCB has optimized its DC layout to five centers in four cities and built elastic AI computing equipment rooms. It has also improved the heterogeneous AI computing cloud to centrally manage multi-architecture computing power and enable fine-grained compute resource allocation. Through computing platform optimization, full-lifecycle model management, dynamic traffic-based AI compute scheduling, and resource reuse, the bank has increased the number of token calls by nearly three times within half

a year, and the average model performance has improved by about 30%.

Network foundation and AI computing interconnection

CCB has built a multi-plane network architecture for financial-grade hyperscale DCs, improving network interconnection within DCs and effectively controlling fault impacts. CCB has also been the first in the finance industry to achieve lossless transmission over long distances, keeping the cross-domain computing communication degradation rate below 5% over a distance of 1,000 kilometers. In addition, CCB has made breakthroughs in cutting-edge technologies like AI compute scheduling under heavy loads, with the network throughput of multi-node AI training exceeding 98%.

Higher quality, efficiency, and secure operations

Currently, more than 20 intelligent application scenarios have been implemented, significantly improving the quality and efficiency of O&M, business operations, security, and compliance. In terms of O&M and business operations, the average fault locating time in typical complex scenarios is expected to be reduced by 50%, and the efficiency of generating resource application solutions is improved by 50%. In terms of compliance management, the time required for manual contract payment review is reduced by more than 95%, and the approval period for outsourcing personnel to enter the site is shortened by 81%. In terms of security, intelligent filtering and noise reduction are performed on massive security alarm logs, reducing the proportion of invalid alarms by 90%. AI-assisted alarm analysis reduces the time required for analyzing a single alarm by 80%. The ability to detect attacks targeting agents is improved by three to five times.

4.3 Zhongshan Hospital's AI DC Practice in the Healthcare Industry

Zhongshan Hospital Affiliated to Fudan University is tasked with building China's national AI application pilot base (for healthcare). Positioned to focus on clinical research, resolve industry pain points, and build an ecosystem benchmark, the hospital is committed to creating a national innovation support platform that covers the entire chain from R&D and testing to verification and application. So far, Zhongshan Hospital has built a pilot and verification platform that integrates the computing infrastructure, corpus knowledge base, healthcare large models and agents, and multi-scenario clinical verification and evaluation, laying a solid foundation for the large-scale application of AI in clinical medical research.

Challenges

• AI infrastructure construction and O&M

AI iterates on a quarterly basis, and the infrastructure has to be a continuously changing system. Requirements in different scenarios are uncertain, making it difficult to accurately predict the computing demand scale and dynamically allocate resources. The evolution of cabinet power from 16 kW to 200 kW brings disruptive challenges to power supply and distribution, as well as cooling in equipment rooms. Furthermore, the healthcare industry typically lacks O&M staff and tools that can manage large-scale AI clusters and cloud-based infrastructure.

• Massive amounts of multimodal data, model training, and inference

Multimodal data from whole-slide pathology images, genomic sequences, and long-text medical records is growing exponentially, creating exceptionally large files that use diverse protocols (NFS/CIFS/S3). This requires high efficiency in data storage and transfers. In addition, long-context inference in the healthcare domain demands extremely short TTFT and high token throughput. The storage and compute architecture must satisfy capacity, interoperability, and performance requirements.

• Security, compliance, and network construction in the healthcare domain

Privacy data related to personal health is highly sensitive. Cluster management is performed based on domain and

rights, making fault locating difficult and resources inflexible. In addition, cross-DC O&M costs are high, service provisioning efficiency remains low, and traditional private lines deliver limited bandwidth, making it difficult to support the rapid iteration and pilot verification of clinical services.

Innovation

Zhongshan Hospital has built its AI infrastructure based on a "1+3+1" system: one computing foundation, three capability domains, and a Super Doctor matrix.

A unified computing foundation resource platform that integrates AI compute, general-purpose compute, storage, network, cloud platform, and toolchain resources has been built, using advanced chips to deliver world-class computing performance. This unified platform supports the development of three key capabilities: model optimization tools, the healthcare skills workshop, and the agent factory. Ultimately, Zhongshan Hospital has created a Super Doctor agent matrix for clinical scenarios, covering services like intelligent customer service, Electronic Medical Record (EMR) generation, and pathology imaging-assisted diagnosis. This allows the computing infrastructure to efficiently contribute to AI-powered healthcare value creation.

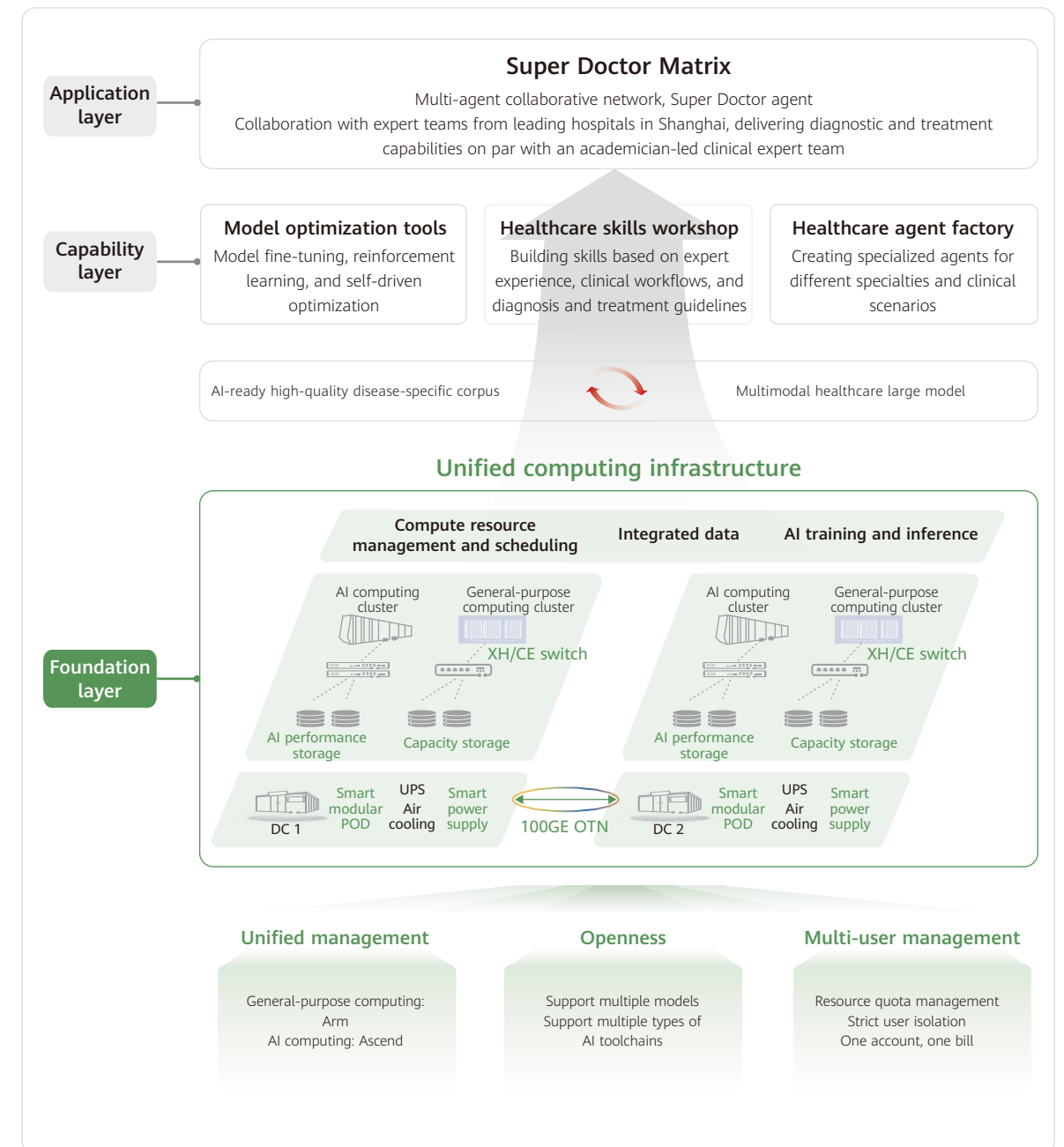


Figure 4-4 Zhongshan Hospital's AI DC solution architecture

Zhongshan Hospital has planned in advance based on innovative concepts like compute-network integration, compute-storage collaboration, and intelligent management and O&M. At the foundation layer, to enable unified management, openness, and unified multi-user operations, the pilot base has created a seamless pipeline across four layers: data management, compute scheduling, storage and compute efficiency improvement, and intelligent O&M.

- **Data management:** Multi-protocol lossless interworking enables the efficient transfer of multimodal data, such as pathology, imaging, genomics, and textual data, on a unified storage foundation. Scenario-based compression for healthcare data maximizes the business value of each byte of storage.

- **Compute scheduling:** Inference by day and training by night, along with dynamic scaling, enables on-demand reuse of computing power. The serverless XPU technology significantly reduces the time required for deploying large models, achieving zero wait times for computing power access and improving compute utilization. Heterogeneous compute scheduling matches the demands of different scenarios, such as scientific research and clinical applications, maximizing compute resource utilization efficiency.

- **Storage and compute efficiency improvement:** The Unified

Cache Manager (UCM) is adopted to optimize inference for ultra-long medical text. This shortens the TTFT and improves the throughput of models, while also increasing the number of users served by AI applications, maximizing the compute utilization of AI clusters.

- **Intelligent O&M:** An observable, predictable, and intelligent O&M system is established, covering everything from model application observability to prevention and identification of optical module contamination. Intelligent O&M analysis based on the AI large model enables intelligent alarm clustering, automated root cause analysis, and accurate push notifications of troubleshooting suggestions, improving O&M efficiency while shortening the time needed.

Achievements

- **Rapid construction, efficient utilization, and intelligent O&M of computing infrastructure:** The inference by day and training by night approach, dynamic scaling, and the serverless XPU technology improve compute utilization efficiency by 20%. Full-stack unified O&M and resource provisioning enable efficient data flows within the cloud.

- **The flywheel effect for AI data and models:** A unified data view is provided for the entire base, building a data warehouse for AI training and inference. It supports fast access and aggregation of all types of multimodal data, and offers knowledge base, memory, and KV cache acceleration capabilities to achieve millisecond-level retrieval response. The pilot base's AI platform enables the intelligent labeling of all types of data and supports multimodal AI corpus generation, supporting rapid model iteration so that models deliver higher accuracy the more they are used.

- **Full-stack security and compliance system, and an AI DC and security network with high computing efficiency:** The AI DC emphasizes on-premises data storage, privacy protection, and compliance. The lossless RoCE network, combined with the proprietary Network Scale Load Balancing (NSLB) algorithm, achieves nearly 98% communication throughput, supports 200G/400G high-speed interconnection, and ensures stable communication in clusters with tens of thousands of cards. In addition, ultra-high bandwidth and ultra-low latency optical transmission technologies improve cross-DC service provisioning efficiency, striking a balance between technological innovation and data privacy protection. This provides deterministic assurance for innovation service development at the pilot base.



| Outlook

The rise of world models, the large-scale adoption of agents, and the continuous evolution of trillion-parameter MoE models are profoundly reshaping what AI DC means and how it works.

Foundation models are transitioning from one-time training and static deployment to continuous online evolution. Agentic AI is ushering in a new era of recursive self-improvement, driving exponential leaps in productivity.

AI infrastructure is undergoing rapid and systematic restructuring. Leveraging high-speed interconnection buses and global memory pooling, we are building 10,000-card-scale SuperPoDs that integrate general-purpose and AI computing. Optical-electrical integration shatters communication bottlenecks, allowing for the construction of hyperscale clusters containing hundreds of thousands of cards. These are all redefining the physical boundaries of AI DCs.

Energy and cooling systems face an equally radical reinvention as DCs pivot toward sustainability and intelligence. Cabinets boasting over 200 kW of power density are driving the popularization of liquid cooling. Cold plate

and immersion cooling technologies are working in tandem to push the PUE close to 1.1. The integration of AI into the energy system is enabling compute-electricity collaboration, direct green power supply, and waste heat recovery, supporting zero-carbon operations.

The operations model is also shifting from mere automation to higher levels of autonomy driven by agentic AIOps. Agents monitor node health in real time, and orchestrate power and compute resources at fine granularity, enabling self-healing networks and adaptive scheduling.

AI DCs are no longer just physical spaces that host computing power. Instead, they are the bedrock for advancing toward AGI.

The challenges we face today are the very waves we will ride. Through architectural, systemic, theoretical, and engineering innovation across the full industry chain, we will set sail into a new era of intelligence.

